01

门电路

1 - 4 分立元件门电路

1 - 5 IC门电路

门电路按逻辑功能分：与门、或门、非门、与非门、或非门、与或非门、异或门等。

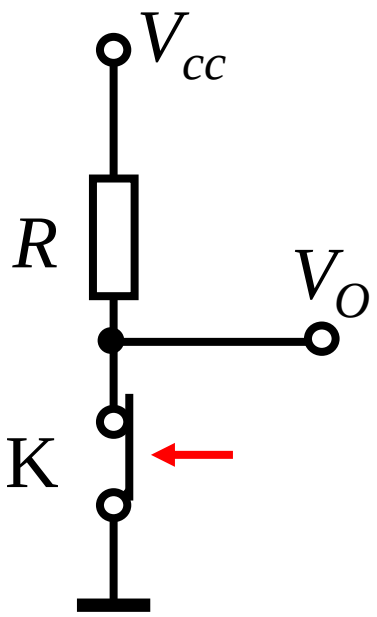
数字电子电路中二值逻辑的1和0两种逻辑状态用两个不同的可以截然区分的电压范围来表示：高电平和低电平

门电路的输出状态与逻辑变量赋值对应关系：

一般采用正逻辑：

高电平 → 1

低电平 → 0



K开------V_O输出高电平，对应“1”

K合------V_O输出低电平，对应“0”

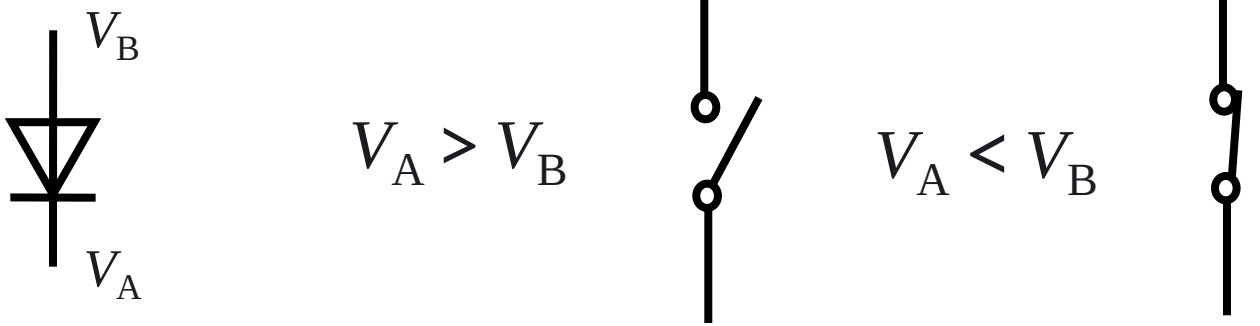
导通电阻为0

断开电阻为∞

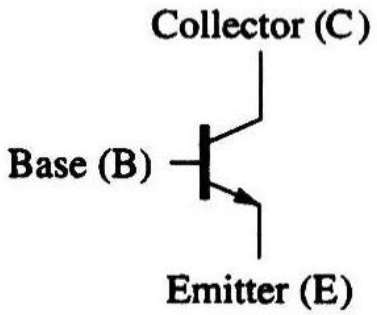
开关延迟为0

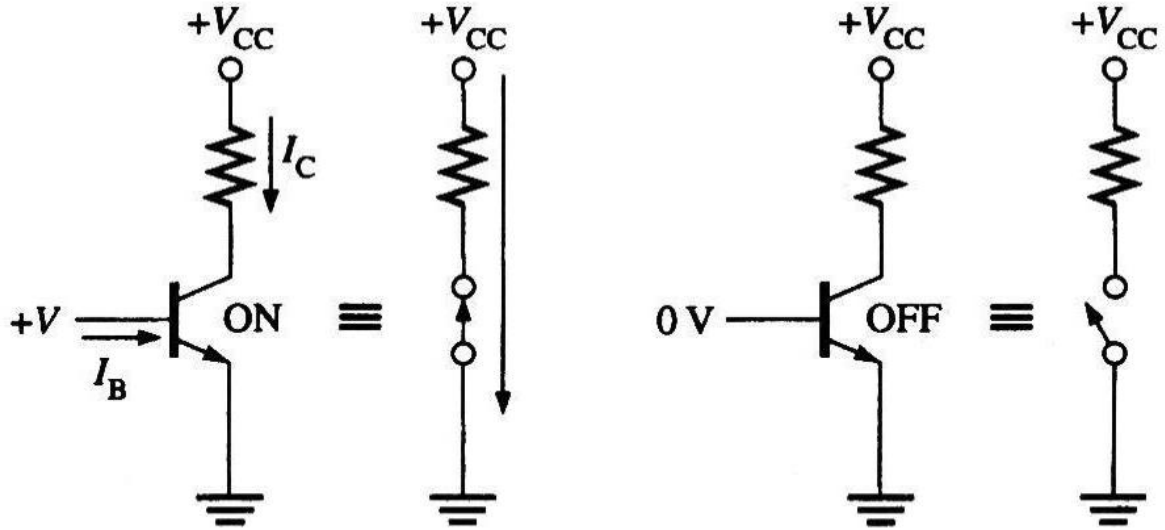
半导体二极管、三极管和MOS管都是构成电子开关的基本元件

1. 二极管的开关作用



2. 三极管的开关作用





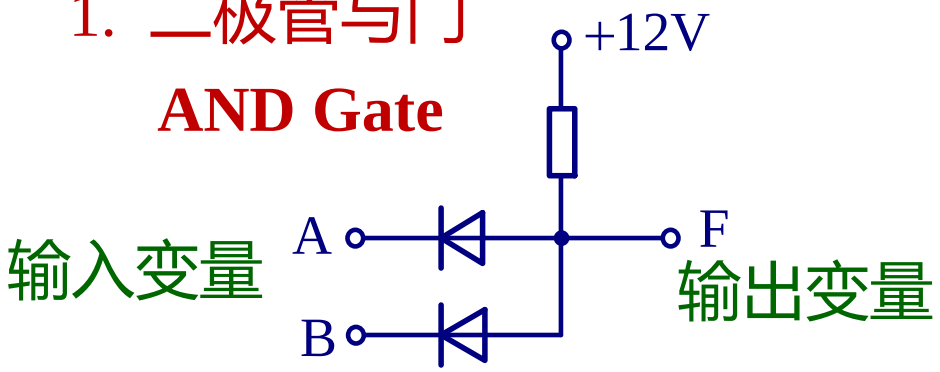
(a) Saturated (ON) transistor and ideal switch equivalent

(b) OFF transistor and ideal switch equivalent

一、分立元件逻辑门电路

1. 二极管与门

AND Gate



$(u_D = 0.7\text{V})$

u_A	u_B	u_F
0V	0V	0.7V
0V	3V	0.7V
3V	0V	0.7V
3V	3V	3.7V

3V以上为逻辑1，0.7V以下为逻辑0（正逻辑）

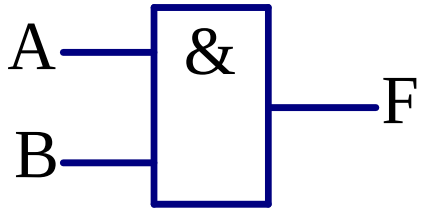
Truth Table

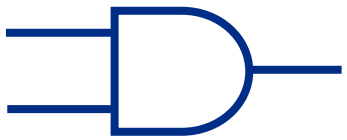
A	B	F
0	0	0
0	1	0
1	0	0
1	1	1

Logic expression：

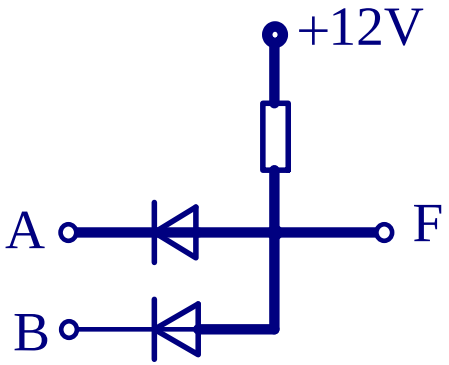
$$F = A \cdot B$$

Symbol：





1. 二极管与门



u_A	u_B	u_F
0V	0V	0.7V
0V	3V	0.7V
3V	0V	0.7V
3V	3V	3.7V

若：3V以上为逻辑0，
0.7V以下为逻辑1
（负逻辑）

真值表（正逻辑）

A	B	F
0	0	0
0	1	0
1	0	0
1	1	1

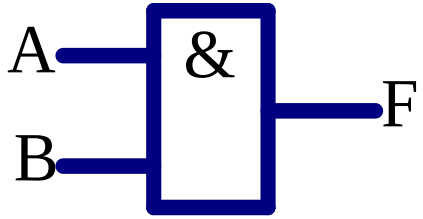
真值表（负逻辑）

A	B	F
1	1	1
1	0	1
0	1	1
0	0	0

逻辑表达式：

$F=A \cdot B$

逻辑符号：

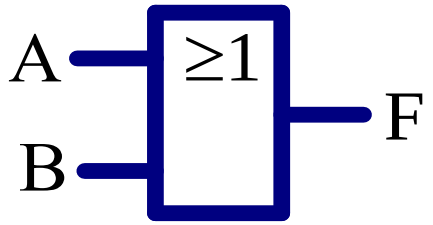


正逻辑与门就是负逻辑或门

逻辑表达式：

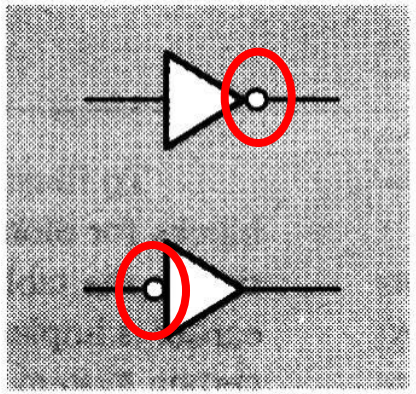
$F=A+B$

逻辑符号：

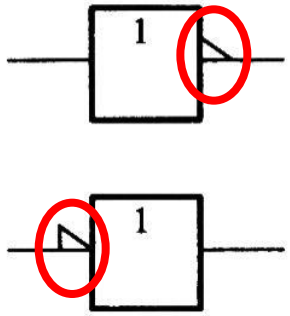


关于反相器的符号说明

bubble



(a) Distinctive shape symbols with negation indicators



(b) Rectangular outline symbols with polarity indicators

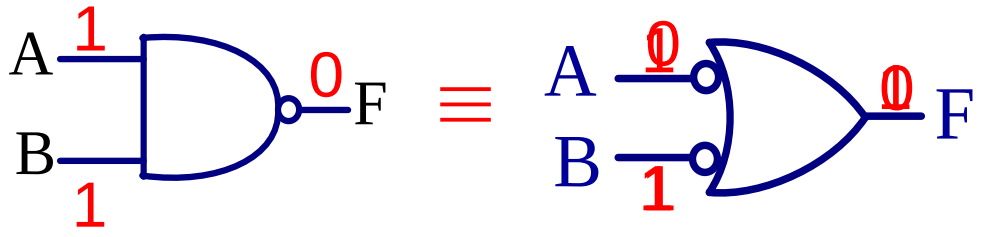
triangle

When appearing on the input/output, it means that a 0 is the active or asserted input/output state

When appearing on the input/output, it means that a LOW level is the active or asserted input/output state

与非门 NAND Gate (NOT-AND)

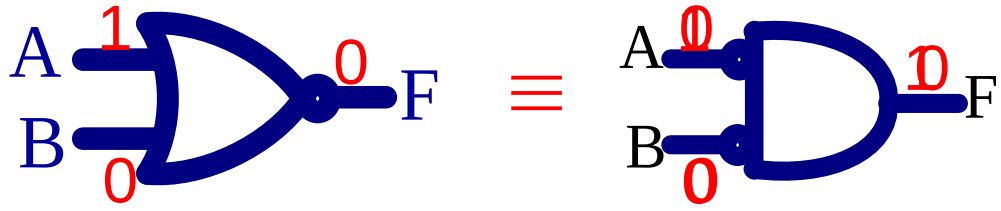
$$F = \overline{A \cdot B} = \bar{A} + \bar{B}$$



A 2-input NAND gate functions as a negative-OR gate

或非门 NOR Gate (NOT-OR)

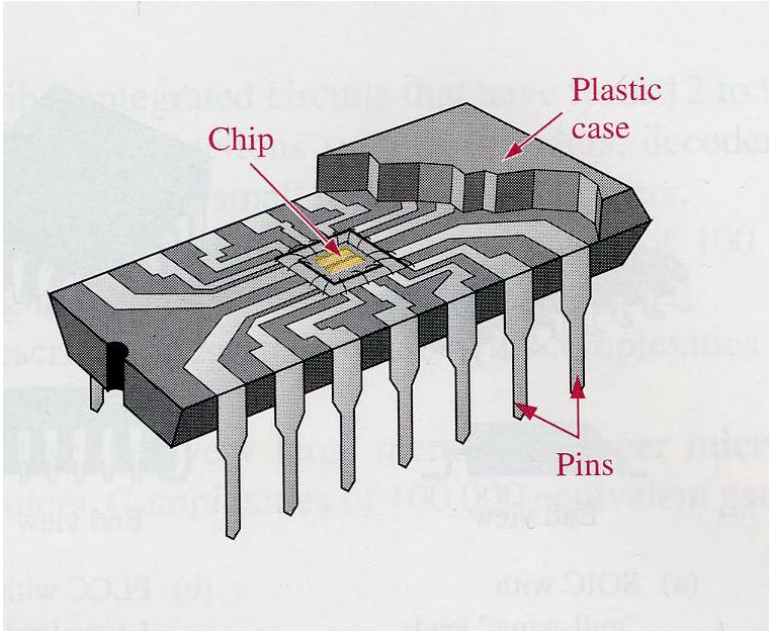
$$F = \overline{A + B} = \bar{A}\bar{B}$$



A 2-input NOR gate functions as a negative-AND gate

分立元件门电路的缺点：

1. 体积大、工作不可靠。
2. 需要不同电源。
3. 各种门的输入、输出电平不匹配。



数字集成电路：在一块半导体基片上制作出一个完整的逻辑电路所需要的全部元件和连线。使用时接：电源、输入和输出。数字集成电路具有体积小、可靠性高、速度快、而且价格便宜的特点。

二、集成电路

按其规模分类

SSI (small-scale integration) <12 gates

基本逻辑门、触发器

MSI (medium-scale integration) 12-100 gates

编码器、解码器、计数器等

LSI (large-scale integration) 100-1,000 gates

内存

VLSI (very large-scale integration) 1000-10,000 gates

digital logic systems, single-chip computers

ULSI (ultra large-scale integration) >100,000 gates

Microprocessor (DSP, CPU)

IC technologies

可用于实现基本逻辑运算的数字集成电路技术：

TTL(bipolar)、CMOS、BiCMOS

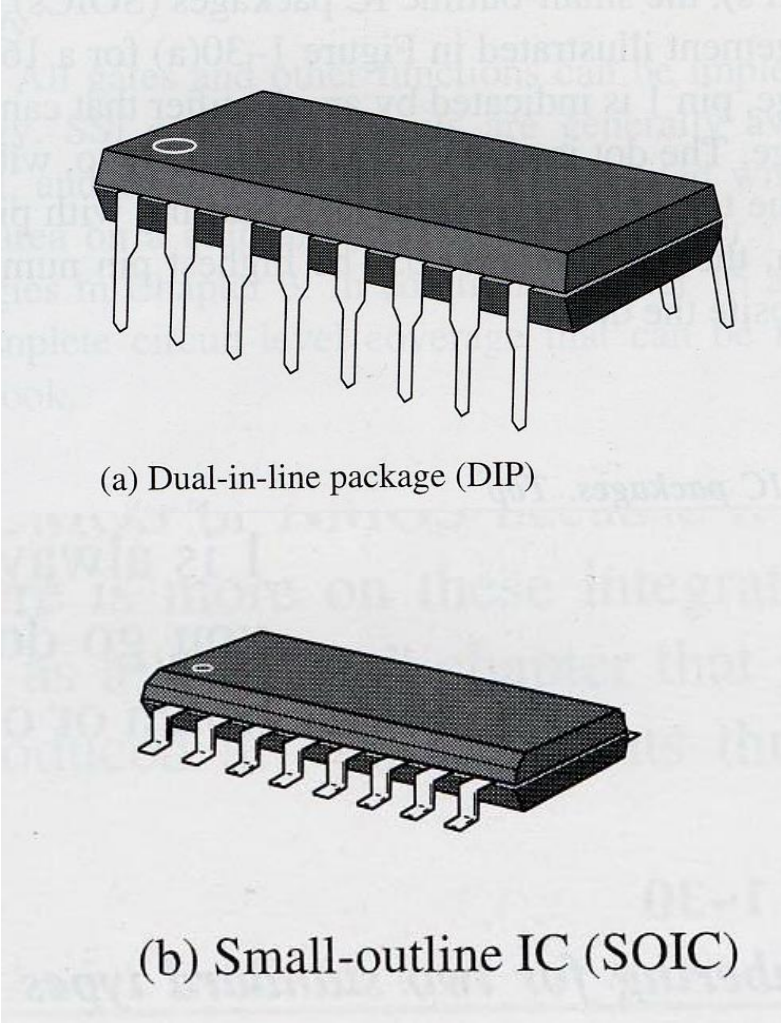
TTL型电路：输入和输出端结构都采用了双极型晶体管，称之为：

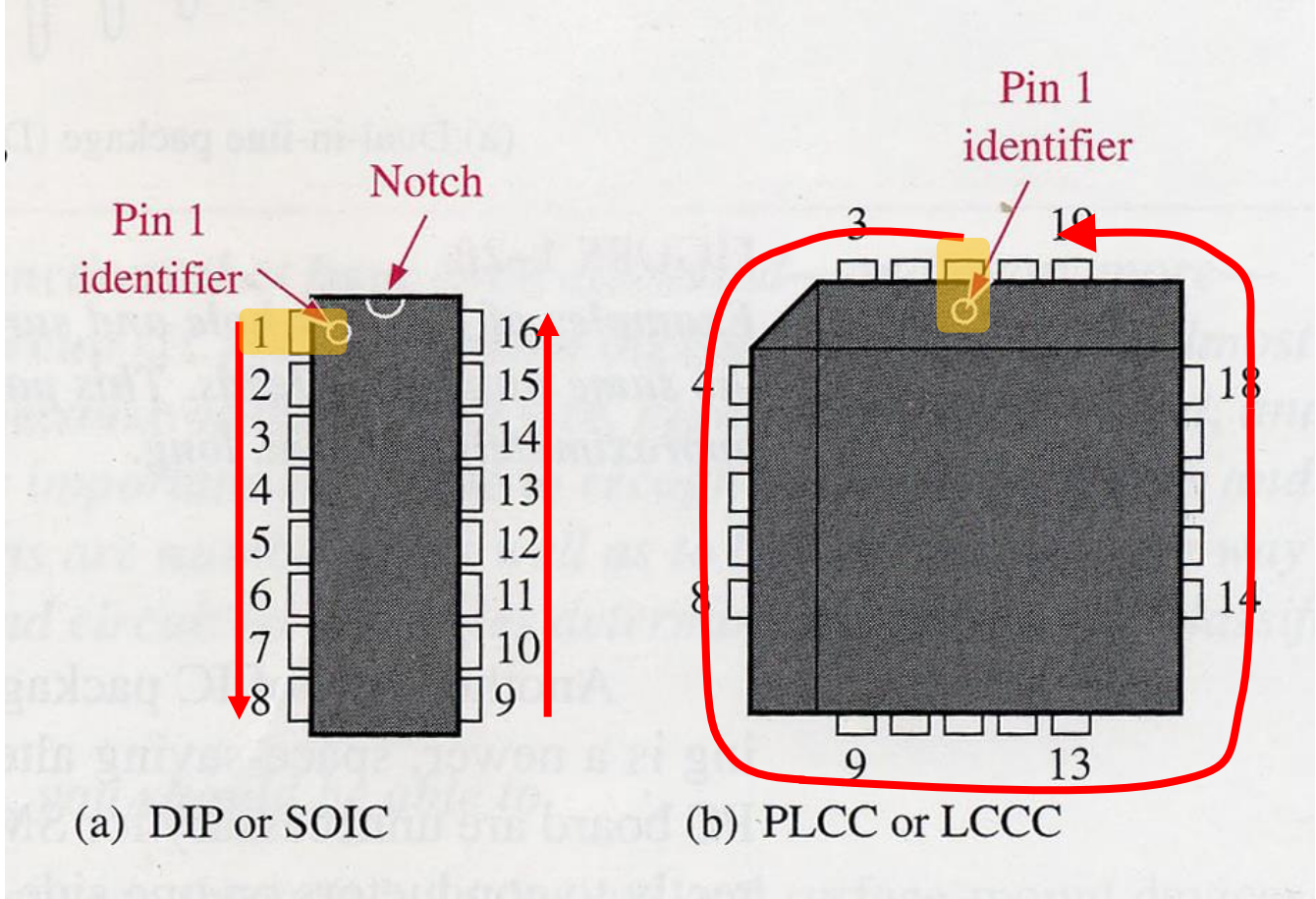
Transistor— Transistor Logic。

CMOS型电路：Complementary Metal-Oxide Semiconductor

互补金属氧化物半导体。

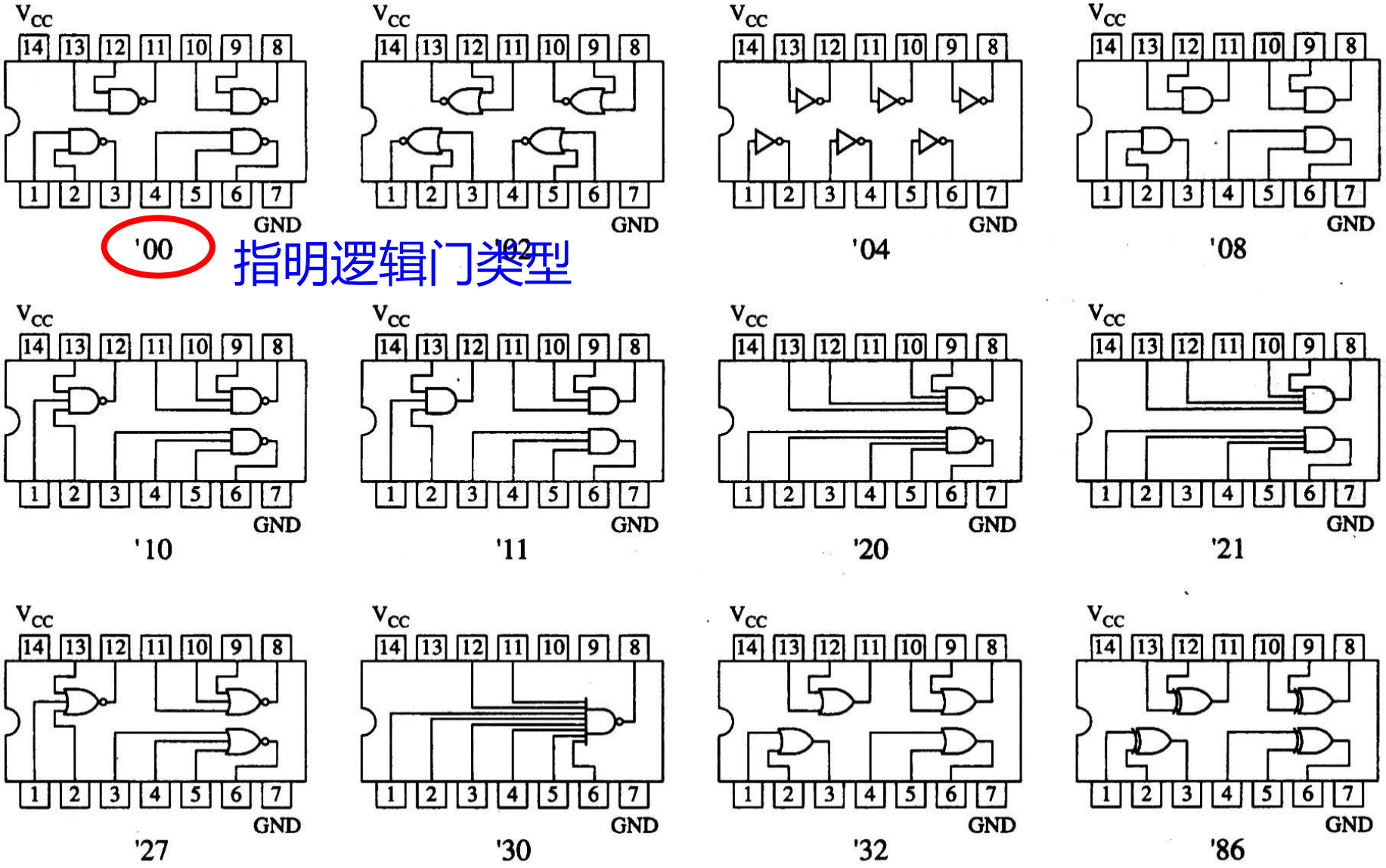
1. 集成电路封装 IC Packages　　2. 引脚编号 Pin Numbering



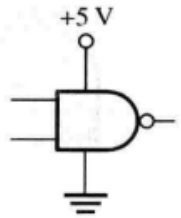


两种集成电路封装的引脚编号格式（俯视图）

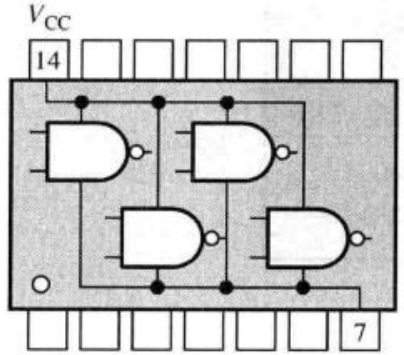
一些常见IC门的引脚配置



指明逻辑门类型

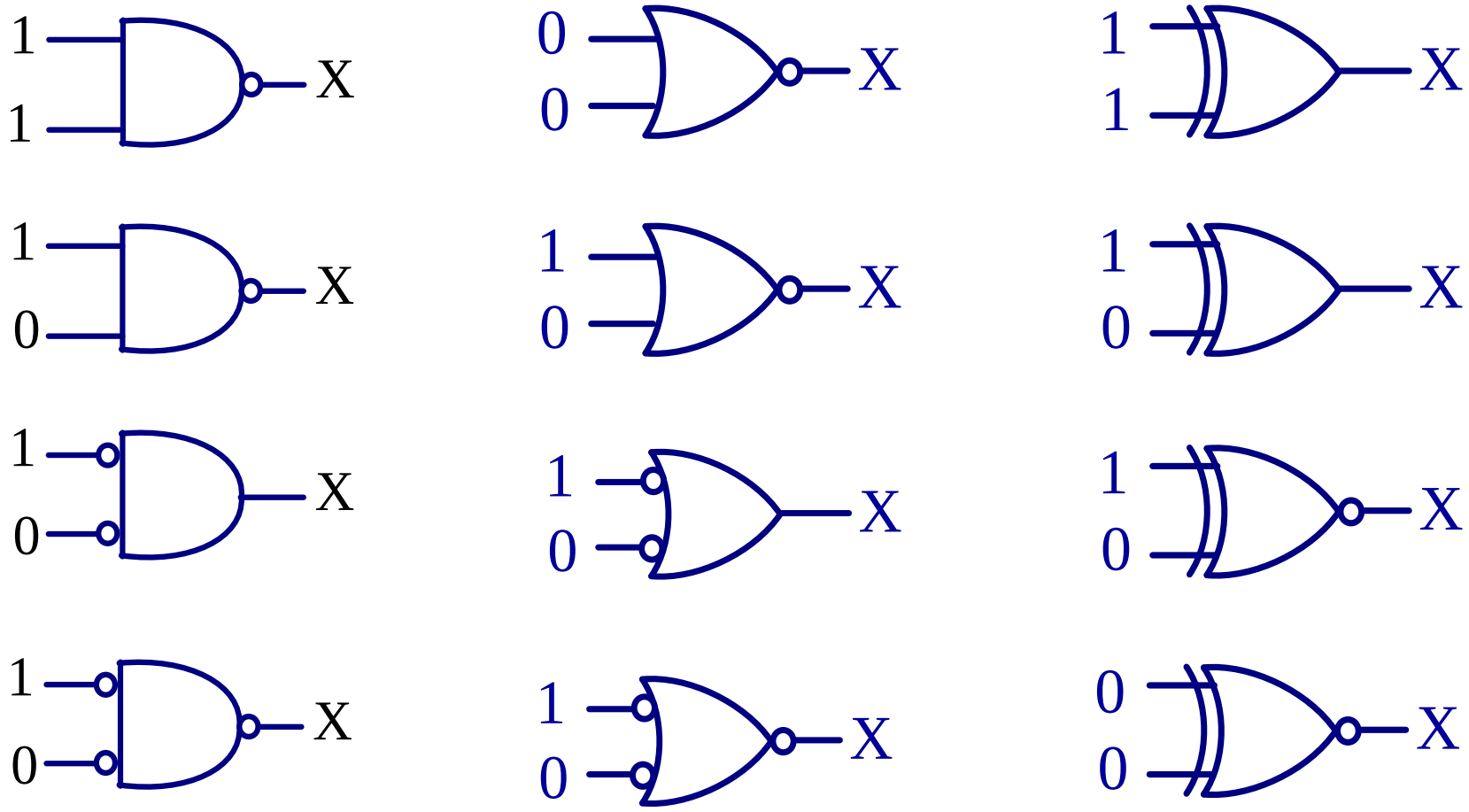


(a) 单个门

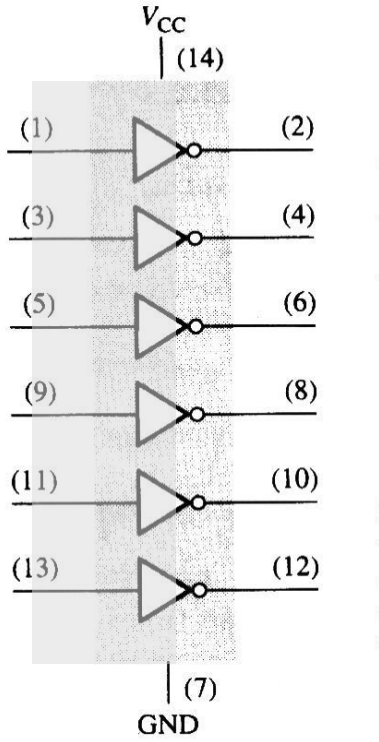


(b) IC 双列直插封装芯片

Determine the output , X , with the inputs shown in following figure

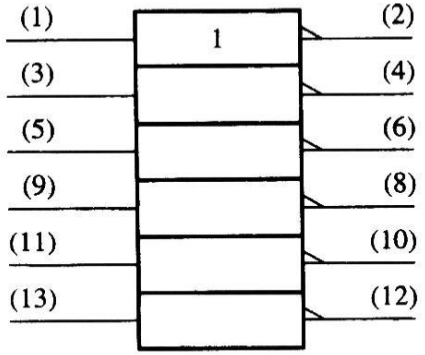


Logic Symbols

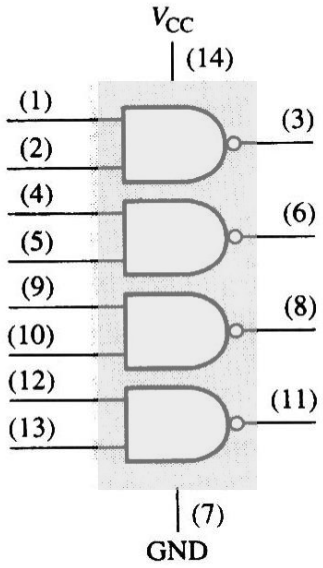


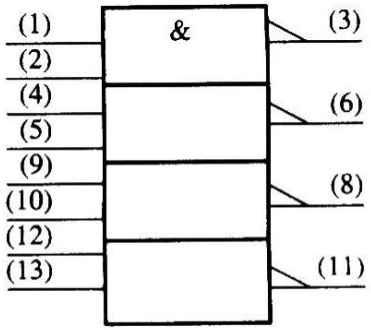
Distinctive shape logic diagram

(a) Hex inverter



Rectangular outline logic symbol with polarity indicators. The inverter qualifying symbol (1) appears in the top block and applies to all blocks below.



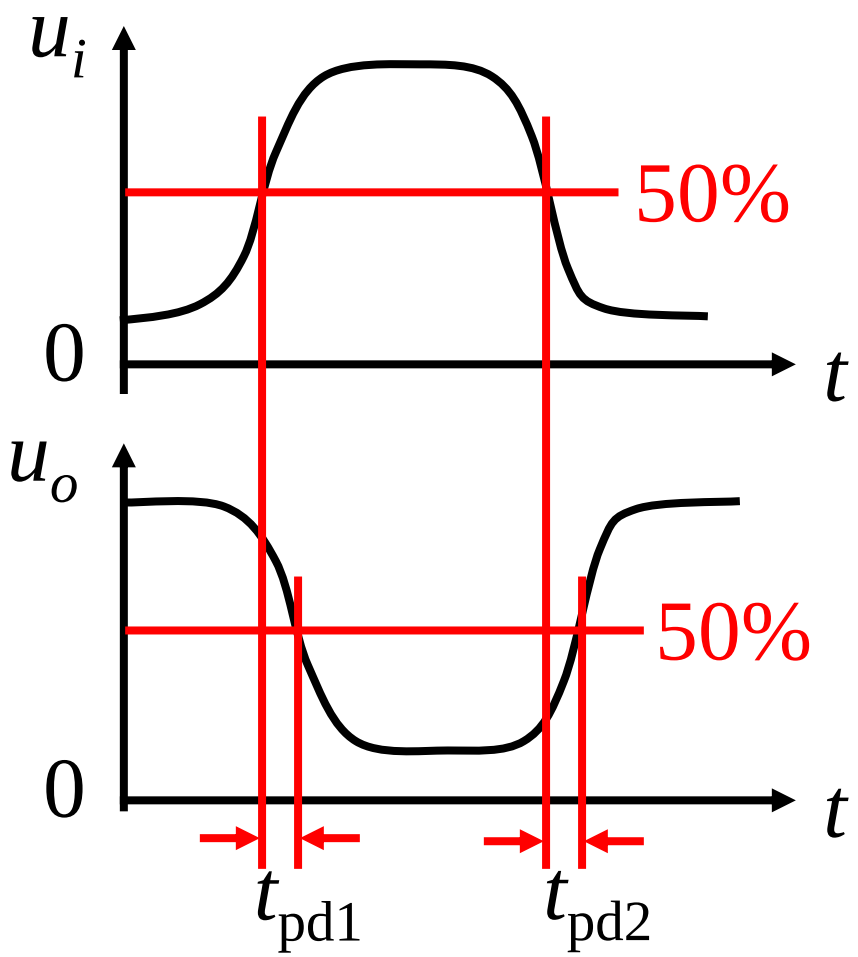


(b) Quad 2-input NAND

逻辑电路的性能参数

门电路的传输延迟限制了它的工作频率，传输延迟小的电路速度较快

1. 传输延迟时间 （以反相器为例）



特定的运行条件下

标准TTL门，典型传输延迟时间为11ns

F系列TTL门，典型传输延迟时间为3.3ns

HCT系列COMS门，典型传输延迟时间为7ns

AC系列COMS门，典型传输延迟时间为5ns

ALVC系列COMS门，典型传输延迟时间为3ns

某种ECL门，典型传输延迟时间为0.22ns

2. 直流电源电压

3. 功耗、速率-功耗乘积（Speed-Power Product）

4. 逻辑电平、噪声容限

5. 负载和扇出系数 Fan-Out

略

02

组合逻辑电路



2 - 1 组合逻辑电路分析

2 - 2 常用的组合逻辑功能模块

数字电路

- 组合逻辑电路
  - 特点：任何时刻的输出只取决于当前的输入。
  - 组成：门电路，输出输入间没有反馈延迟通路，不存在记忆元件。
- 时序逻辑电路
  - 特点：输出取决于当前的输入和原来的状态。
  - 组成：组合电路、记忆元件。

组合逻辑电路的研究内容：

给定逻辑图 —分析→ 得到逻辑功能

给定逻辑功能 —设计→ 画出逻辑图

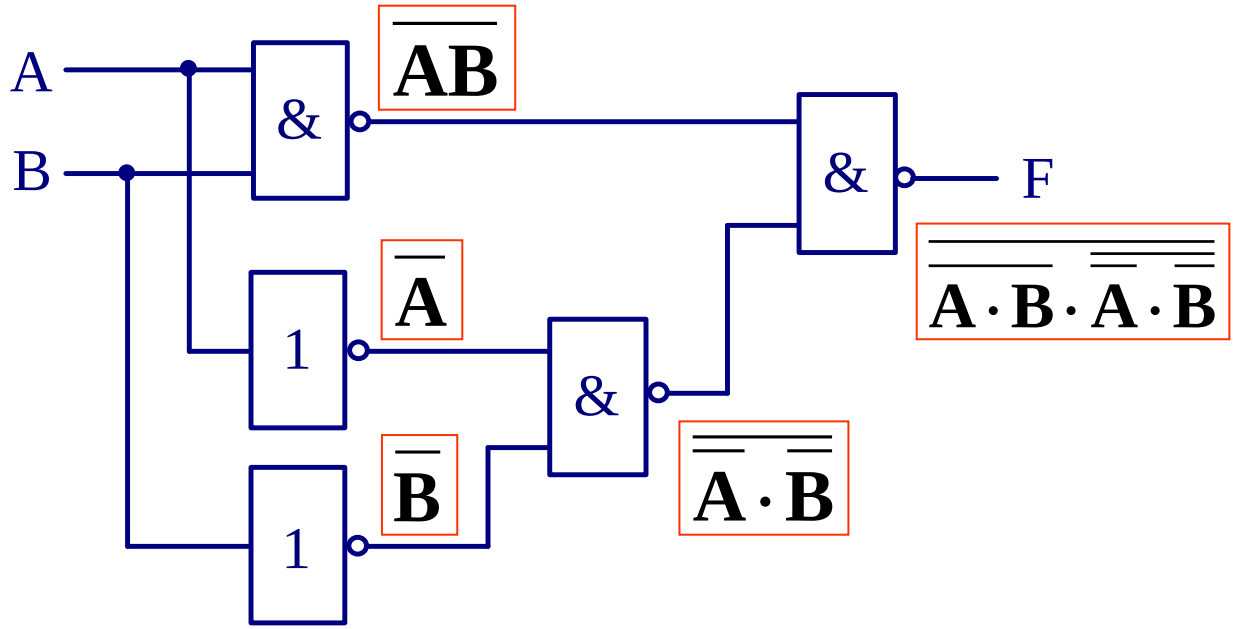
组合逻辑电路分析



分析步骤：

1. 由给定的逻辑图逐级写出逻辑关系表达式；
2. 用逻辑代数或卡诺图化简逻辑表达式；
3. 列出逻辑真值表并得出结论。

例：分析下图的逻辑功能。



$$F = \overline{\overline{A \cdot B} \cdot \overline{A \cdot B}} = \overline{A \cdot B} + \overline{A \cdot B} = A \cdot B + \overline{A} \cdot \overline{B}$$

真值表

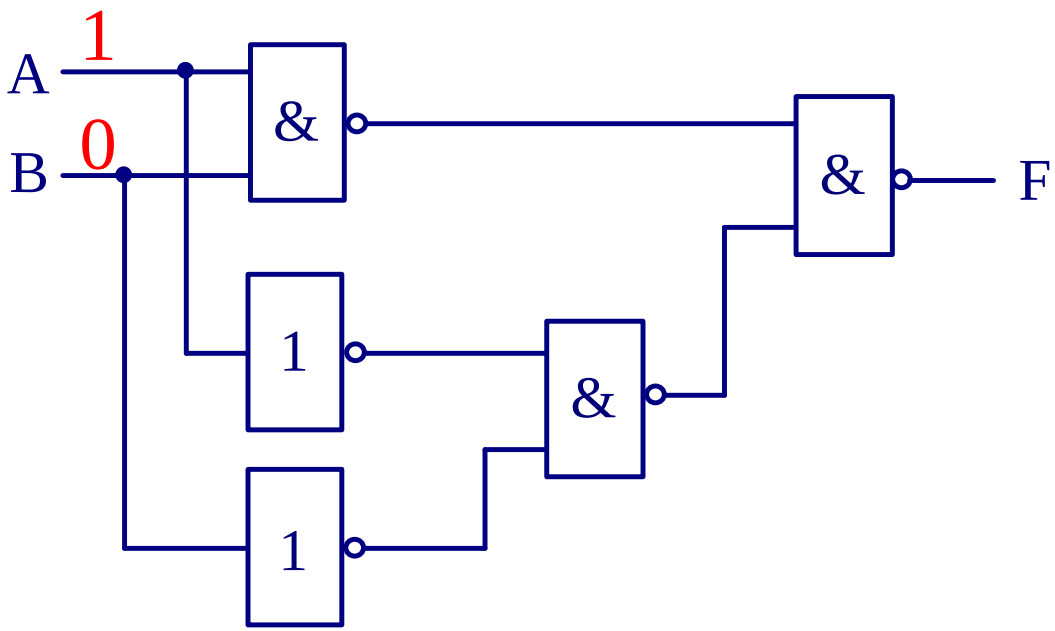
A	B	F
0	0	1
0	1	0
1	0	0
1	1	1

特点：输入相同为“1”；

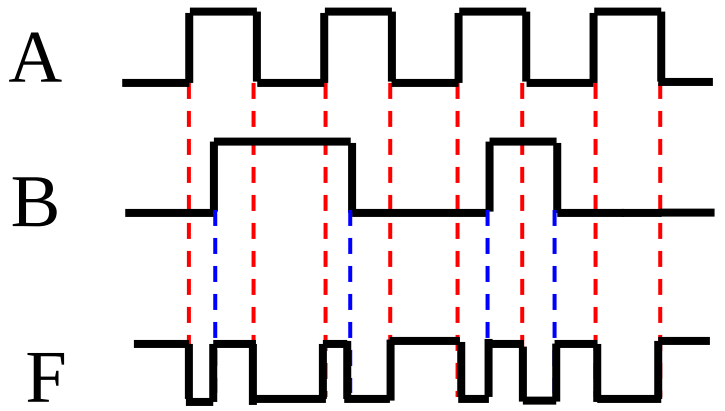
输入不同为“0”。

$$F = \overline{A \oplus B} = A \odot B$$

例：分析下图的逻辑功能，并画出输出波形



$$F = A \cdot B + \overline{A} \cdot \overline{B} = 1 \cdot 0 + \bar{1} \cdot \bar{0} = 0$$



组合逻辑电路设计



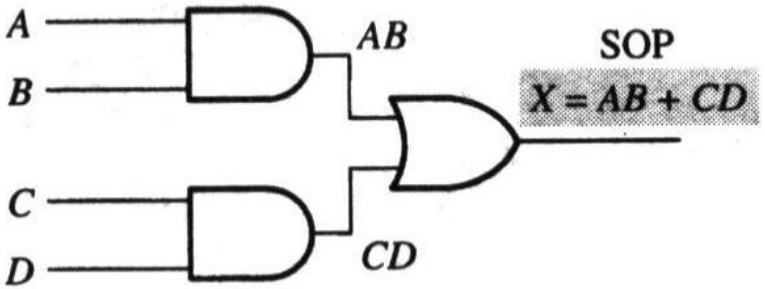
分析步骤：

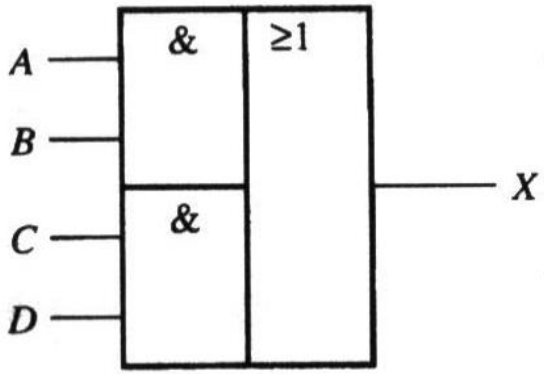
1. 指定实际问题的逻辑含义，列出真值表；
2. 写出逻辑函数并化简或转换；
3. 选择合适的器件，画出逻辑电路图。

基本组合逻辑器件

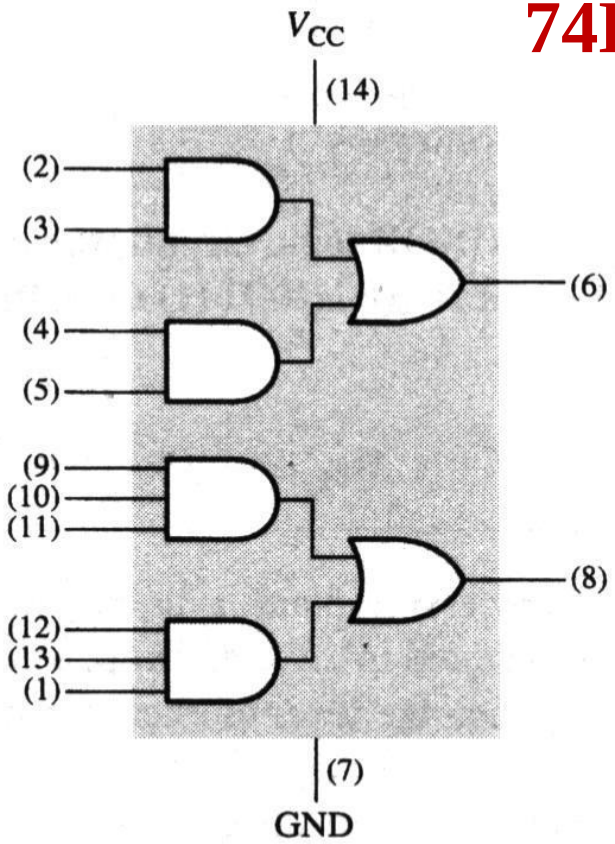
任何逻辑表达式都可以表示成SOP或POS等形式

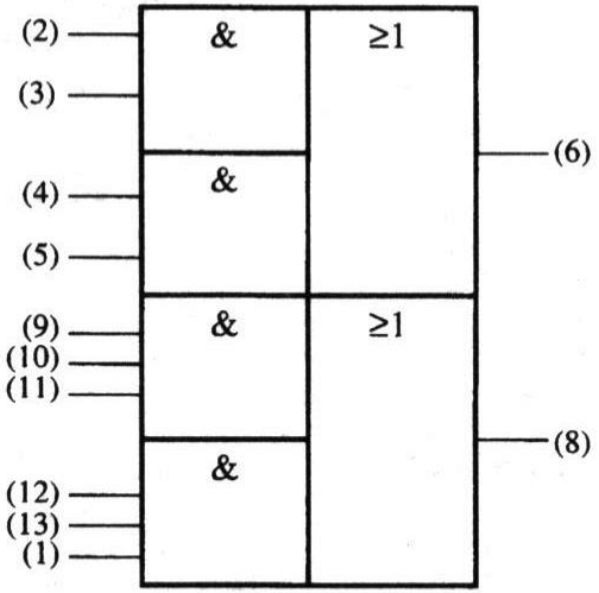
例：Y=AB+CD　　Y=($\bar{A}$+B)(C+D)



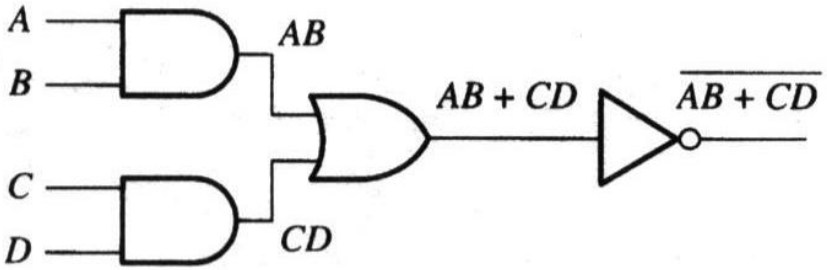


74HC58

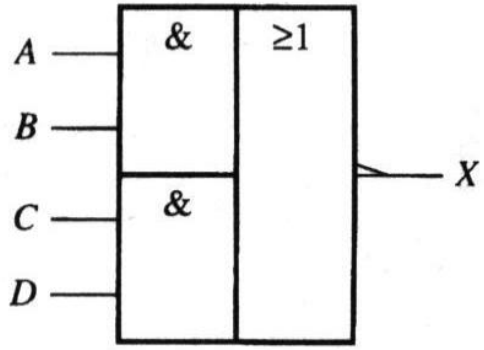




$$Y=\overline{AB+CD}$$

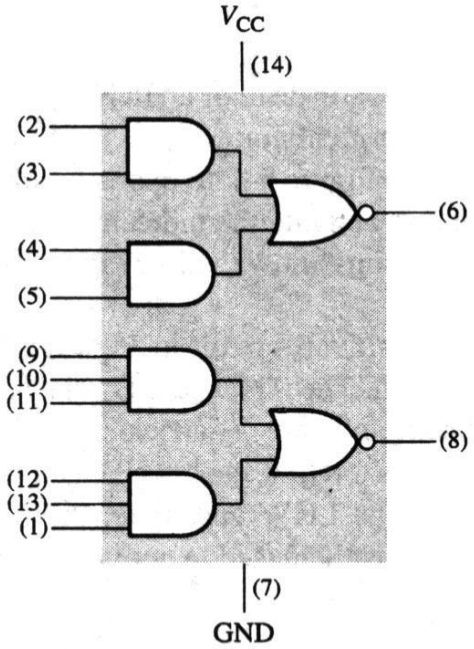


(a)

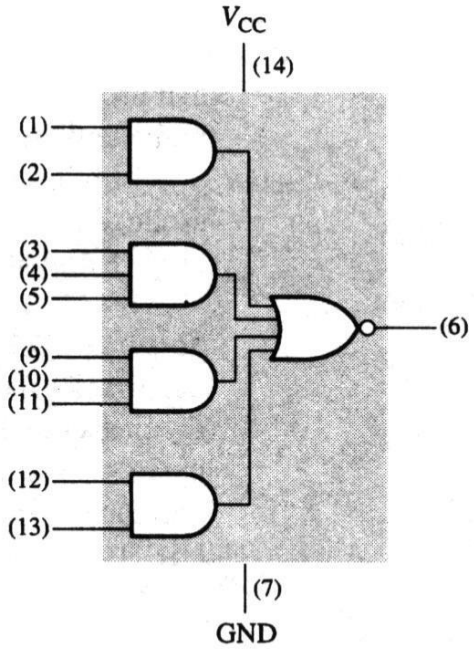


(b)

74LS51、74LS54

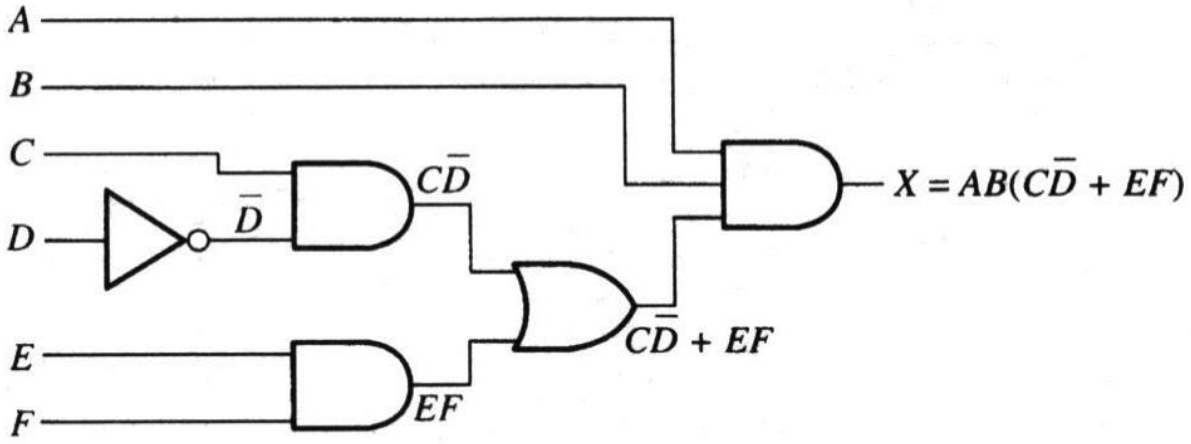


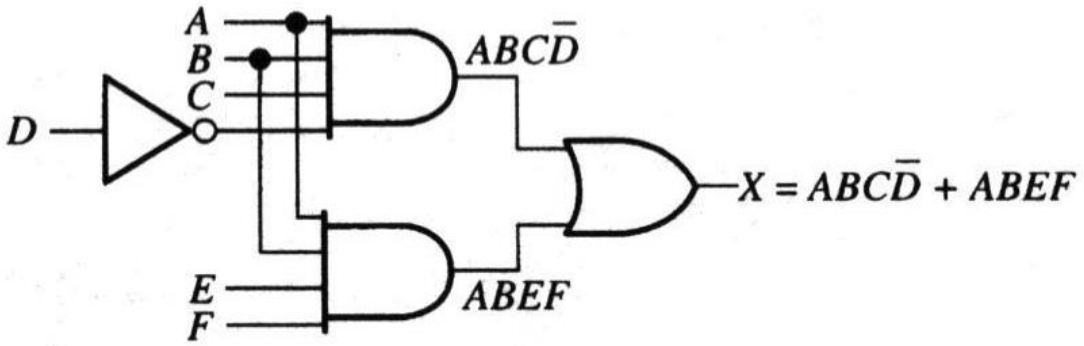
(a) 74LS51



(b) 74LS54

例：用门电路实现逻辑函数 $X = AB(C\overline{D} + EF) = ABC\overline{D} + ABEF$

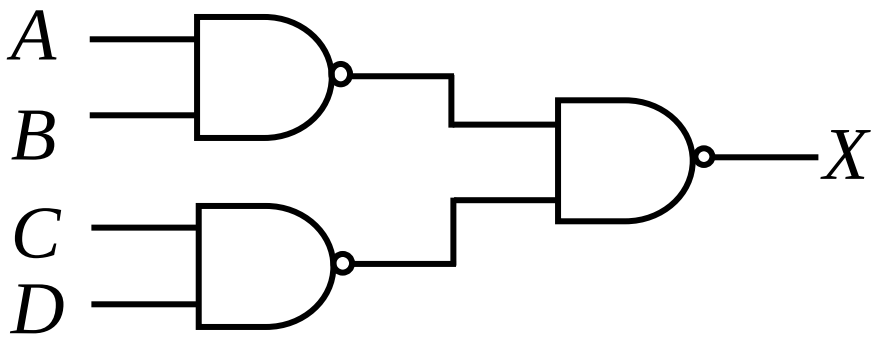




例：用与非门电路实现逻辑函数 $X = AB + CD$

$$X = AB + CD$$

$$= \overline{\overline{AB + CD}} = \overline{\overline{AB} \cdot \overline{CD}}$$

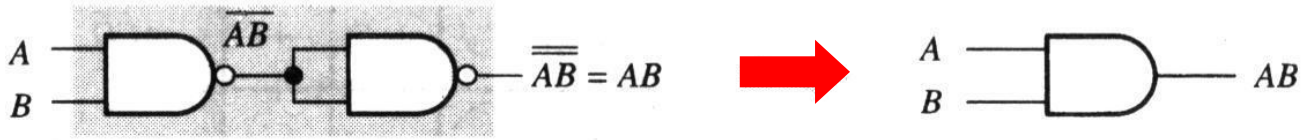


SJTU

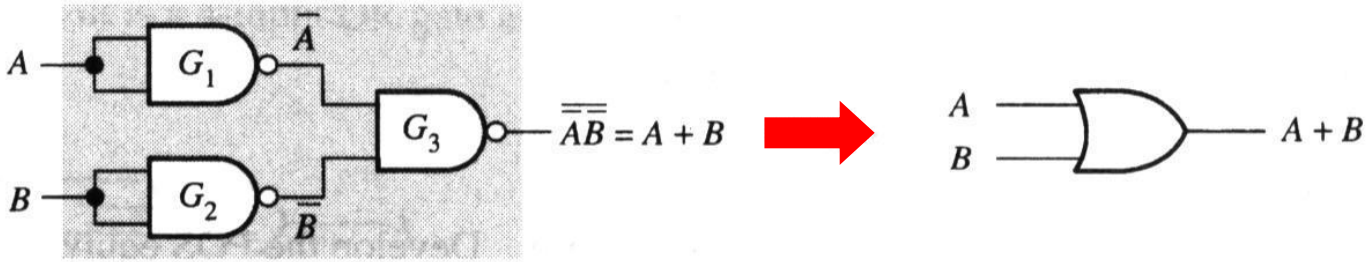
通用门



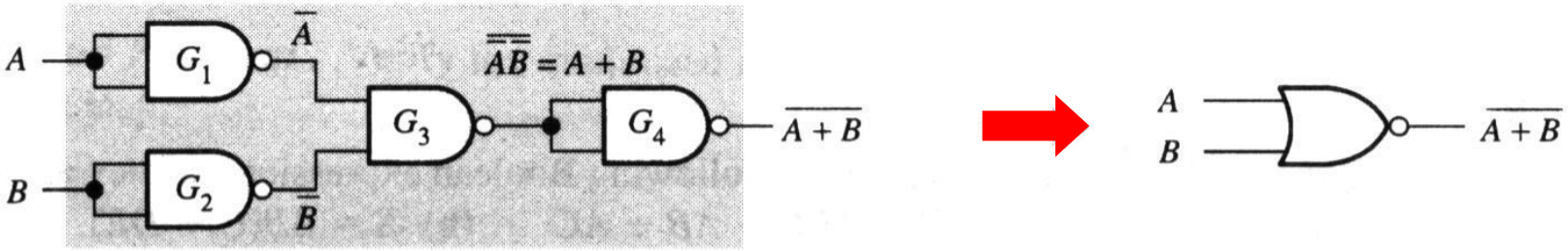
(a) A NAND gate used as an inverter



(b) Two NAND gates used as an AND gate



(c) Three NAND gates used as an OR gate



(d) Four NAND gates used as a NOR gate

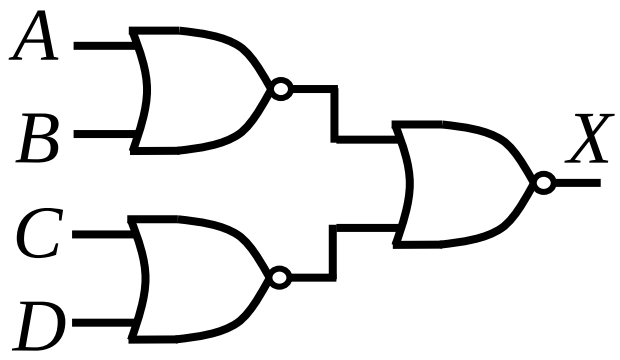
例：用或非门电路实现逻辑函数

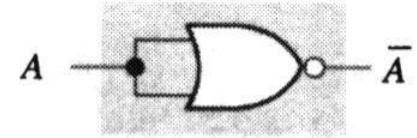
$$X = (A+B)(C+D)$$

$$X = (A+B)(C+D)$$

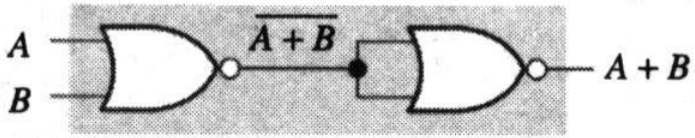
$$= \overline{\overline{(A+B)(C+D)}}$$

$$= \overline{\overline{A+B} + \overline{C+D}}$$

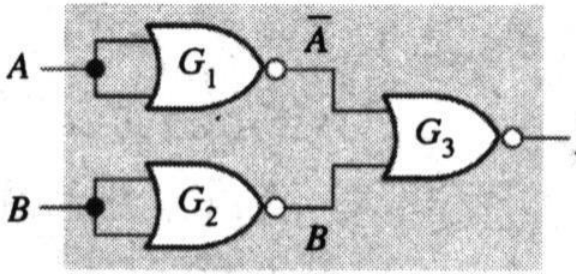




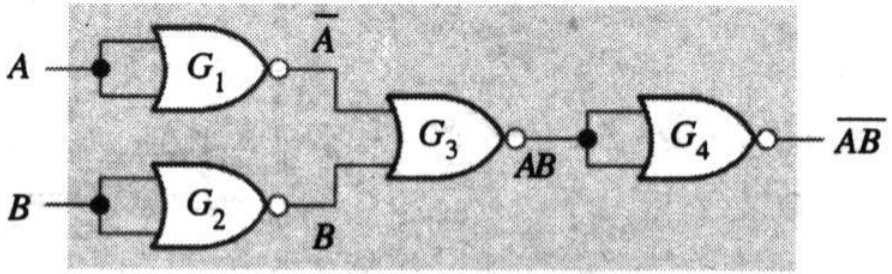
(a) A NOR gate used as an inverter



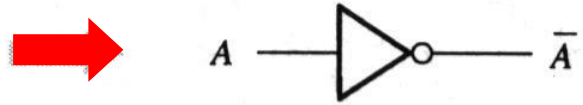
(b) Two NOR gates used as an OR gate

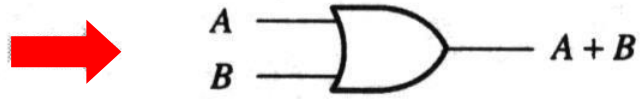


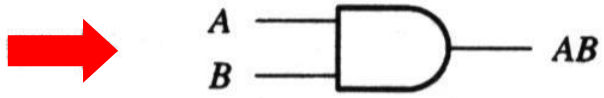
(c) Three NOR gates used as an AND gate

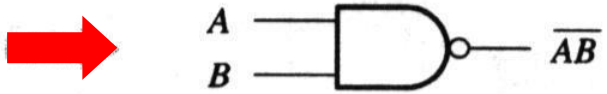


(d) Four NOR gates used as a NAND gate



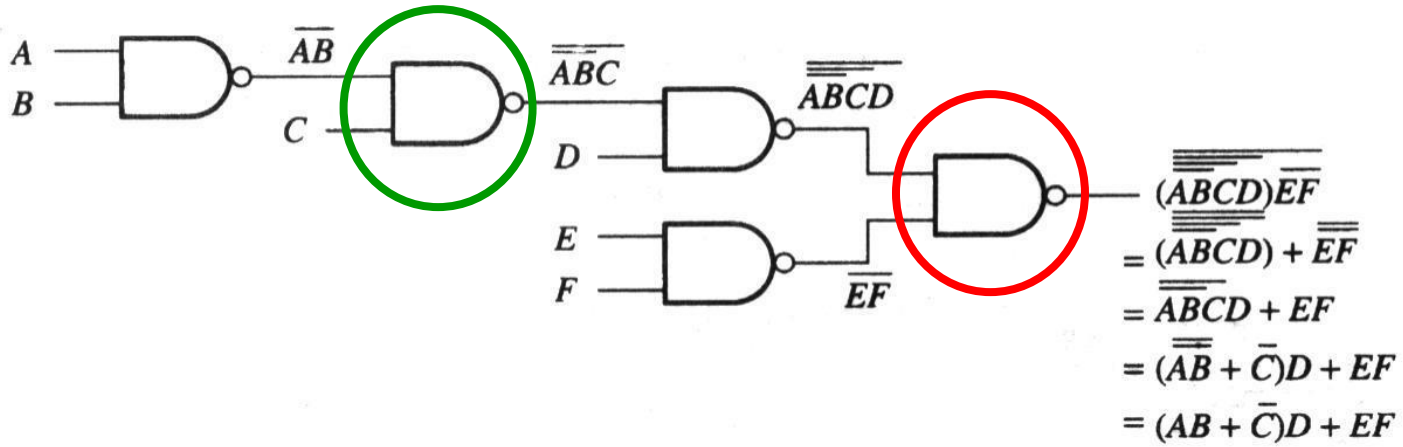


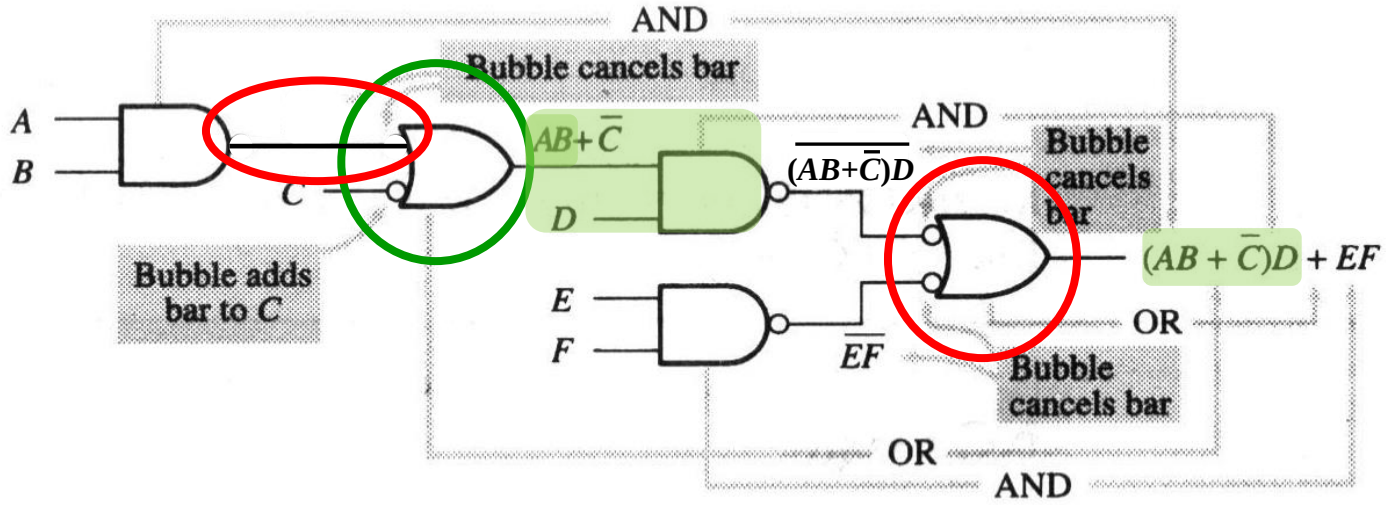




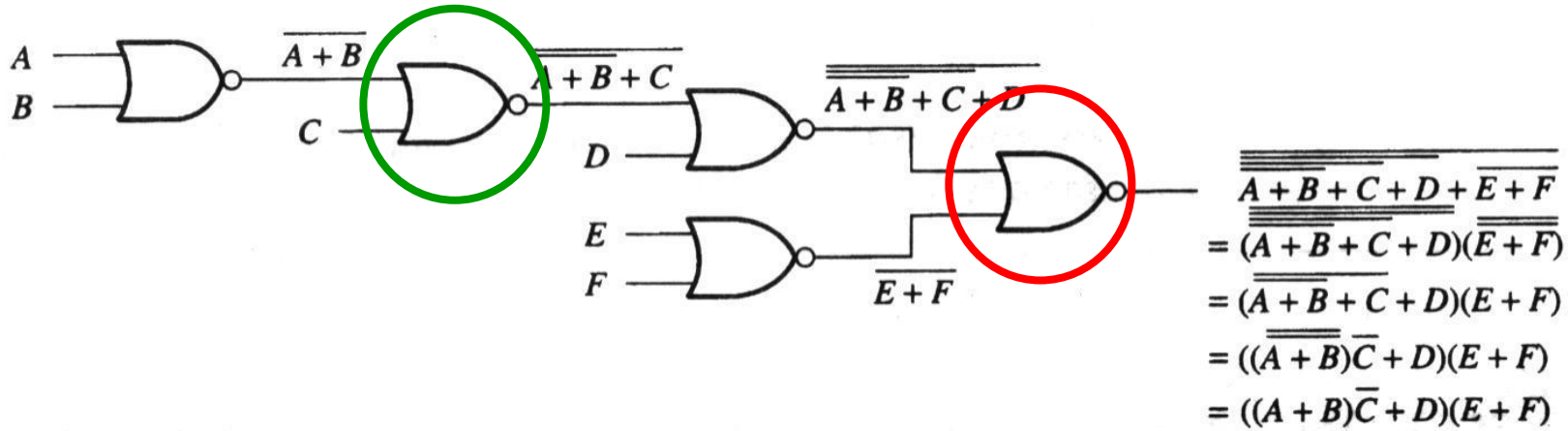
S J T U

$$\overline{AB} = \overline{A} + \overline{B}$$

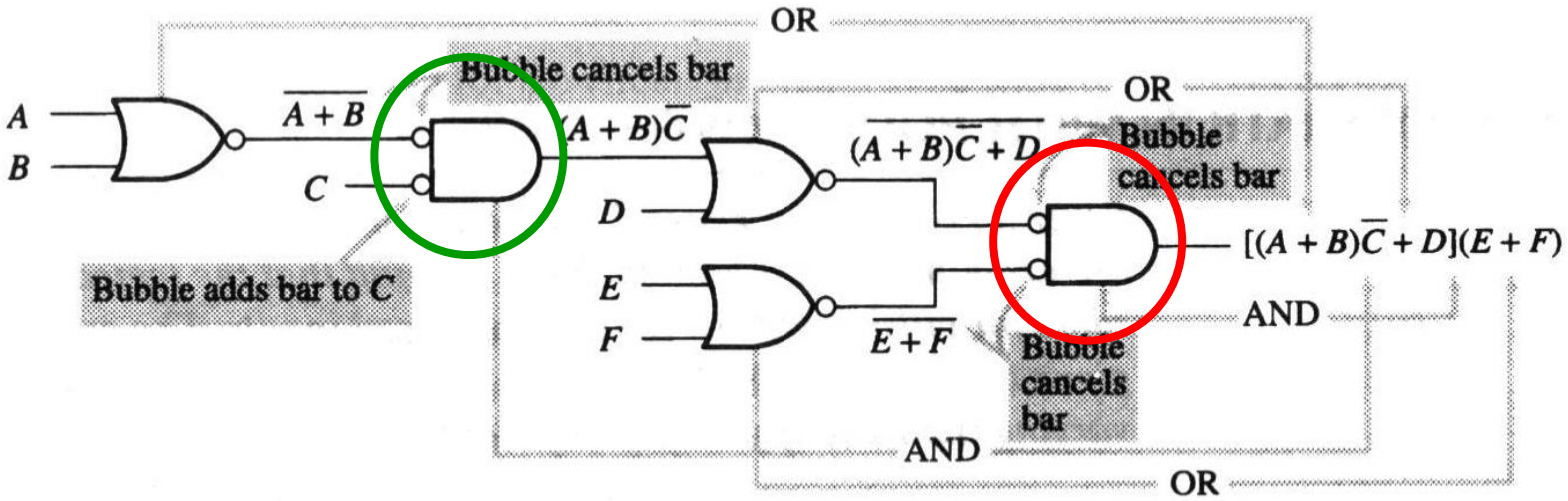




$$\overline{A+B} = \overline{A} \cdot \overline{B}$$



(a) Final output expression is obtained after several Boolean steps.



(b) Output expression can be obtained directly from the function of each gate symbol in the diagram.

例：设计三人表决电路（A、B、C）。每人一个按键，如果同意则按下，不同意则不按。结果用指示灯表示，多数同意时指示灯亮，否则不亮。

1. 首先指明逻辑符号取“0”、“1”的含义；

三个按键A、B、C按下时为1，不按时为0。用逻辑变量F表示表决结果，多数赞成时是1，否则是0。

2. 根据题意列出真值表；

A	B	C	F
0	0	0	0
0	0	1	0
0	1	0	0
0	1	1	1
1	0	0	0
1	0	1	1
1	1	0	1
1	1	1	1

3. 列出逻辑表达式，并化简

$$F = \bar{A}BC + A\bar{B}C + AB\bar{C} + ABC$$

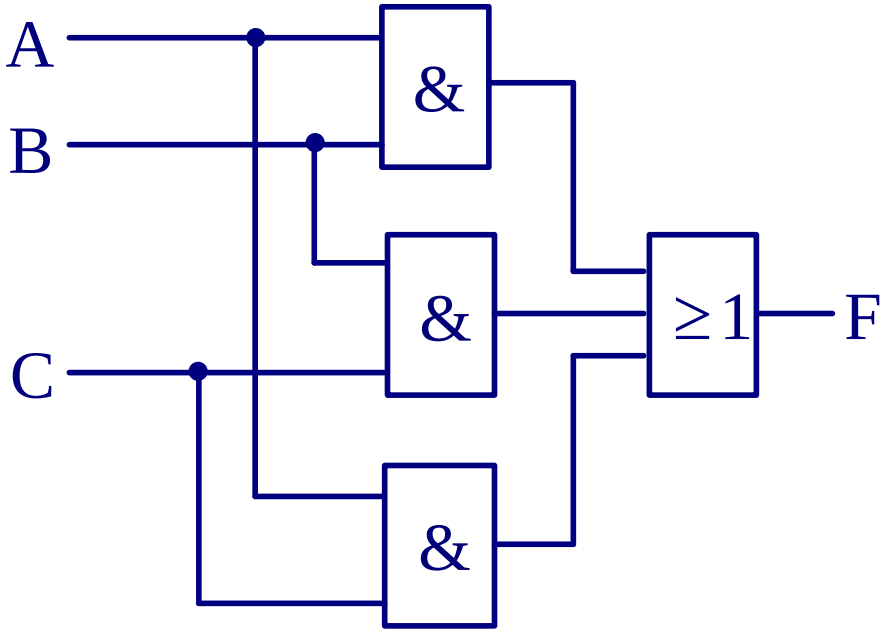
$$= AB + BC + CA$$

A \ BC	00	01	11	10
0	0	0	1	0
1	0	1	1	1

$$F = AB + BC + CA$$

4. 根据逻辑表达式画出逻辑图

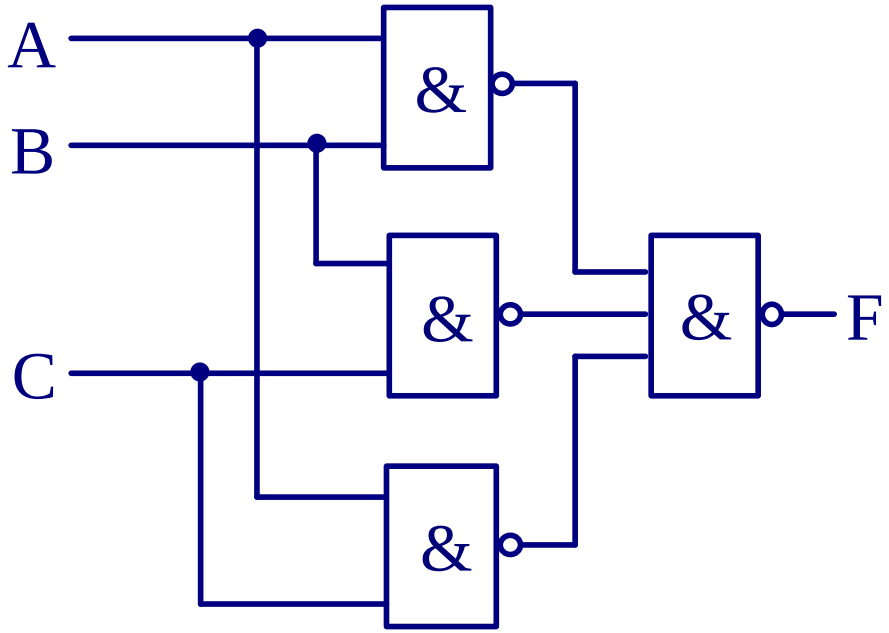
(1) 若用与门和或门实现



(2) 若用与非门实现

$$F = AB + BC + CA$$

$$= \overline{\overline{AB + BC + CA}} = \overline{\overline{AB} \cdot \overline{BC} \cdot \overline{CA}}$$



例： 医院优先照顾重症患者的呼唤电路

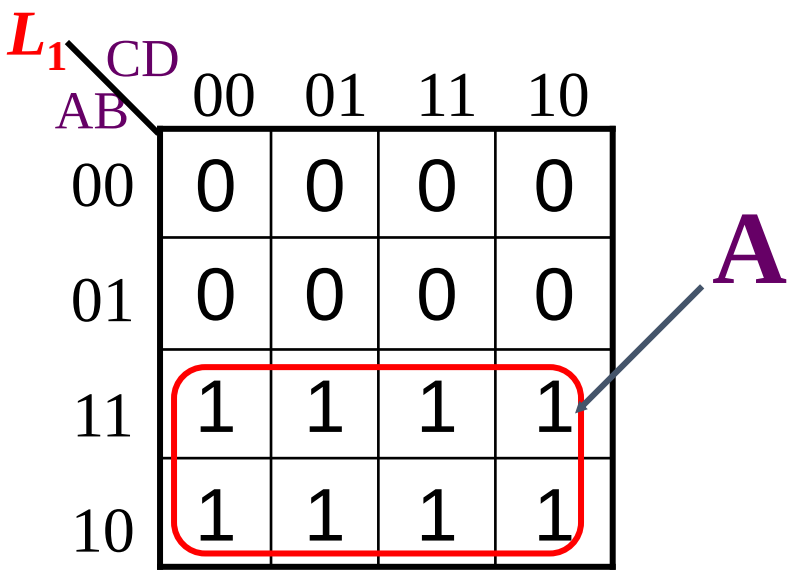
医院1，2，3，4四间病室，分别装有A、B、C、D四个呼唤按钮，值班室里对应的四个灯为L_1、L_2、L_3、L_4，呼唤按钮优先级别由高到低依次为A、B、C、D，设计实现上述功能的逻辑电路

1. 首先指明逻辑符号取“0”、“1”的含义；

   呼唤按钮按下为1，不按为0。灯亮是1，不亮是0。

2. 列出真值表；

A	B	C	D	L_1	L_2	L_3	L_4
1	×	×	×	1	0	0	0
0	1	×	×	0	1	0	0
0	0	1	×	0	0	1	0
0	0	0	1	0	0	0	1



3. 写出逻辑函数式

$$L_1 = A \qquad L_2 = \bar{A}B$$

$$L_3 = \bar{A}\bar{B}C \qquad L_3 = \bar{A}\bar{B}\bar{C}D$$

课堂练习题： 水泵控制逻辑电路

水箱由M_1和M_2 两台水泵供水，水箱中由低到高装有A、B、C三个水位检测元件。要求当水位超过C时停止供水，水位低于C高于B时M_2单独供水，低于B高于A时M_1单独供水， 低于A时两水泵同时供水，试用门电路设计一个控制水泵工作的逻辑电路。

C → 停止供水

C → M_2单独供水

B → M_1单独供水

A → 同时供水

1. 首先指明逻辑符号取“0”、“1”的含义；

水面低于检测元件时用“1”表示，高于检测元件用“0”表示。

水泵供水用“1”表示，停止供水用“0”表示。

注意：水面高于C，则一定高于B和A

2. 列出真值表；

A	B	C	M_1	M_2
1	1	1	1	1
0	1	1	1	0
0	0	1	0	1
0	0	0	0	0

或

A	B	C	M_1	M_2
1	×	×	1	1
0	1	×	1	0
0	0	1	0	1
0	0	0	0	0

3. 写出逻辑函数式

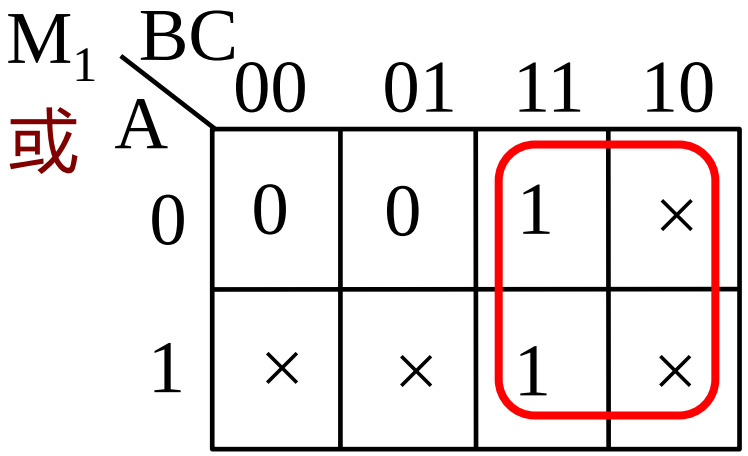
$$M_1 = ABC + \bar{A}BC$$
$$= BC$$

$$M_2 = ABC + \bar{A}\bar{B}C$$

$$M_1 = A + \bar{A}B$$
$$= A + B$$

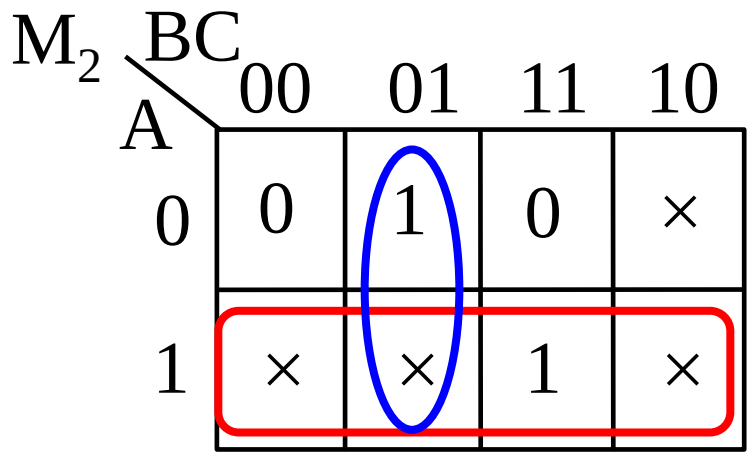
$$M_2 = A + \bar{A}\bar{B}C$$
$$= A + \bar{B}C$$

或



$$M_1 = B$$

$$M_2 = A + \bar{B}C$$



02

常用的组合逻辑器件



内容安排



01-加法器 **Adders**

02-数值比较器 **Comparators**

03-编码器 **Encoders**

04-译码器 **Decoders**

工作原理

功能实现

如何扩展

其它用途

05-代码转换器 **Code converters**

06-数据选择器 **Multiplexers**

07-数据分配器 **De-Multiplexers**

08-奇偶发生器/校验器

Parity generators/checkers

一、加法器

例：A=1101, B=1001， 计算A+B。

```
   1 1 0 1
+  1 0 0 1
-----------
 1 0 1 1 0
```

加法运算的基本规则：

(1) 逢二进一;

(2) 最低位是两个数最低位的叠加，不需考虑低位的进位;

(3) 其余各位都是三个数相加，包括加数、被加数和低位来的进位;

(4) 任何位相加都产生两个结果：本位和、向高位的进位。

用半加器实现

用全加器实现

1. 半加器 Half-adder

半加运算不考虑从低位来的进位。设：

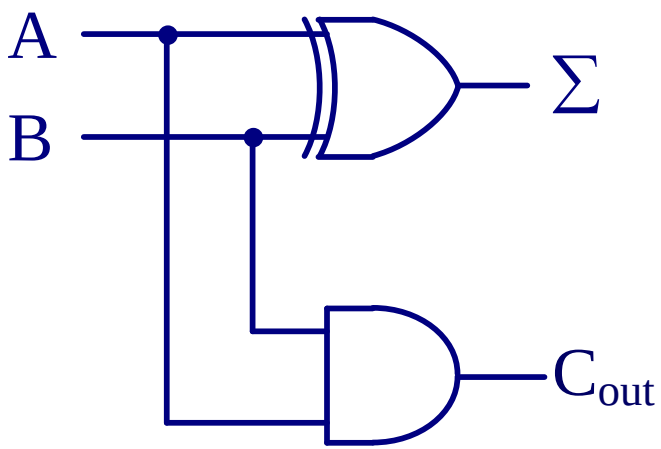
A---加数；B---被加数；∑---本位和；C_{out}---向高位进位。

$$\sum = \bar{A}B + A\bar{B} = A \oplus B \qquad C_{out} = AB$$

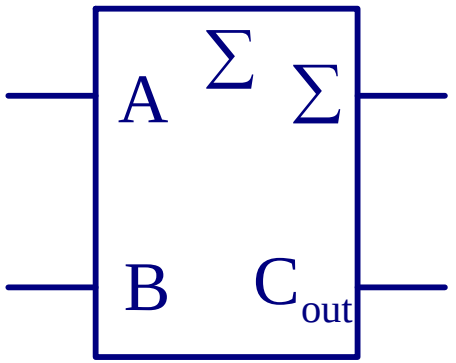
真值表：

A	B	C_{out}	∑
0	0	0	0
0	1	0	1
1	0	0	1
1	1	1	0

逻辑图：



半加器逻辑符号：



2. 全加器 Full-adder

A---加数；B---被加数；c_{in}---低位的进位；∑---本位和；c_{out}---向高位的进位。

A	B	C_{in}	C_{out}	Σ
0	0	0	0	0
0	0	1	0	1
0	1	0	0	1
0	1	1	1	0
1	0	0	0	1
1	0	1	1	0
1	1	0	1	0
1	1	1	1	1

$$\sum = \bar{A}\bar{B}C_{in} + \bar{A}B\bar{C}_{in} + A\bar{B}\bar{C}_{in} + ABC_{in}$$

$$= (\bar{A}\bar{B} + AB)C_{in} + (\bar{A}B + A\bar{B})\bar{C}_{in}$$

$$= (\overline{\bar{A}B + A\bar{B}})C_{in} + (\bar{A}B + A\bar{B})\bar{C}_{in}$$

$$= (\overline{A \oplus B})C_{in} + (A \oplus B)\bar{C}_{in}$$

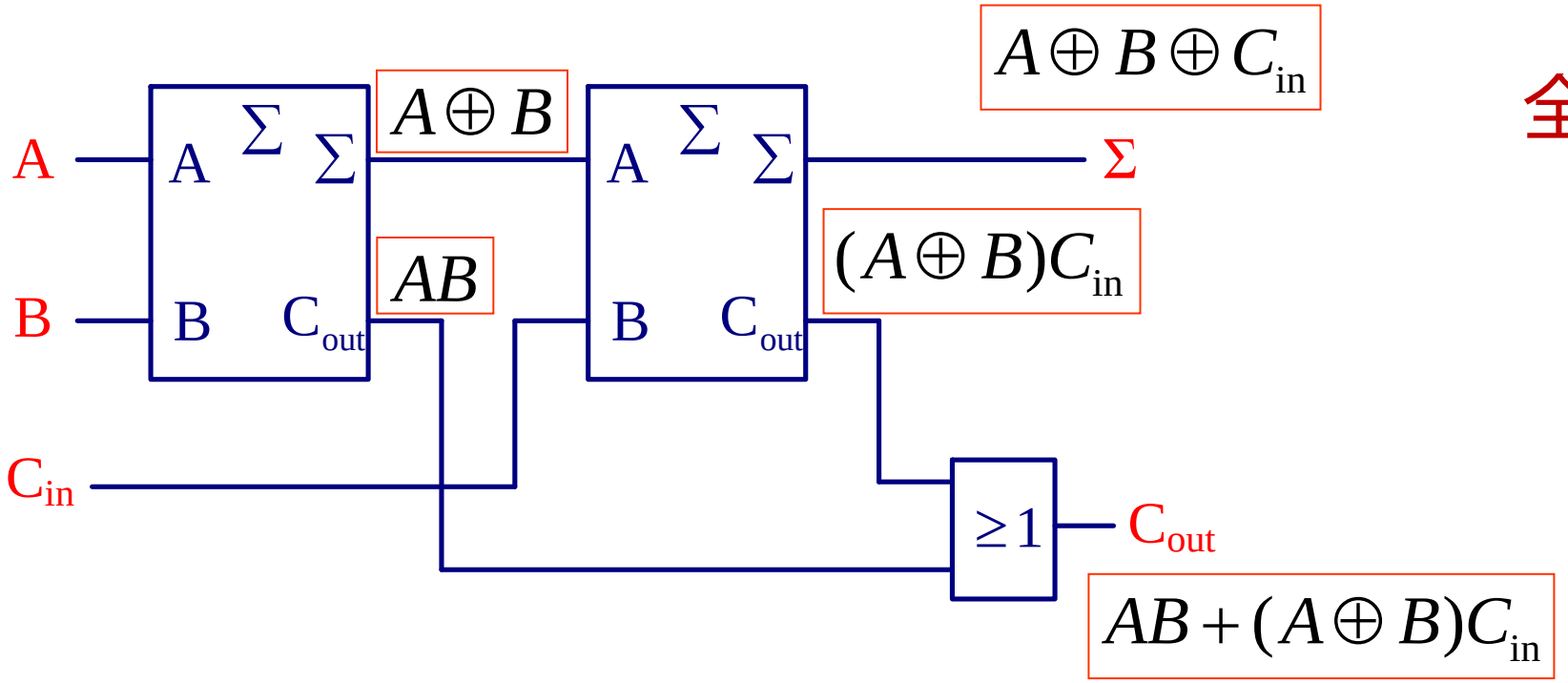
$$= (A \oplus B) \oplus C_{in}$$

$$C_{out} = AB + (A \oplus B)C_{in}$$

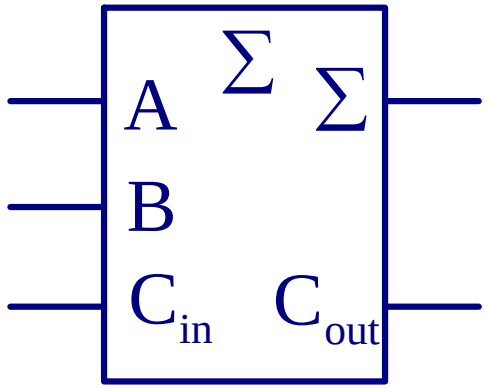
全加器：$\sum = (A \oplus B) \oplus C_{in}$　　$C_{out} = AB + (A \oplus B)C_{in}$

半加器：$\sum = A \oplus B$　$C_{out} = AB$

所以全加器可以用半加器实现



全加器逻辑符号：

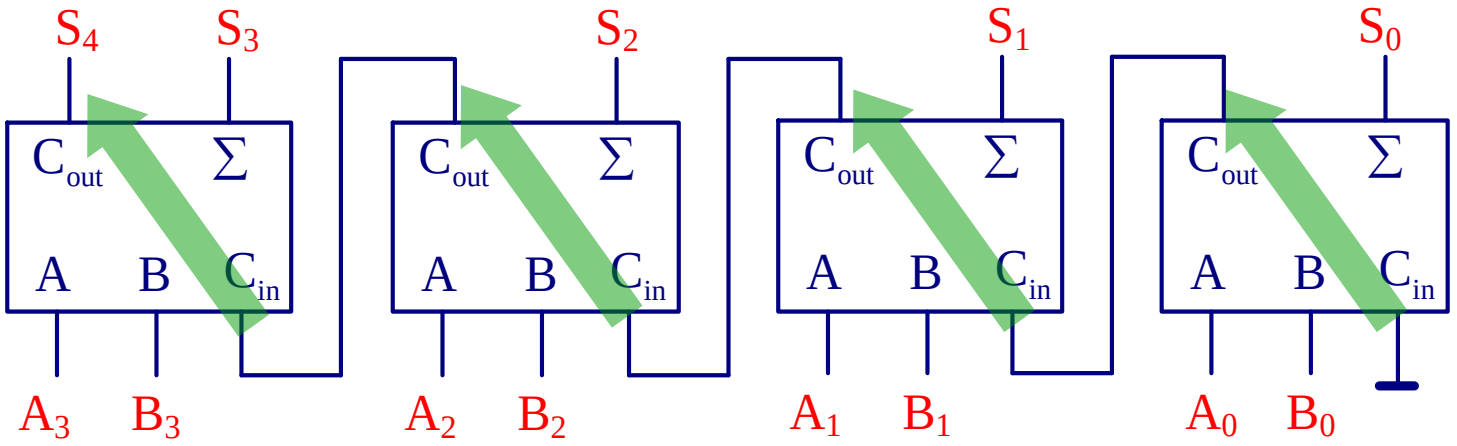


3. 多位加法器 Parallel binary adders

按进位的处理方式可分成：行波进位加法器、超前进位加法器

ripple carry adder **carry look-ahead adder**

例：用四个全加器构成四位行波进位加法器。



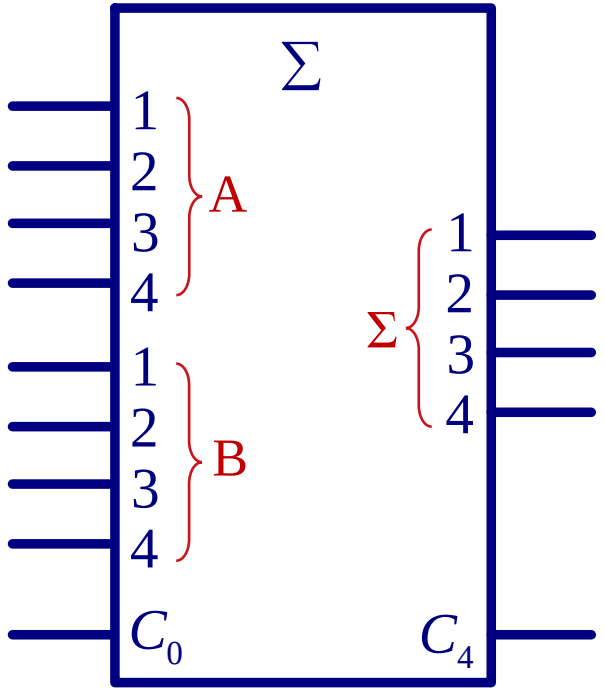
$$\begin{array}{rr} & A_3A_2A_1A_0 \\ + & B_3B_2B_1B_0 \\ \hline & S_4S_3S_2S_1S_0 \end{array}$$

若全加器的进位传送延迟时间为t_p

carry propagation delay

则全部全加器的进位延迟$n \times t_p$

四位加法器逻辑符号



S J T U

4. 超前进位加法器 Carry Look-ahead Adders

基于输入参数，预测每一级的进位输出。是否进位输出，取决于是否发生了:

carry generation 或

carry propagation

A_i	B_i	C_{i-1}	C_i
0	0	0	0
0	0	1	0
0	1	0	0
0	1	1	1
1	0	0	0
1	0	1	1
1	1	0	1
1	1	1	1

$$C_i = A_i B_i + (A_i \oplus B_i) C_{i-1}$$
$$= G_i + P_i C_{i-1}$$
$$= G_i + P_i (G_{i-1} + P_{i-1} C_{i-2})$$

carry propagation

$P_i = A_i + B_i$

$\mathbf{P_i = A_i \oplus B_i}$

carry generation

$G_i = A_i B_i$

$$\Sigma_n = f(A_n, B_n, C_{n-1}) = A_n \oplus B_n \oplus C_{n-1} = P_n \oplus C_{n-1}$$

$$C_n = g(A_n, B_n, C_{n-1}) = G_n + P_n C_{n-1}$$

四位超前进位加法器

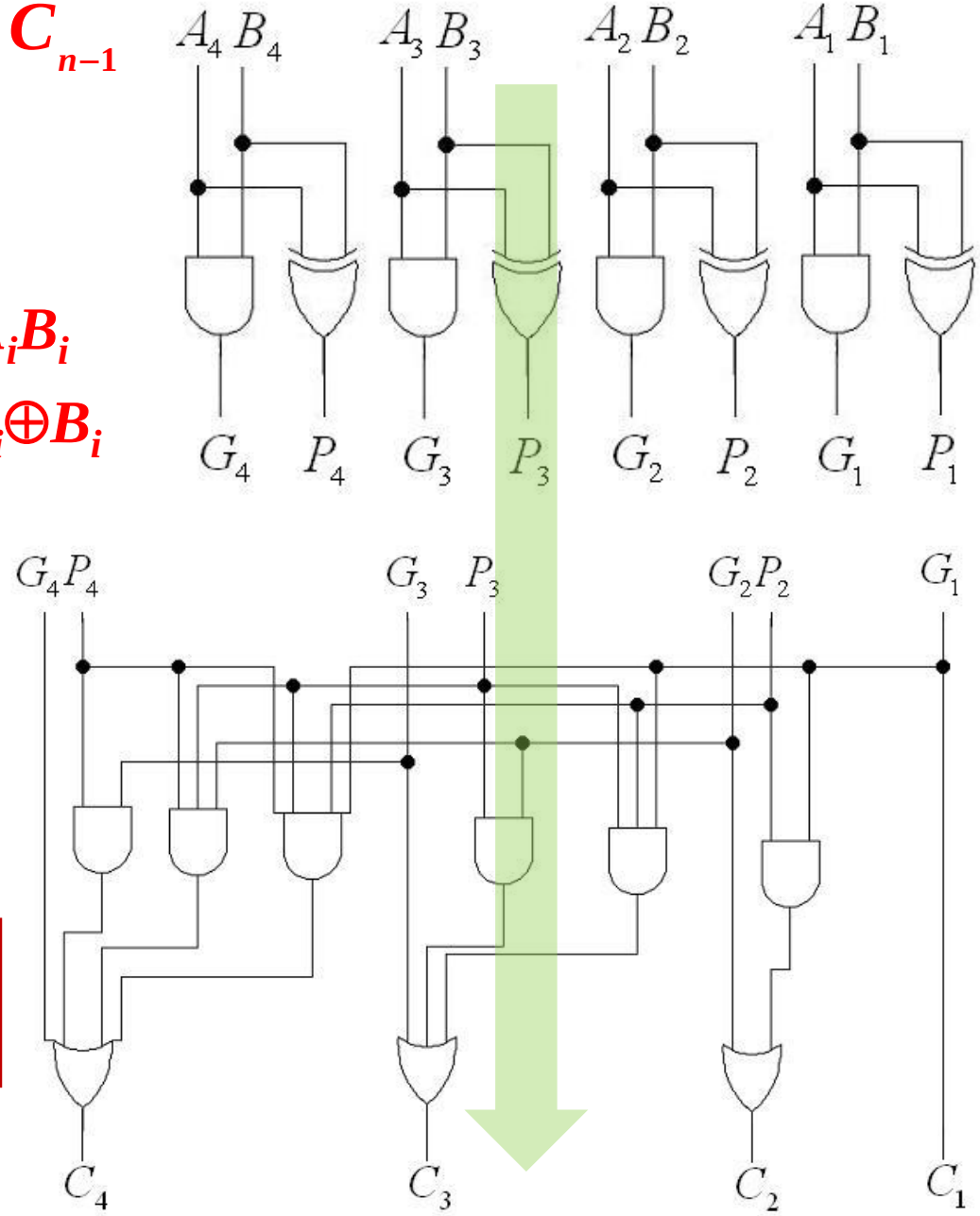
$$\begin{cases} C_1 = G_1 + P_1 C_0 = G_1 \\ C_2 = G_2 + P_2 C_1 = G_2 + P_2 G_1 \\ C_3 = G_3 + P_3 C_2 = G_3 + P_3 G_2 + P_3 P_2 G_1 \\ C_4 = G_4 + P_4 C_3 = G_4 + P_4 G_3 + P_4 P_3 G_2 + P_4 P_3 P_2 G_1 \end{cases}$$

$$\begin{cases} \Sigma_1 = P_1 \oplus C_0 \\ \Sigma_2 = P_2 \oplus C_1 \\ \Sigma_3 = P_3 \oplus C_2 \\ \Sigma_4 = P_4 \oplus C_3 \end{cases}$$

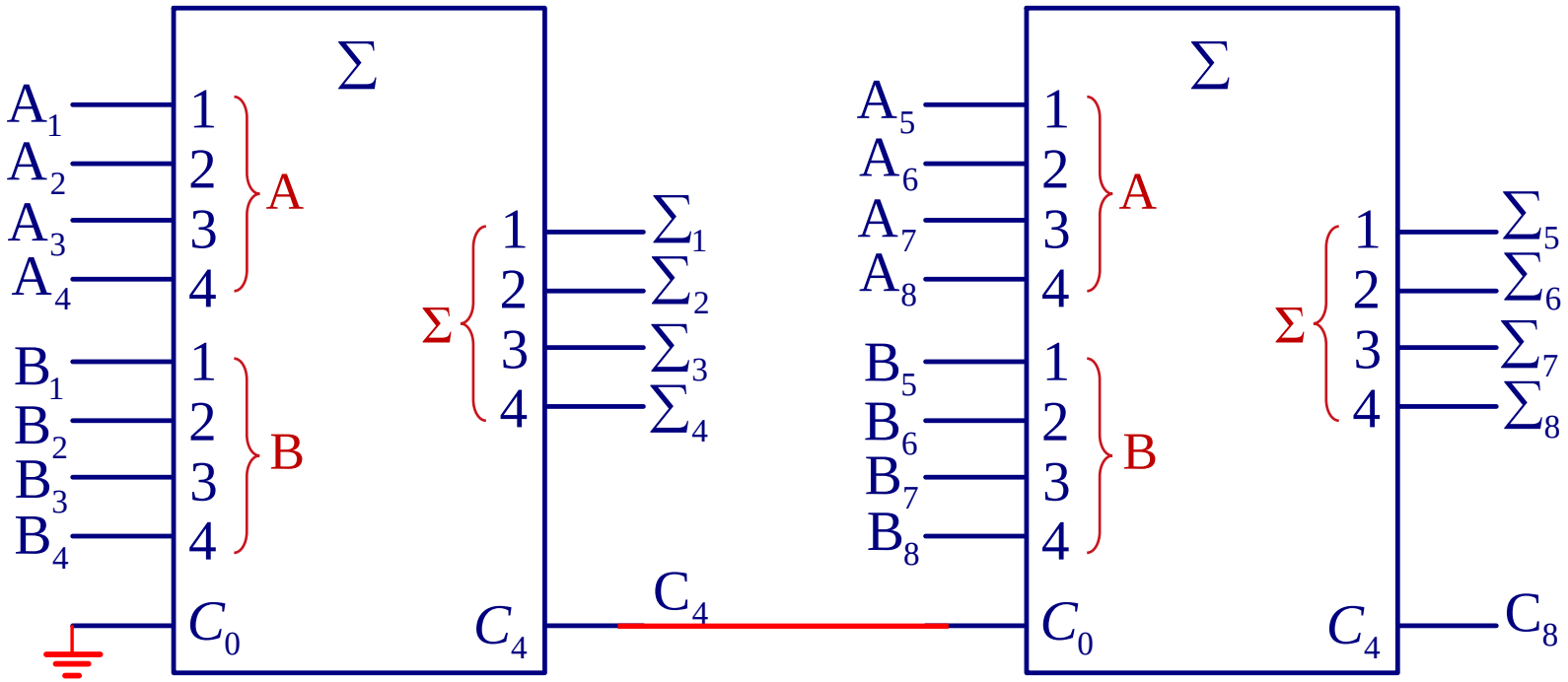
完成加法运算所需的运算时间缩短，但电路的复杂程度增加

$G_i = A_i B_i$

$P_i = A_i \oplus B_i$



5. 加法器的扩展 Adder Expansion



True/False

1. The look-ahead-carry adder is slower than the ripple-carry adder, since it requires additional logic circuits.

Multiple Choice： Using 4 bit adders to create a 16 bit adder

A. requires 16 adders

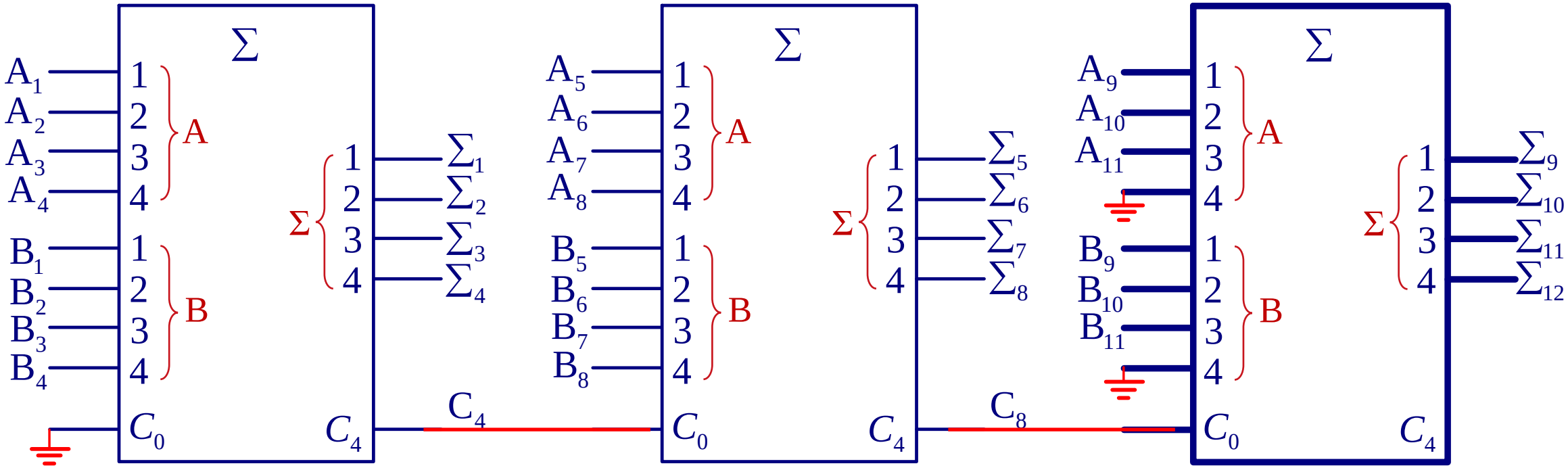
B. requires 4 adders

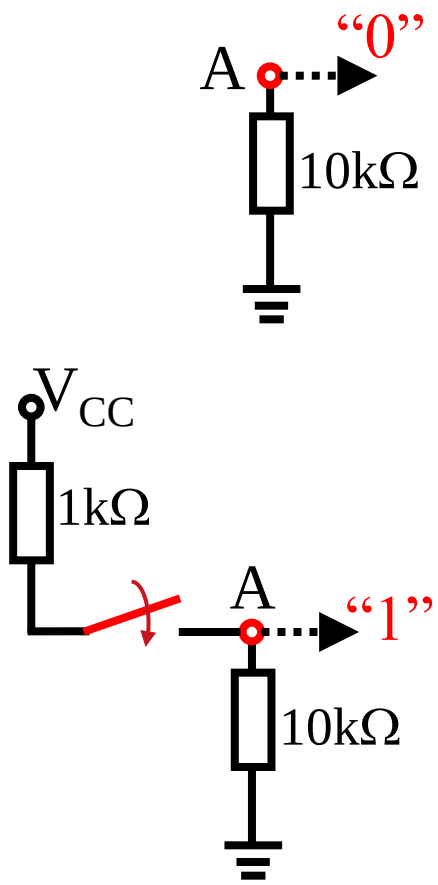
C. requires the carry out of the less significant adder to be connected to the carry in of the next significant adder

D. both b and c

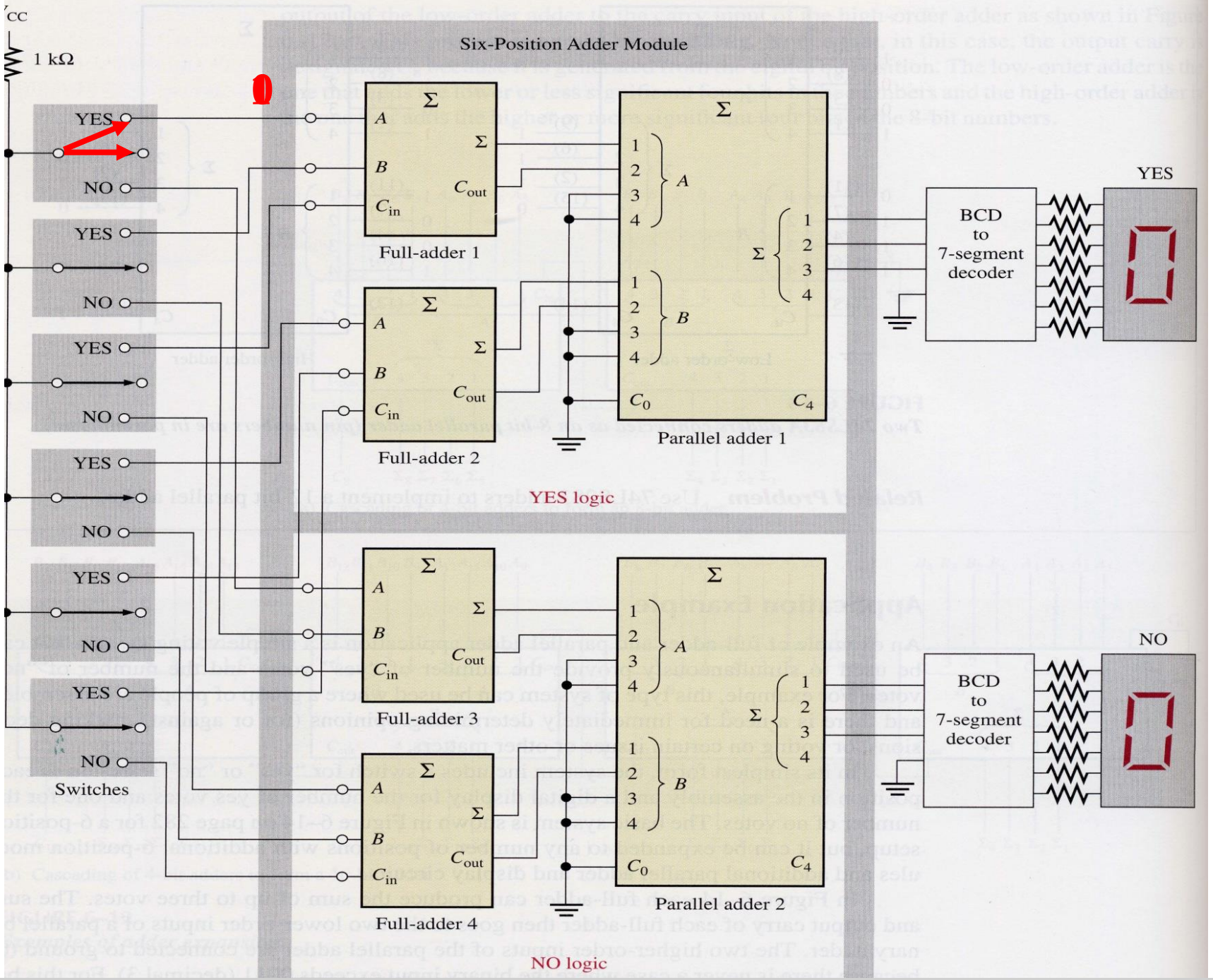
Word Problems： Using 4-bit adders, design an eleven bit adder. Label all critical inputs and outputs.

Using 4-bit adders, design an eleven bit adder. Label all critical inputs and outputs.




A
"0"
10kΩ
VCC
1kΩ
A
"1"
10kΩ

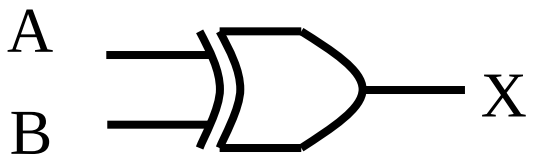
投票表决


VCC
1 kΩ
Six-Position Adder Module
YES
NO
Switches
Full-adder 1
Full-adder 2
Full-adder 3
Full-adder 4
Parallel adder 1
Parallel adder 2
YES logic
NO logic
BCD to 7-segment decoder
YES
NO

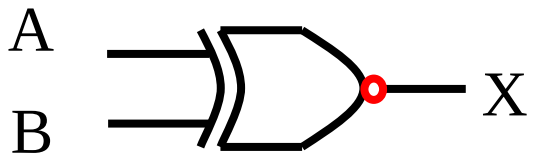
二、比较器 Comparators

1、基本比较器：比较两个输入位是否相等

A	B	X
0	0	0
1	1	0
0	1	1
1	0	1



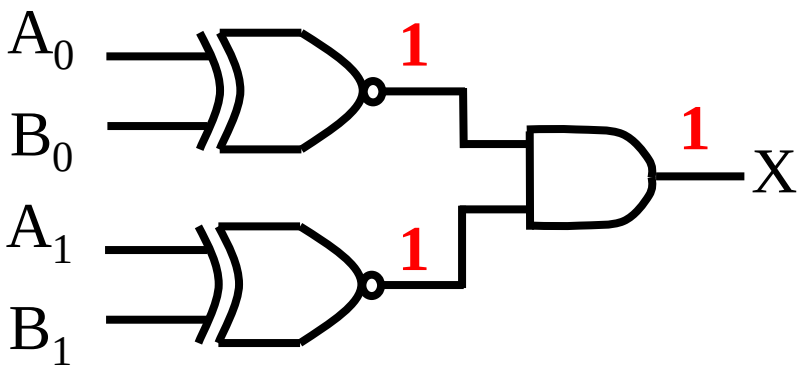
X=0 则输入位相等



X=1 则输入位相等

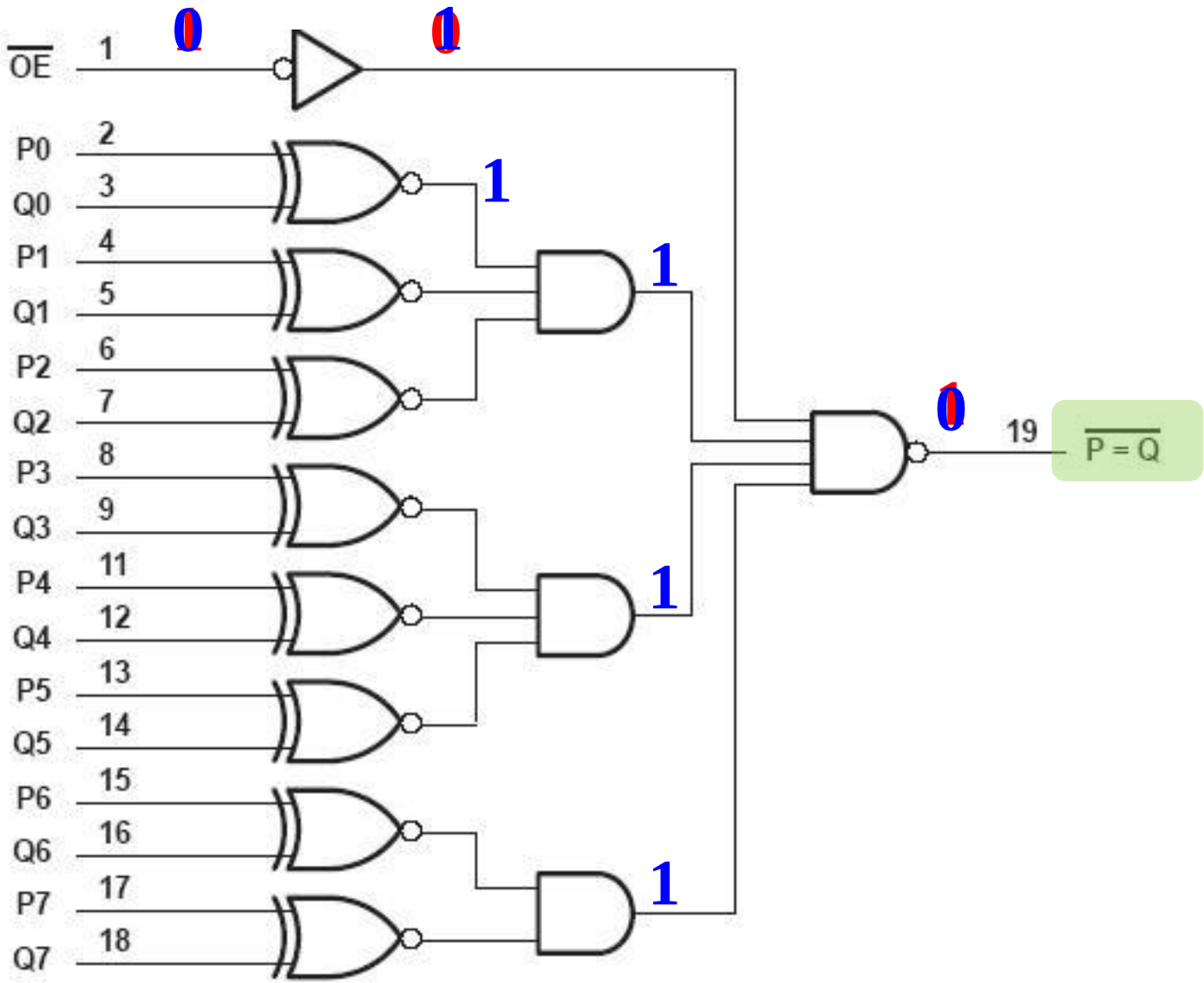
基本比较器可以扩展到任意位数

例：A_1A_0与B_1B_0若相等



74HC688

相等


OE
1
0
1
P0
2
Q0
3
P1
4
Q1
5
P2
6
Q2
7
P3
8
Q3
9
P4
11
Q4
12
P5
13
Q5
14
P6
15
Q6
16
P7
17
Q7
18
1
1
1
1
0
1
19
P = Q

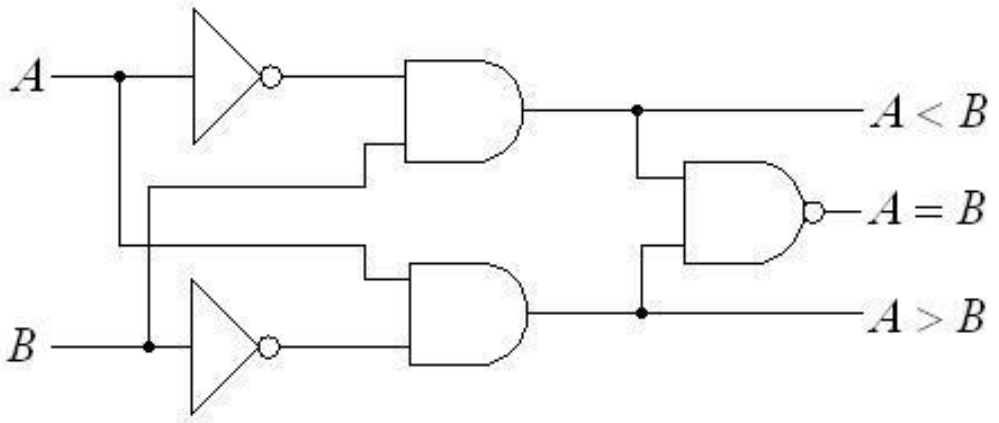
2、一位数值比较器

A	B	$Y_{A>B}$	$Y_{A<B}$	$Y_{A=B}$
0	0	0	0	1
0	1	0	1	0
1	0	1	0	0
1	1	0	0	1

$$Y_{A>B} = A\overline{B}$$

$$Y_{A=B} = AB + \overline{A}\,\overline{B}$$

$$Y_{A<B} = \overline{A}B$$



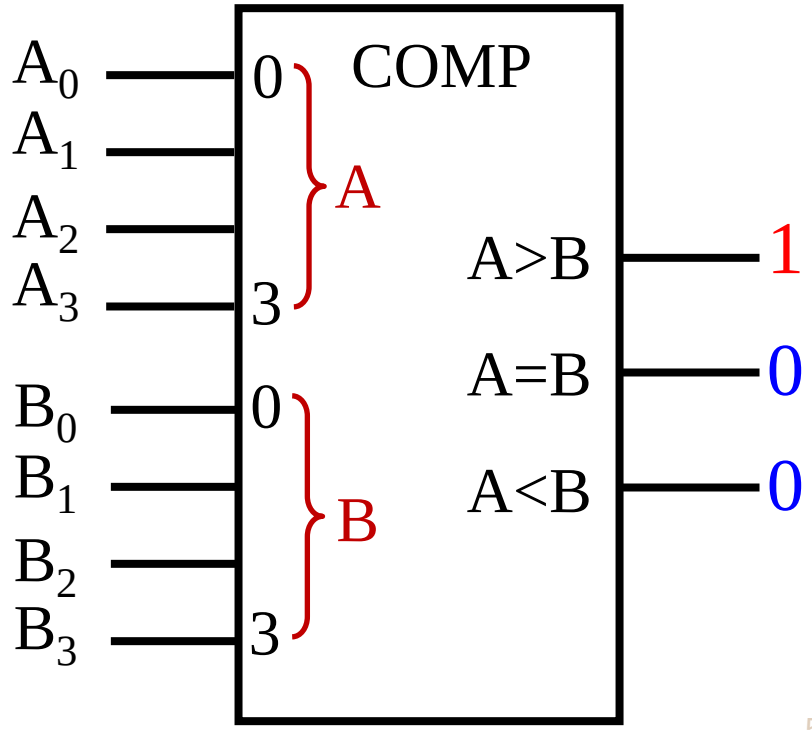
3. 多位数值比较器

比较两个多位数的大小时，自高而低地逐位比较，而且只有在高位相等时，才需要比较低位。

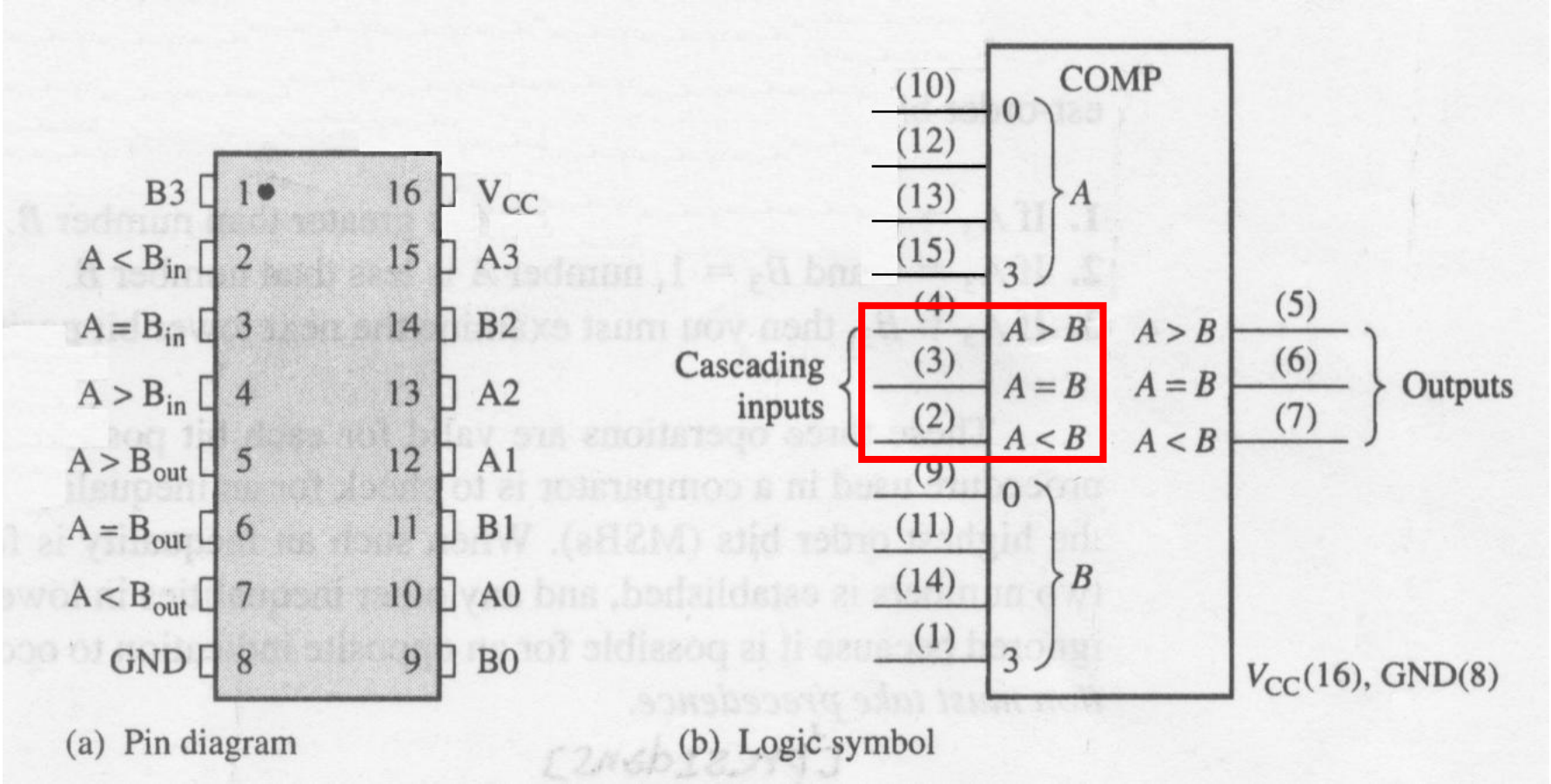
例：四位数值比较器

$A_3A_2A_1A_0$：1001

$B_3B_2B_1B_0$：0101

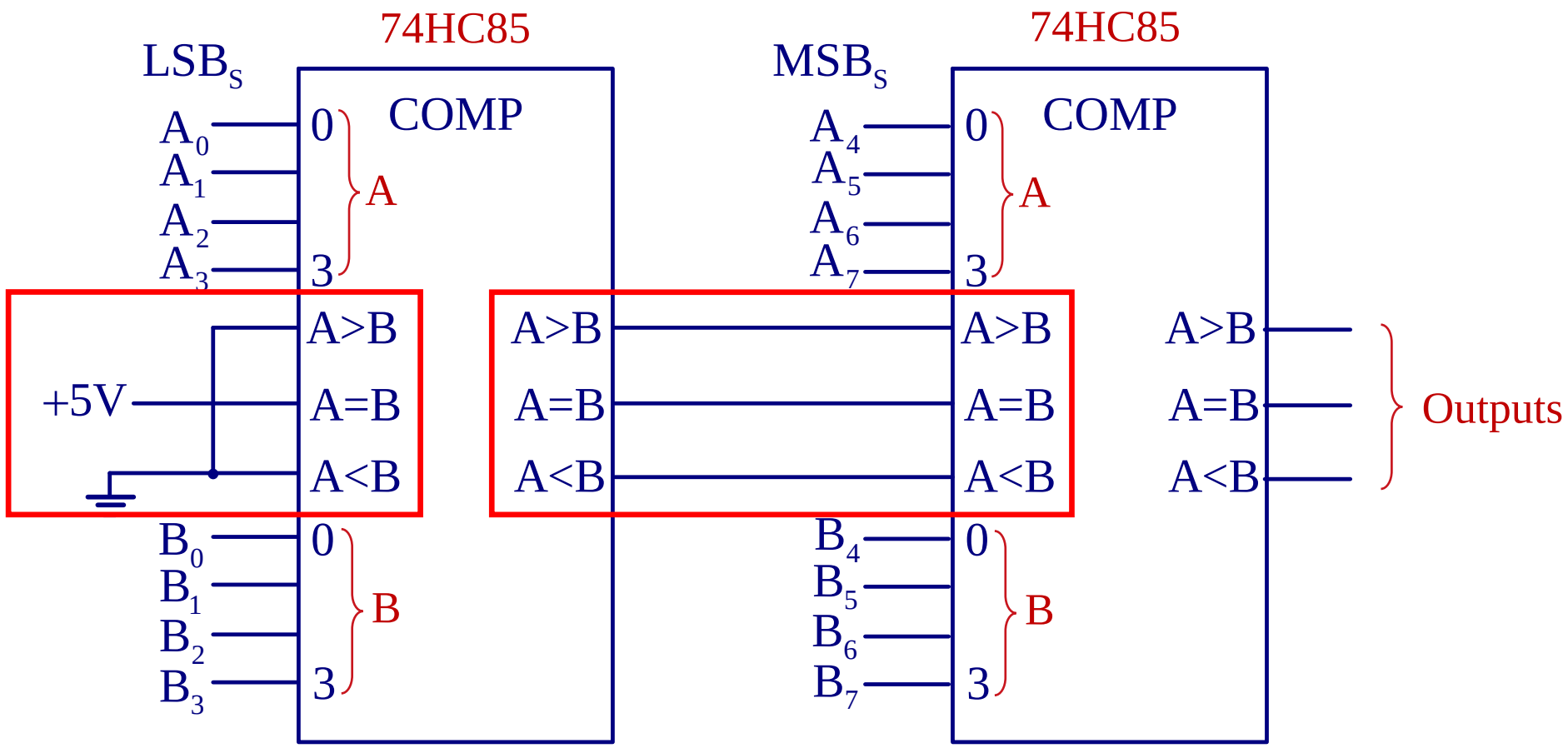


74HC85

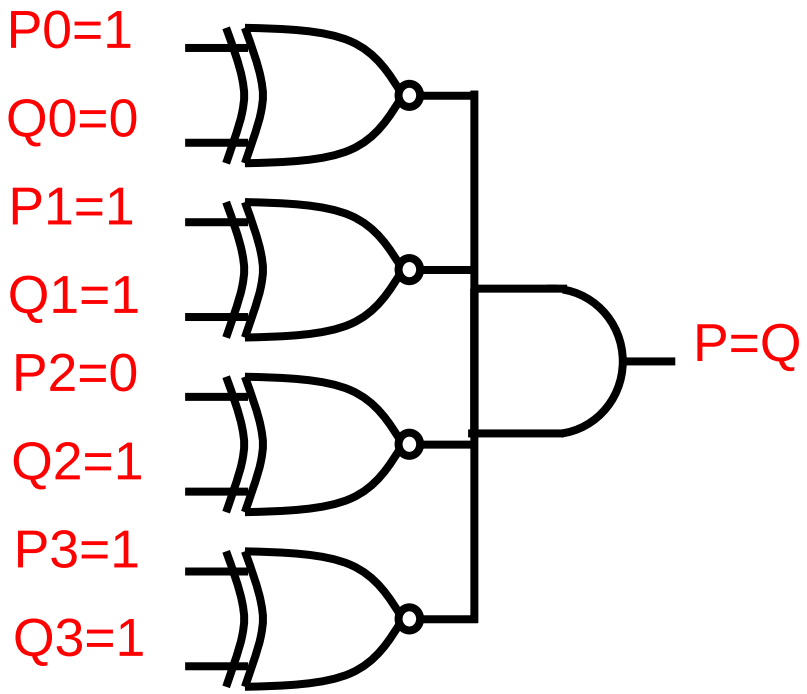


用于扩展的级联输入，接收来自低位比较器的输出

What must be done in following figure in order to use two 74HC85 4-bit comparators to compare two 8-bit numbers?

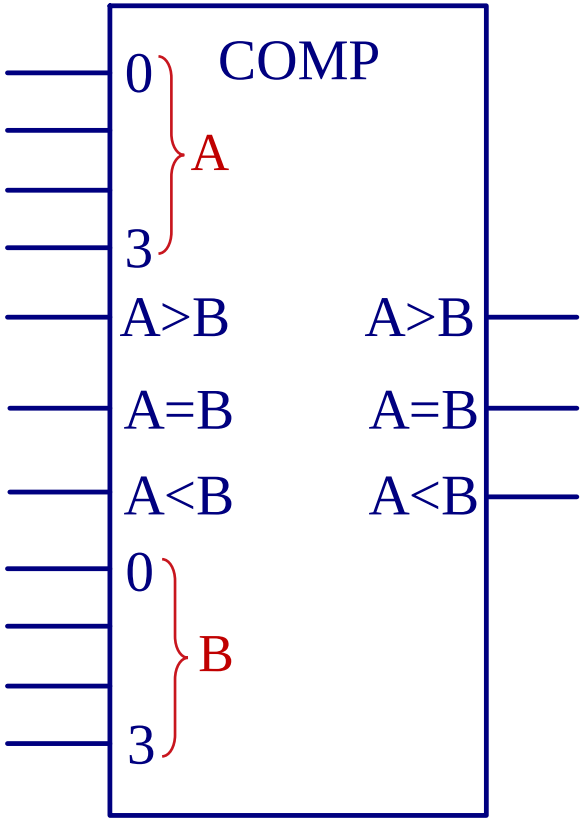


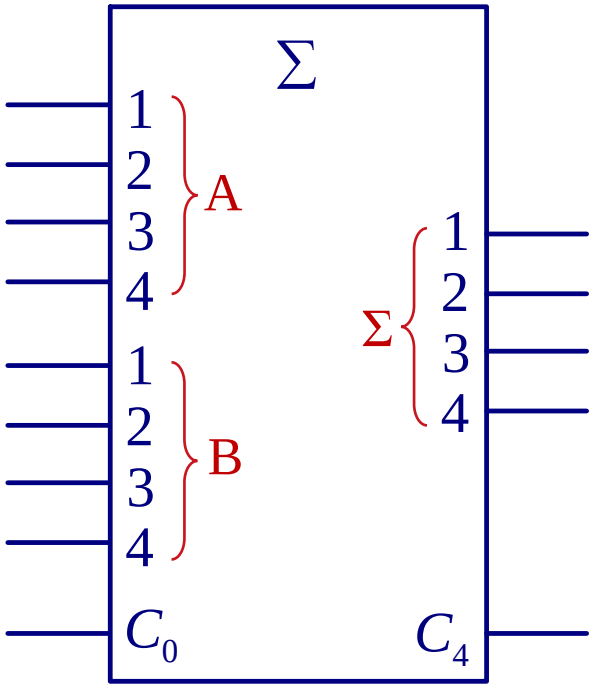
2. The circuit is a/an_____.



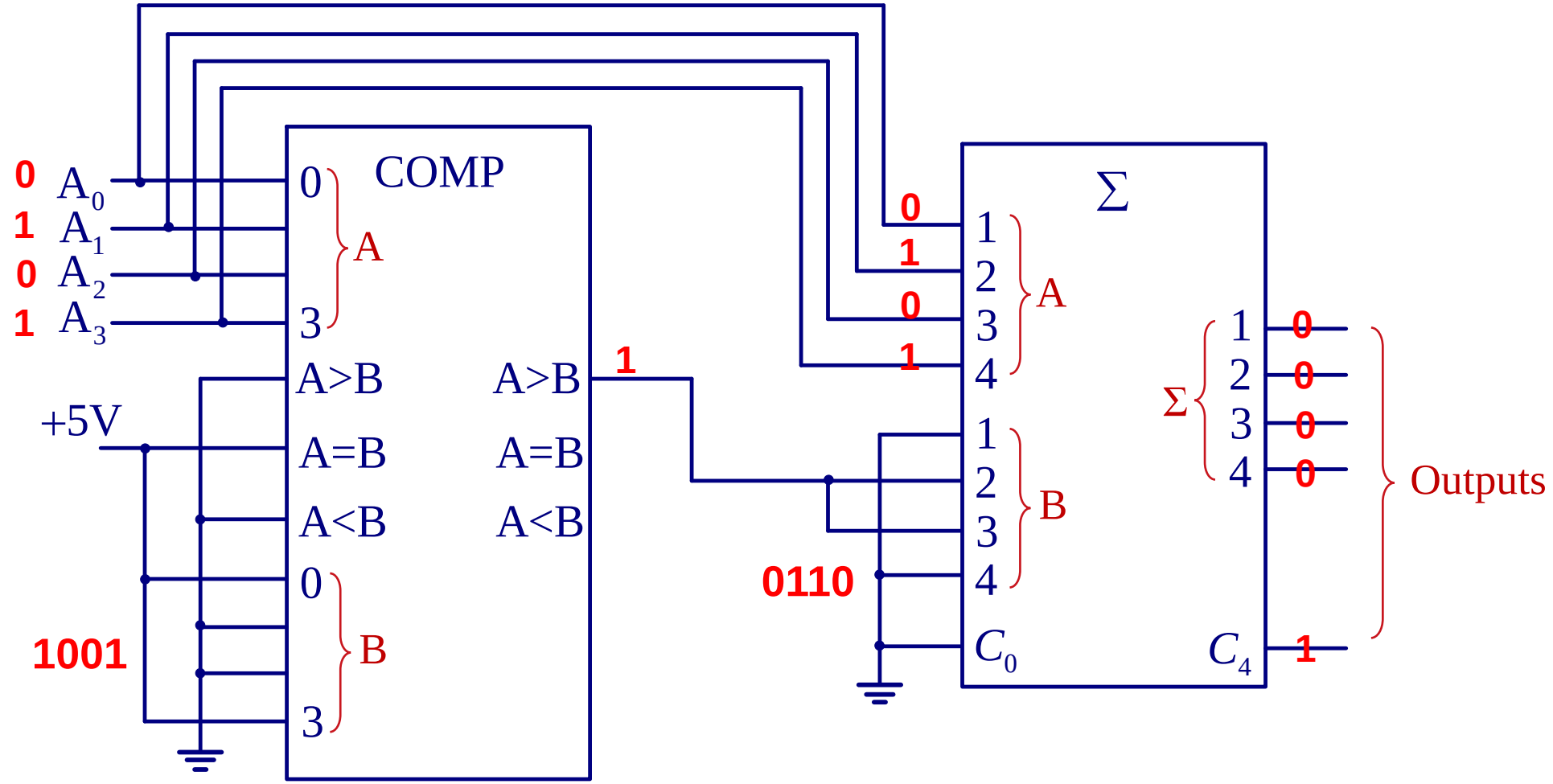
A. 4-bit magnitude comparator
B. 4-bit half adder
C. 8-bit comparator
D. 8-bit binary adder

Design a circuit that detects if a 4 bit binary number is greater than 9, add 6 to the number.

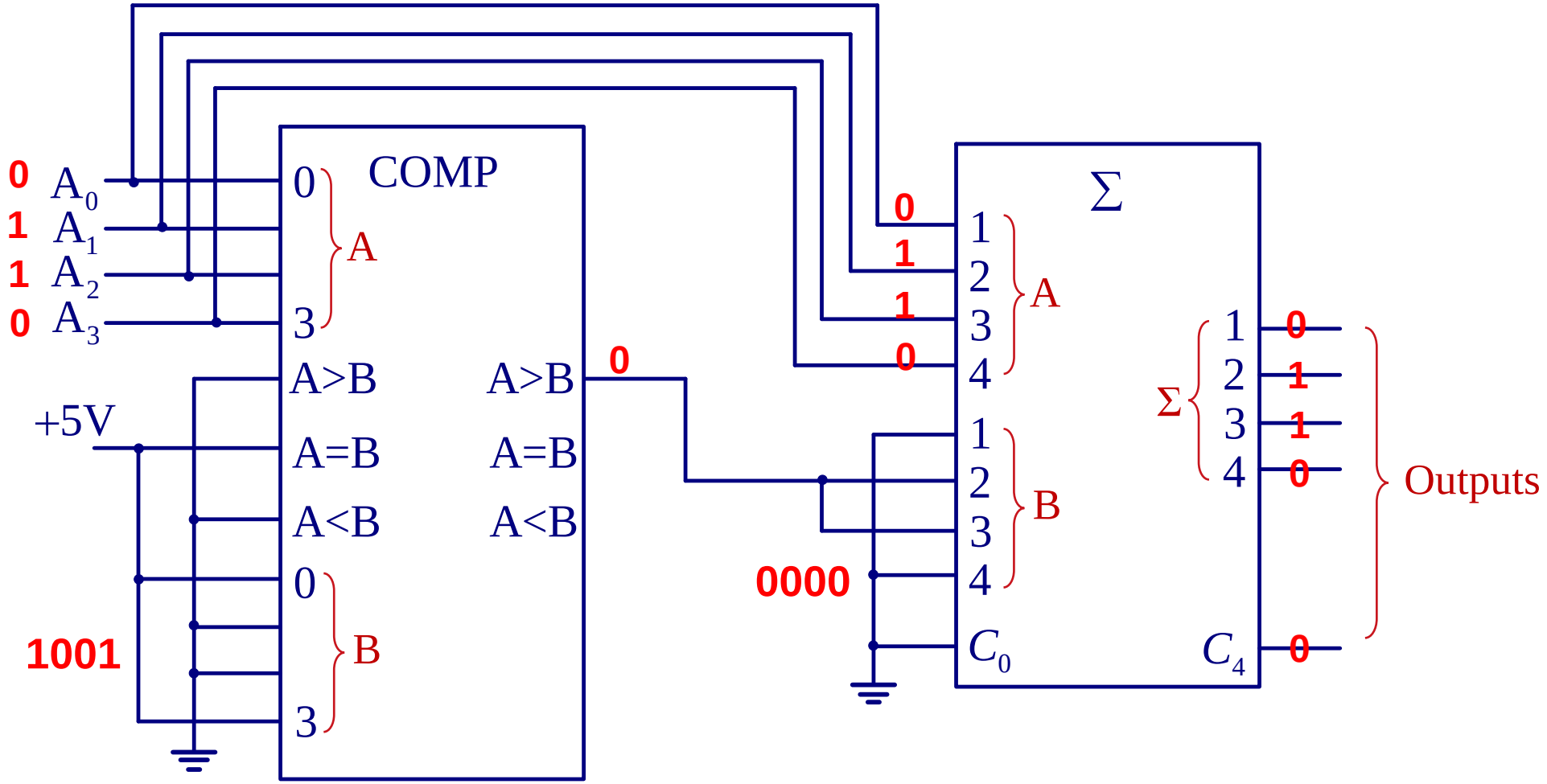




Design a circuit that detects if a 4 bit binary number is greater than 9, add 6 to the number.



Design a circuit that detects if a 4 bit binary number is greater than 9, add 6 to the number.



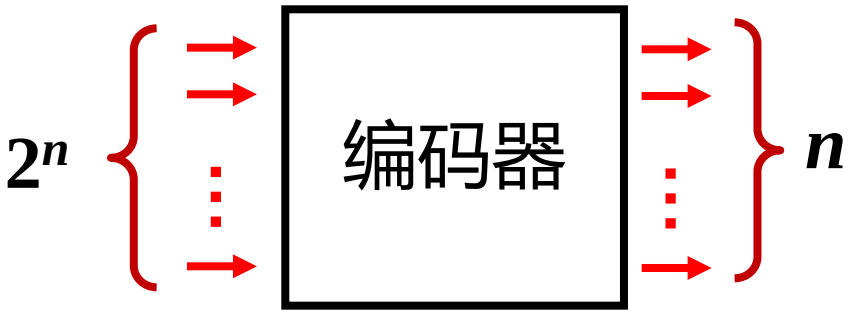
三、编码器 Encoders

所谓编码就是赋予选定的一系列二进制代码以固定的含义。

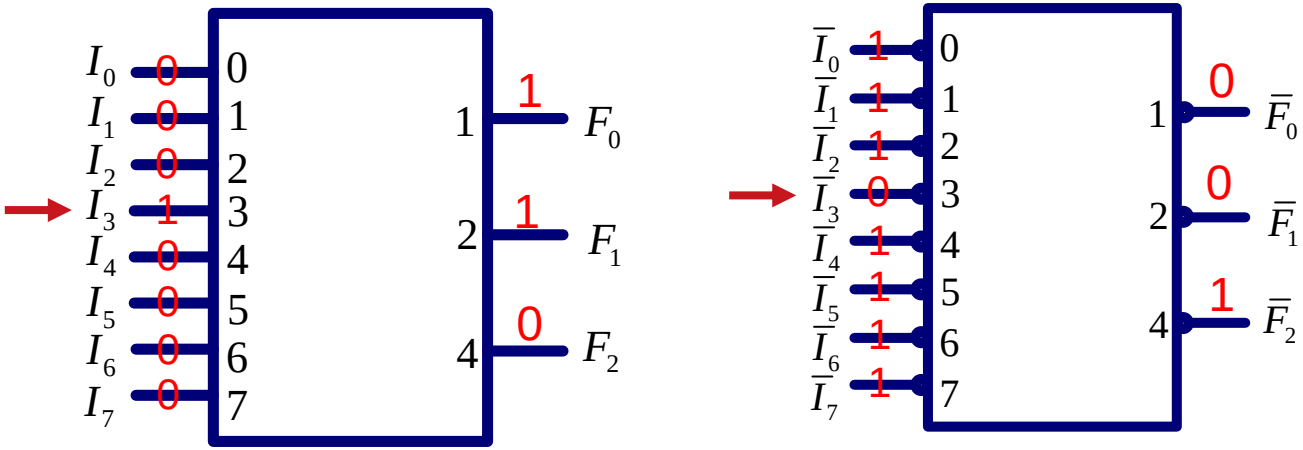
1、二进制编码器

将一系列信号状态编制成二进制代码

n 位二进制代码有 2^n 种不同的组合，可以表示 2^n 个信号



例：三位二进制编码器（8线-3线编码器）



例：设计一个4线-2线二进制编码器

设四个输入端为I_0~I_3，输出为Y_1Y_0， 任何时刻仅有一个输入端的状态“与众不同”（有效）。

设计编码器的过程与设计一般的组合逻辑电路相同，首先要列出逻辑状态表（即真值表），然后写出逻辑表达式并进行化简，最后画出逻辑图。

(1) 输入高电平有效、原码输出

I_0	I_1	I_2	I_3	Y_1	Y_0
1	0	0	0	0	0
0	1	0	0	0	1
0	0	1	0	1	0
0	0	0	1	1	1

(2) 输入低电平有效、反码输出

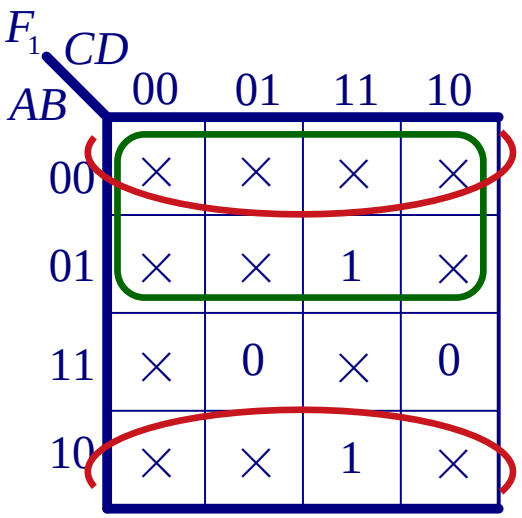
$\overline{I}_0$	$\overline{I}_1$	$\overline{I}_2$	$\overline{I}_3$	$\overline{Y}_1$	$\overline{Y}_0$
0	1	1	1	1	1
1	0	1	1	1	0
1	1	0	1	0	1
1	1	1	0	0	0

例：设计一个4线-2线二进制编码器

任何时刻只有一个输入有效，则输出无误

低电平有效、反码输出

$\bar{I}_0$	$\bar{I}_1$	$\bar{I}_2$	$\bar{I}_3$	$\bar{Y}_1$	$\bar{Y}_0$
0	1	1	1	1	1
1	0	1	1	1	0
1	1	0	1	0	1
1	1	1	0	0	0

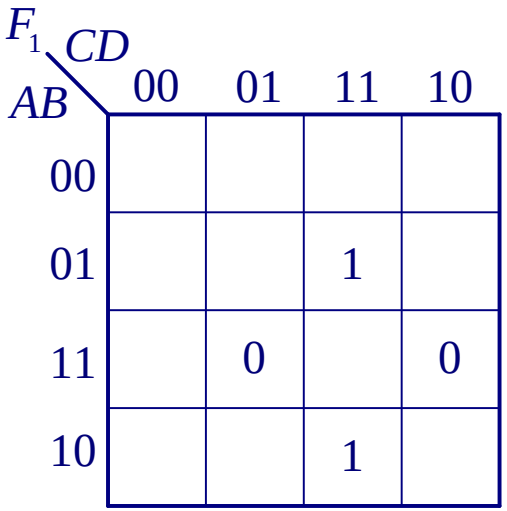


$$F_1 = \bar{A} + \bar{B}$$

A	B	C	D	F_1	F_0
0	1	1	1	1	1
1	0	1	1	1	0
1	1	0	1	0	1
1	1	1	0	0	0

$$F_1 = \bar{A}BCD + A\bar{B}CD$$

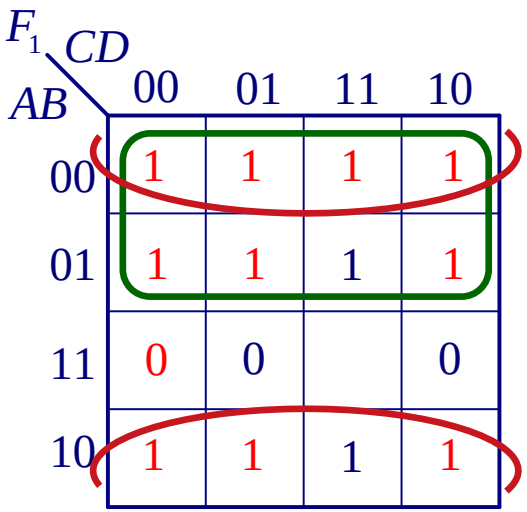
$$F_0 = \bar{A}BCD + AB\bar{C}D$$



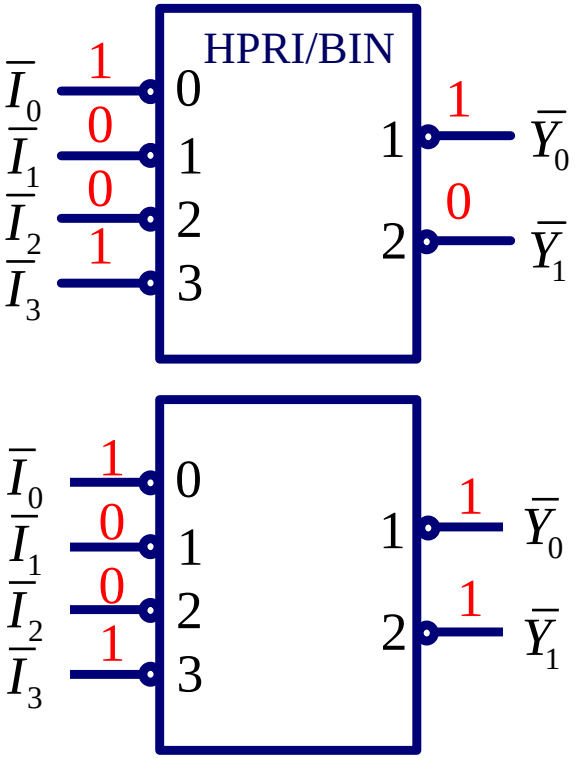
$\bar{I}_0$	$\bar{I}_1$	$\bar{I}_2$	$\bar{I}_3$	$\bar{Y}_0$	$\bar{Y}_1$
0	×	×	×	1	1
×	0	×	×	1	0
×	×	0	×	0	1
×	×	×	0	0	0

$$F_1 = \bar{A} + \bar{B}$$

$$F_0 = \bar{A} + \bar{C}$$

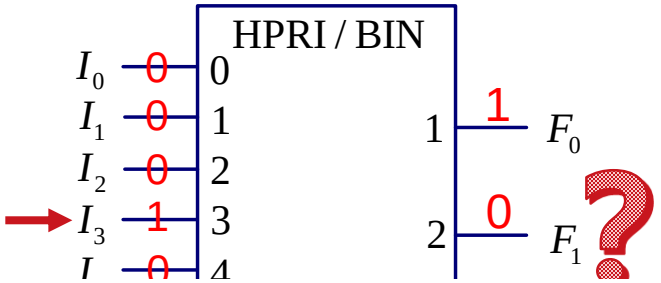


$\bar{I}_0$	$\bar{I}_1$	$\bar{I}_2$	$\bar{I}_3$	$\bar{Y}_0$	$\bar{Y}_1$
0	1	1	1	1	1
×	0	1	1	1	0
×	×	0	1	0	1
×	×	×	0	0	0



S J T U

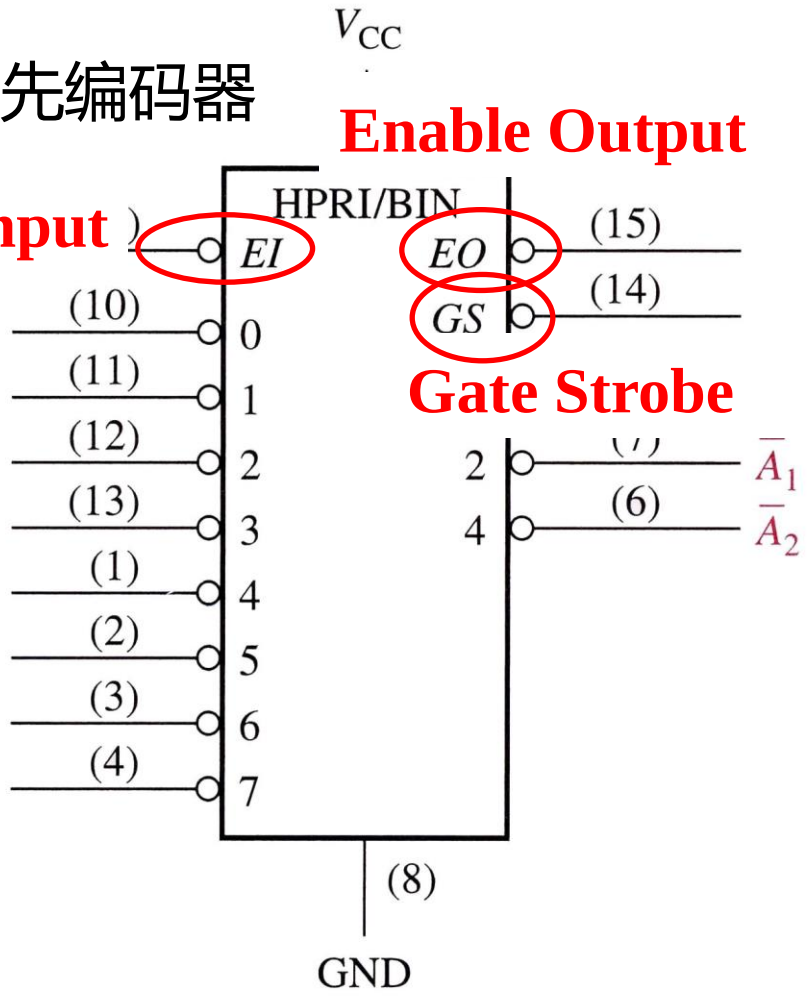
练习：二进制编码器有多个输入信号有效



功能表

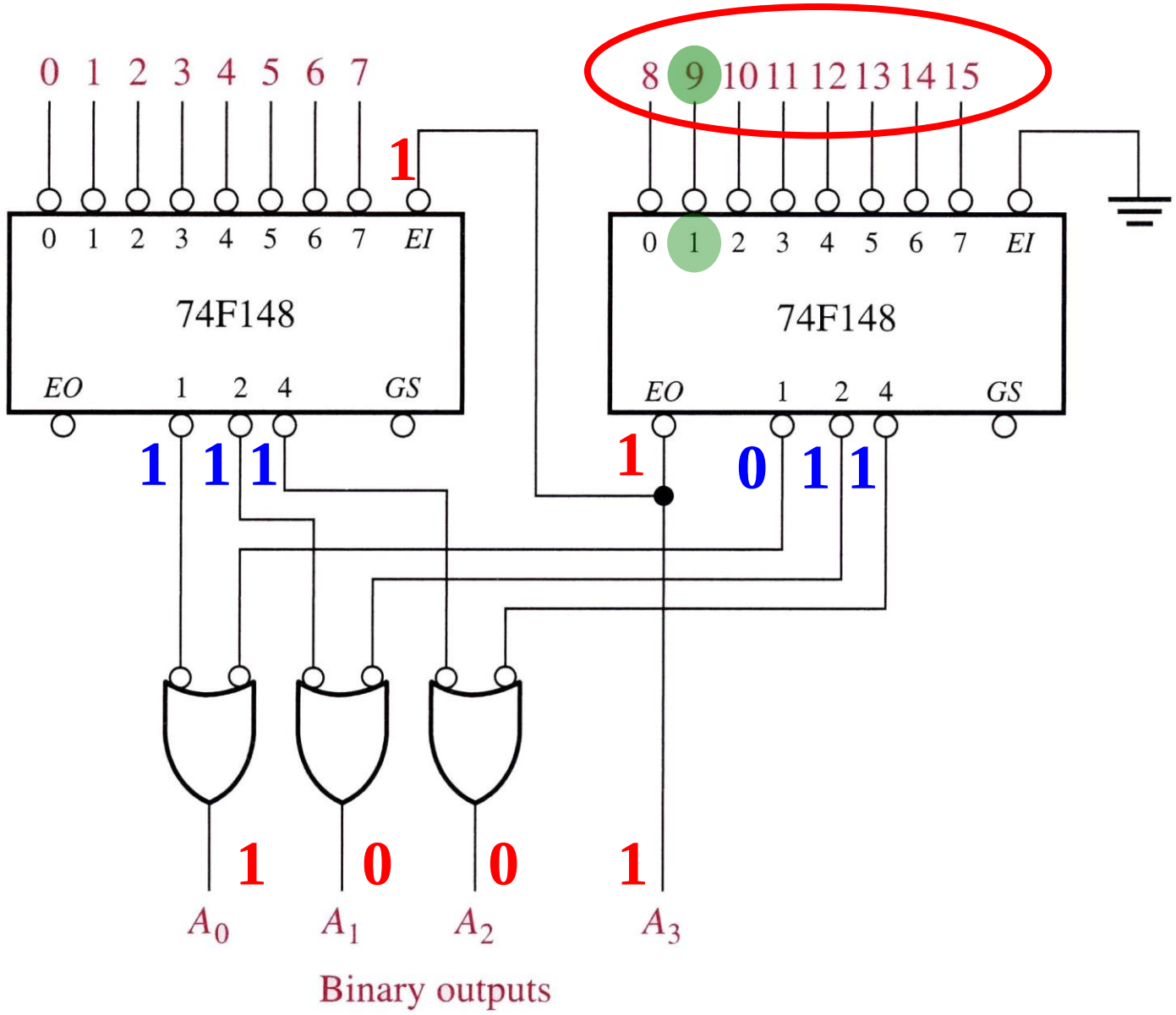
输入									输出				
$\overline{EI}$	$\overline{0}$	$\overline{1}$	$\overline{2}$	$\overline{3}$	$\overline{4}$	$\overline{5}$	$\overline{6}$	$\overline{7}$	$\overline{A}_2$	$\overline{A}_1$	$\overline{A}_0$	$\overline{GS}$	$\overline{EO}$
1	×	×	×	×	×	×	×	×	1	1	1	1	1
0	1	1	1	1	1	1	1	1	1	1	1	1	0
0	×	×	×	×	×	×	×	0	0	0	0	0	1
0	×	×	×	×	×	×	0	1	0	0	1	0	1
0	×	×	×	×	×	0	1	1	0	1	0	0	1
0	×	×	×	×	0	1	1	1	0	1	1	0	1
0	×	×	×	0	1	1	1	1	1	0	0	0	1
0	×	×	0	1	1	1	1	1	1	0	1	0	1
0	×	0	1	1	1	1	1	1	1	1	0	0	1
0	0	1	1	1	1	1	1	1	1	1	1	0	1

74F148 优先编码器



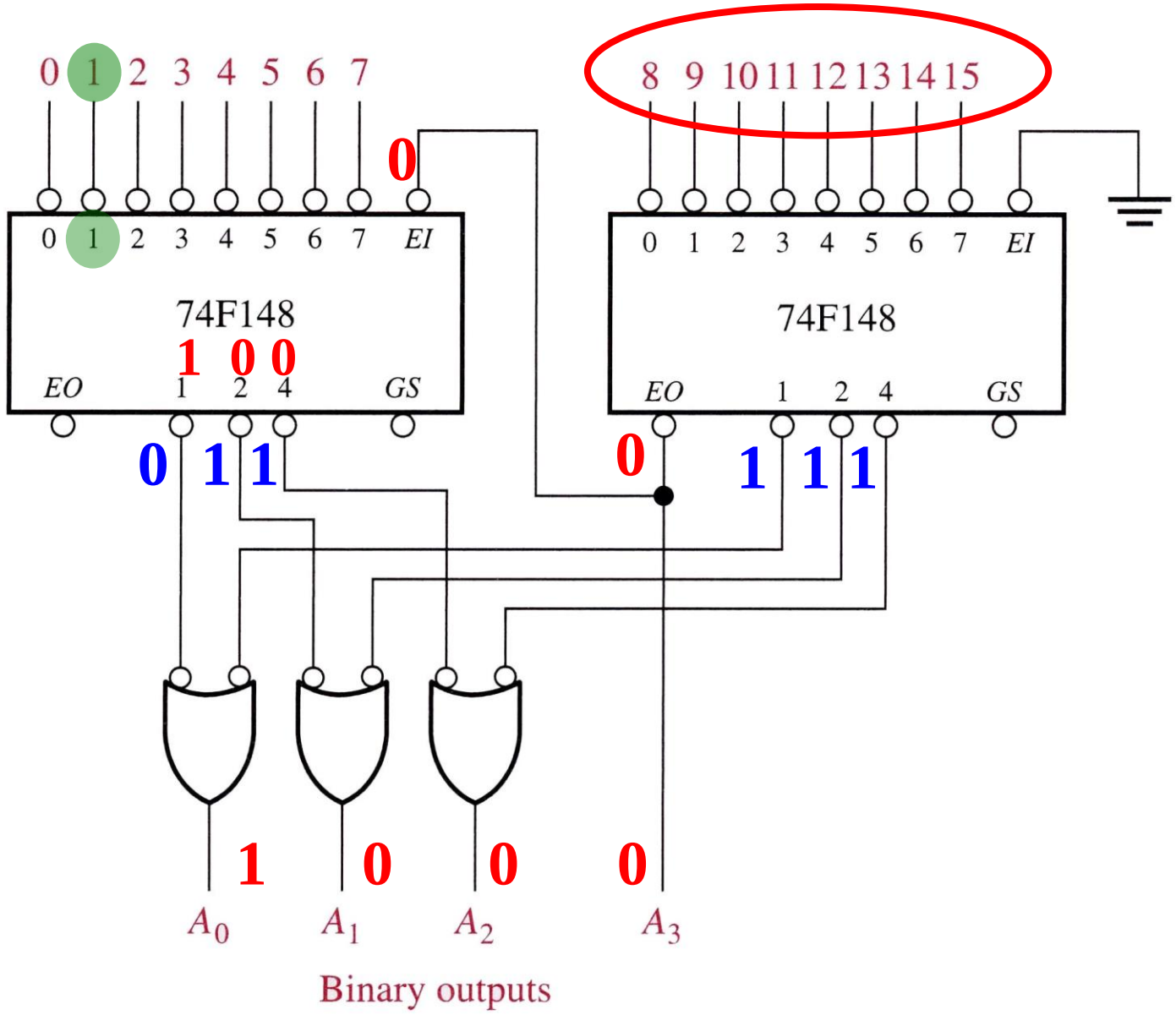
16线-4线优先编码器

有有效输入信号时



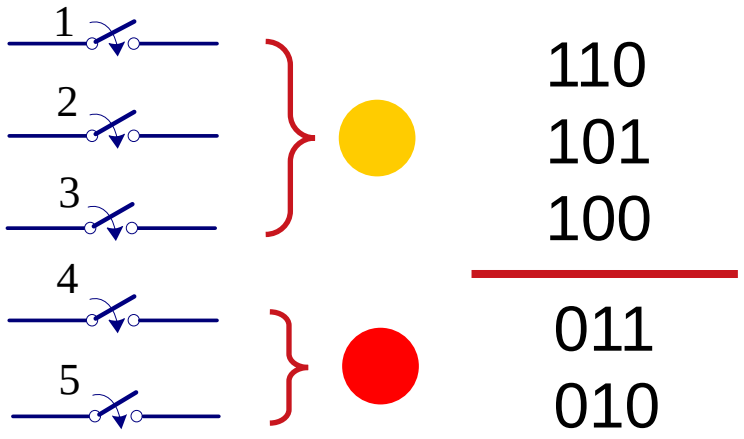
16线-4线优先编码器

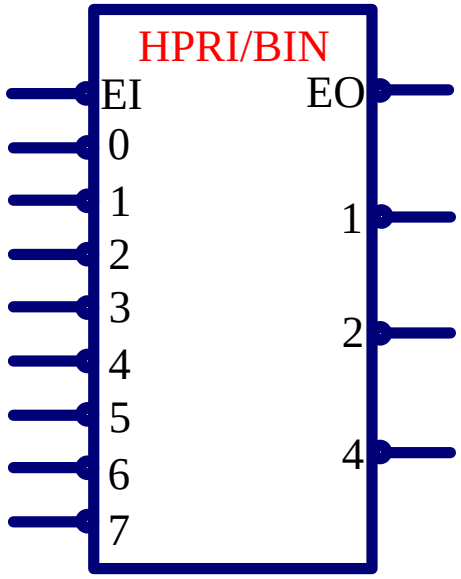
无有效输入信号时

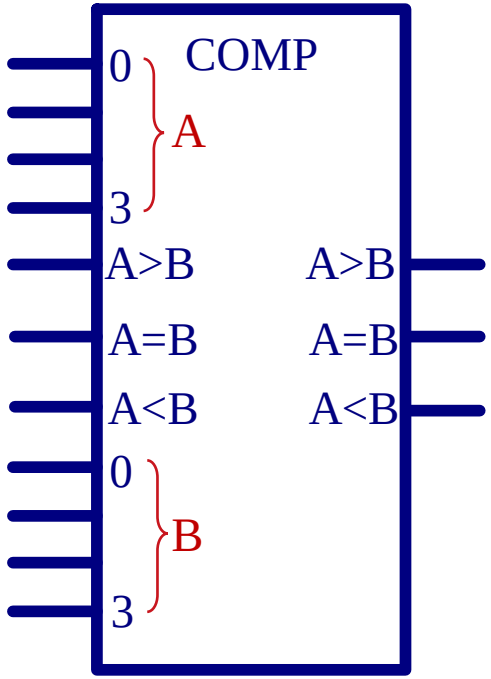


Example： Design a circuit that accepts an input from 5 different switches. If the switch number pressed is 1, 2, or 3, a yellow LED is illuminated. If the switch is 4 or 5, a red LED is illuminated. If 2 switches are pressed at the same time, the higher number has priority.

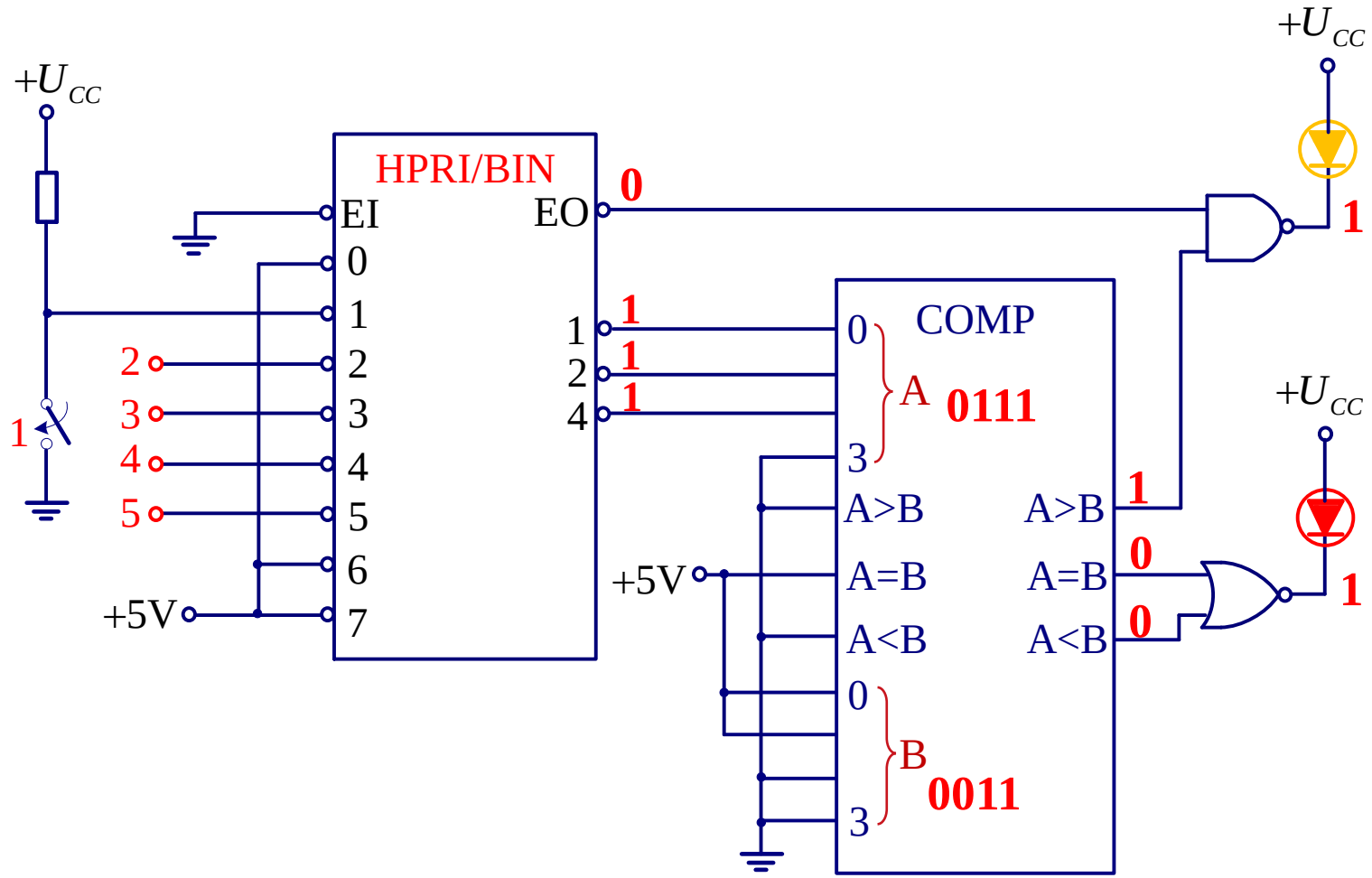
(Using MSI combinational logic circuits)



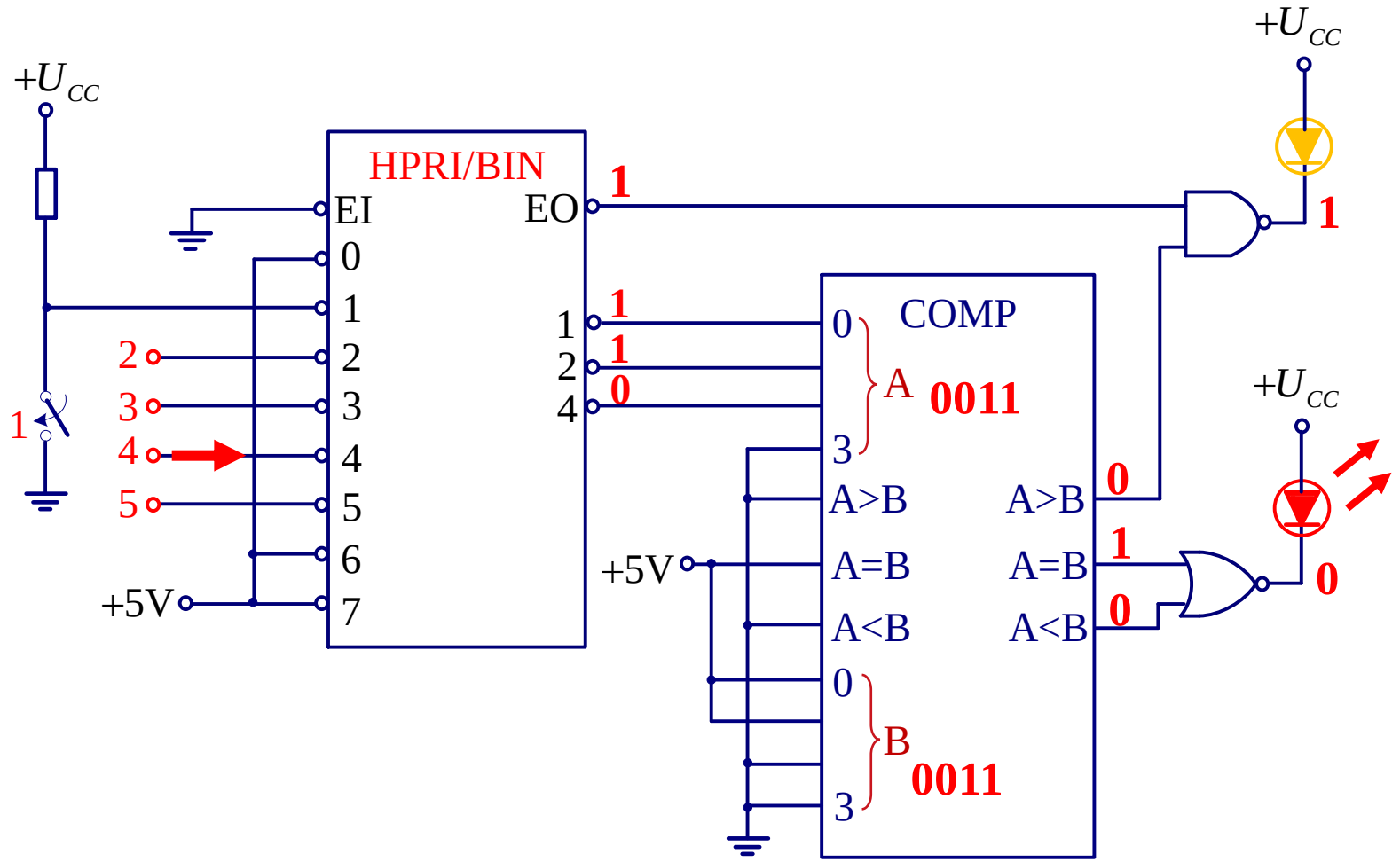




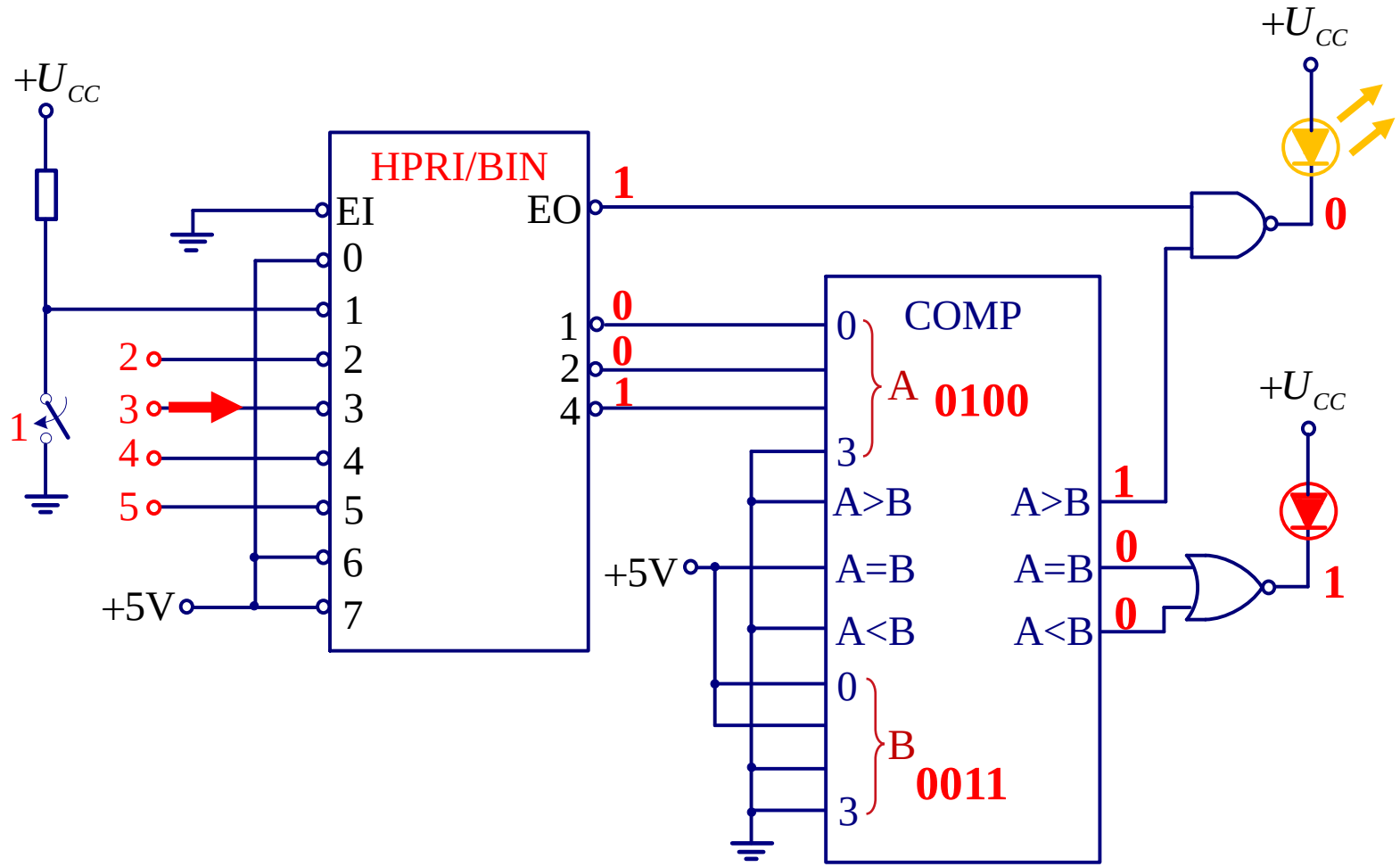
When none of the inputs is active, EO is low



When one of the inputs is active, EO is high

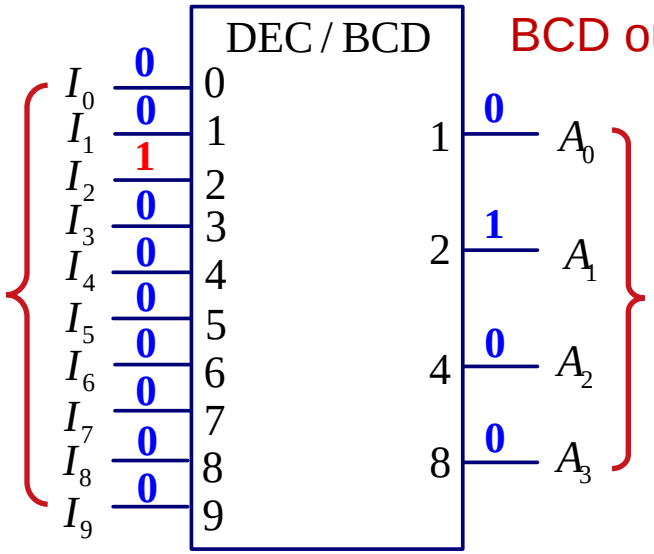


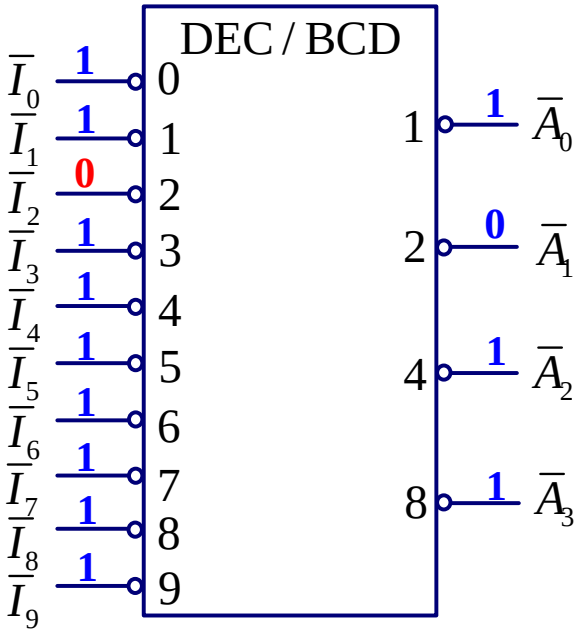
When one of the inputs is active, EO is high



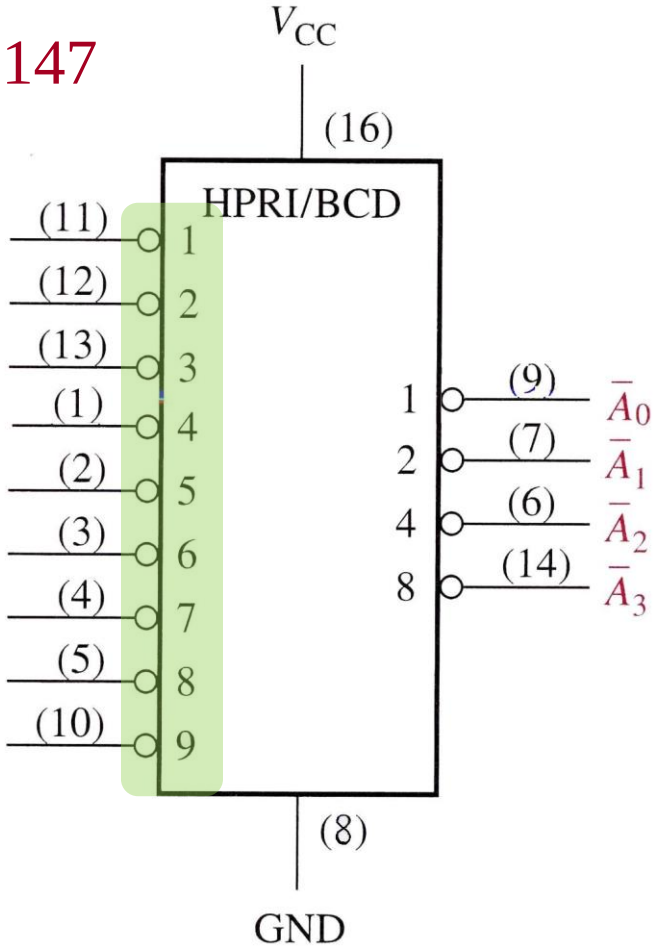
2、二-十进制编码器 Decimal-to-BCD Encoder

将十个状态（对应于十进制的0～9）编制成BCD码

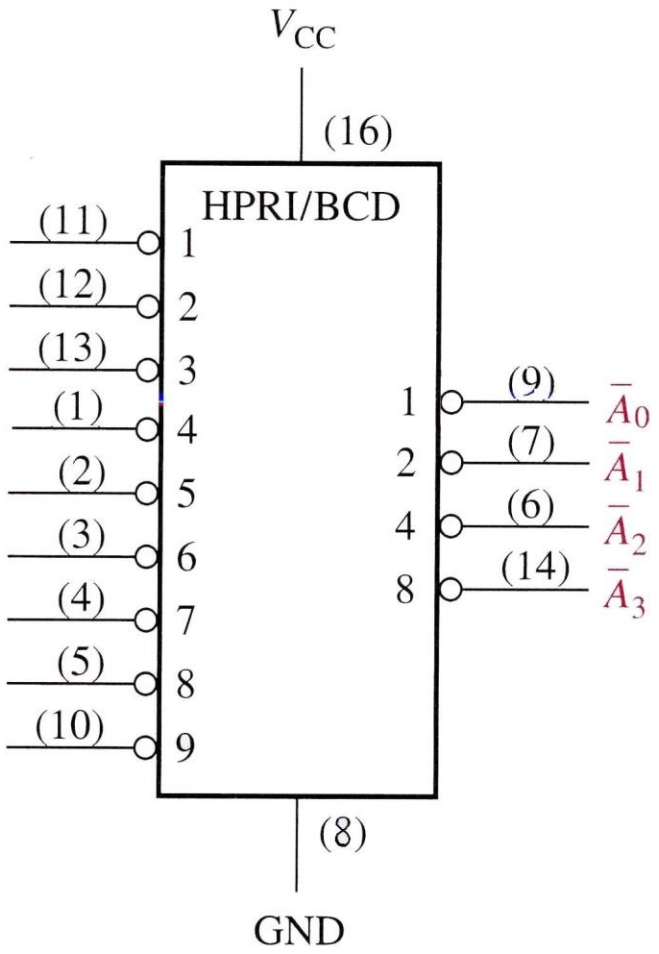




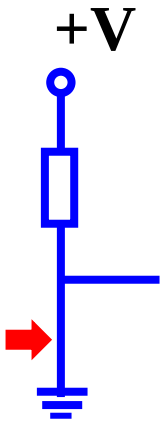
例：74HC147

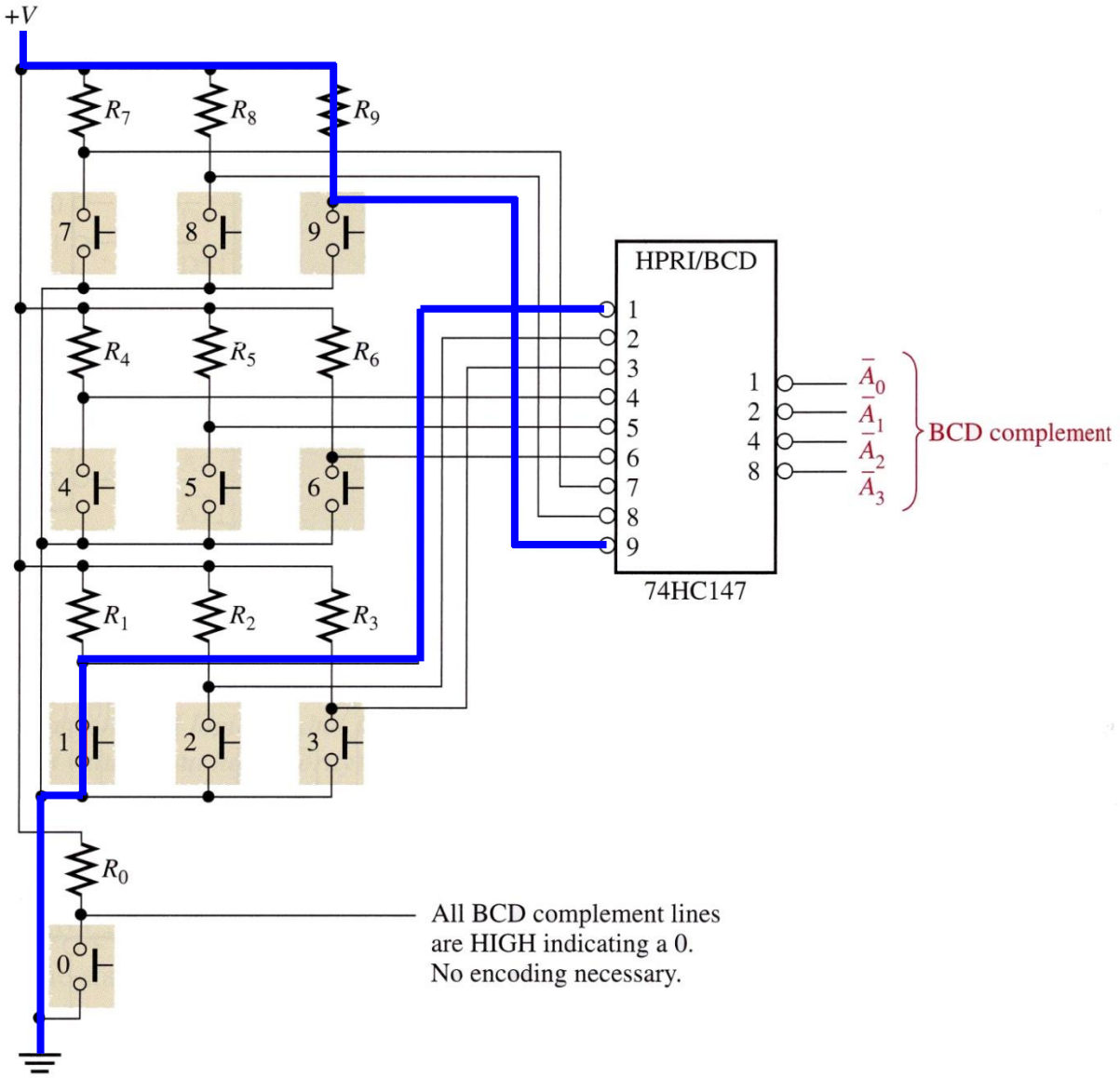


$\overline{I}_1$	$\overline{I}_2$	$\overline{I}_3$	$\overline{I}_4$	$\overline{I}_5$	$\overline{I}_6$	$\overline{I}_7$	$\overline{I}_8$	$\overline{I}_9$	$\overline{A}_3$	$\overline{A}_2$	$\overline{A}_1$	$\overline{A}_0$
1	1	1	1	1	1	1	1	1	1	1	1	1
1	1	1	1	1	1	1	1	1	1	1	1	1
0	1	1	1	1	1	1	1	1	1	1	1	0
×	0	1	1	1	1	1	1	1	1	1	0	1
×	×	0	1	1	1	1	1	1	1	1	0	0
×	×	×	0	1	1	1	1	1	1	0	1	1
×	×	×	×	0	1	1	1	1	1	0	1	0
×	×	×	×	×	0	1	1	1	1	0	0	1
×	×	×	×	×	×	0	1	1	1	0	0	0
×	×	×	×	×	×	×	0	1	0	1	1	1
×	×	×	×	×	×	×	×	0	0	1	1	0



应用：键盘编码


+V


+V
R7
R8
R9
7
8
9
R4
R5
R6
4
5
6
R1
R2
R3
1
2
3
R0
0
HPRI/BCD
1
2
3
4
5
6
7
8
9
74HC147
1
2
4
8
$\bar{A}_0$
$\bar{A}_1$
$\bar{A}_2$
$\bar{A}_3$
BCD complement
All BCD complement lines
are HIGH indicating a 0.
No encoding necessary.

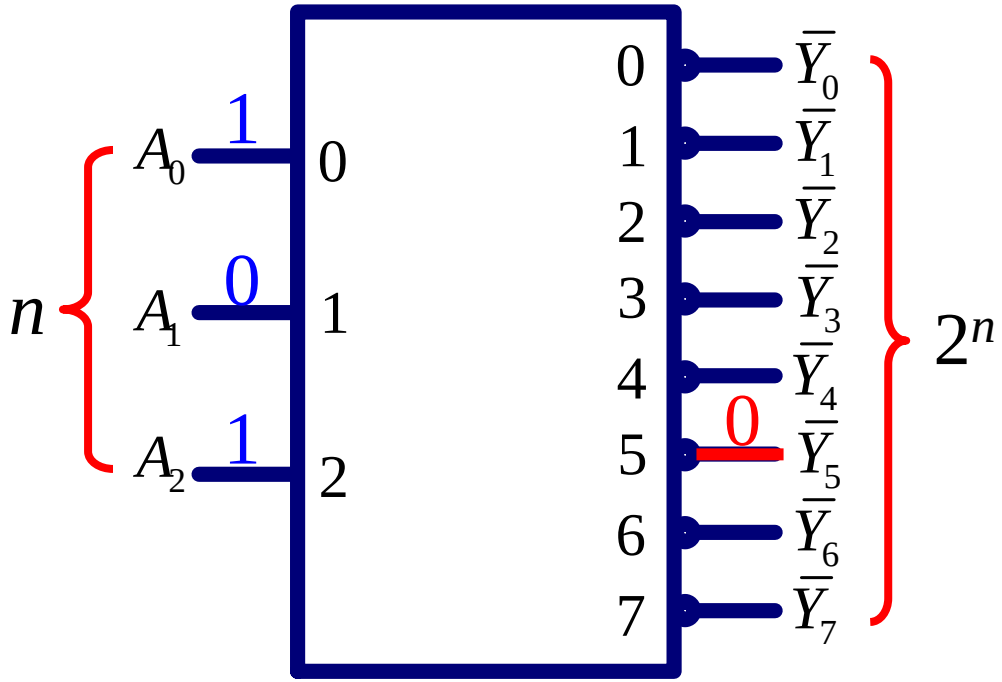
四、译码器 Decoders

译码是编码的逆过程，即将某二进制码翻译成电路的某种状态。

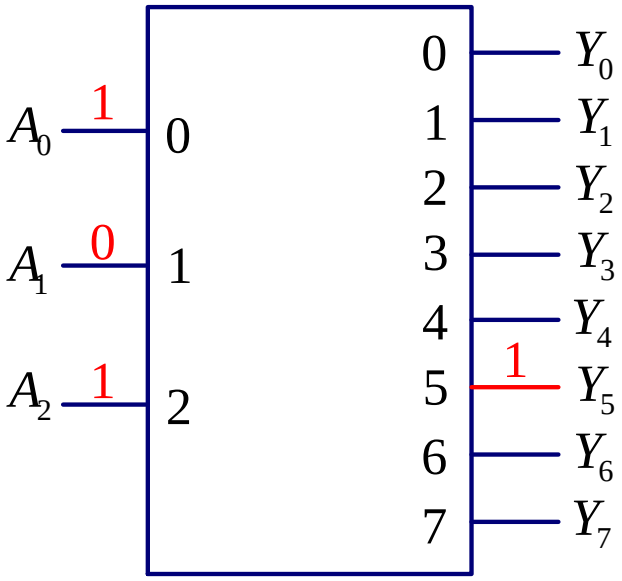
1、二进制译码器

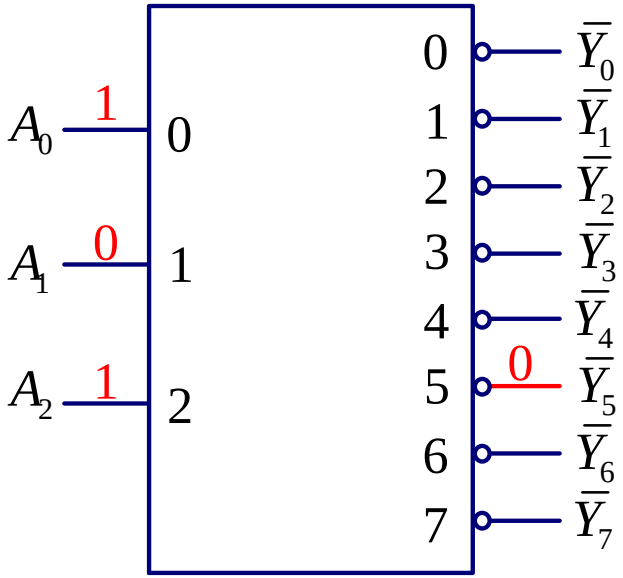
译码器的输入 —— 一组二进制代码

译码器的输出 —— 一组高低电平信号

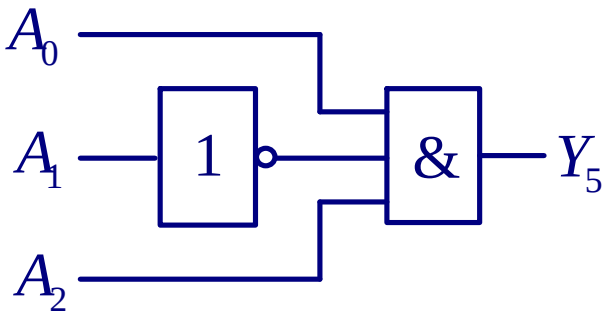


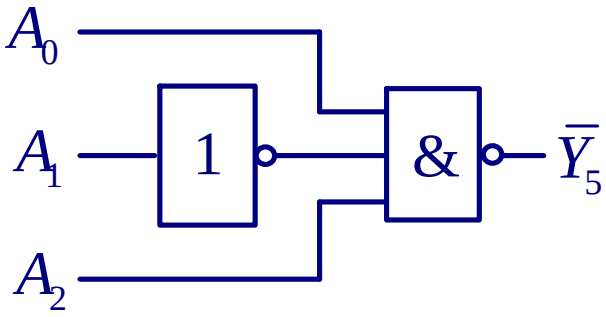
1. 二进制译码器 Minterm Generator——输入变量的所有最小项的译码输出





A_2	A_1	A_0	Y_5	$\bar{Y}_5$
0	0	0	0	1
0	0	1	0	1
0	1	0	0	1
0	1	1	0	1
1	0	0	0	1
1	0	1	1	0
1	1	0	0	1
1	1	1	0	1



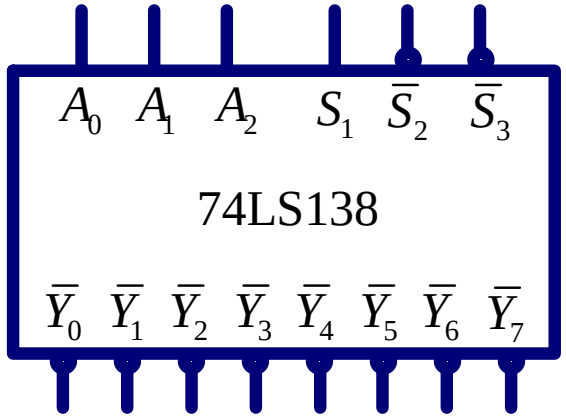


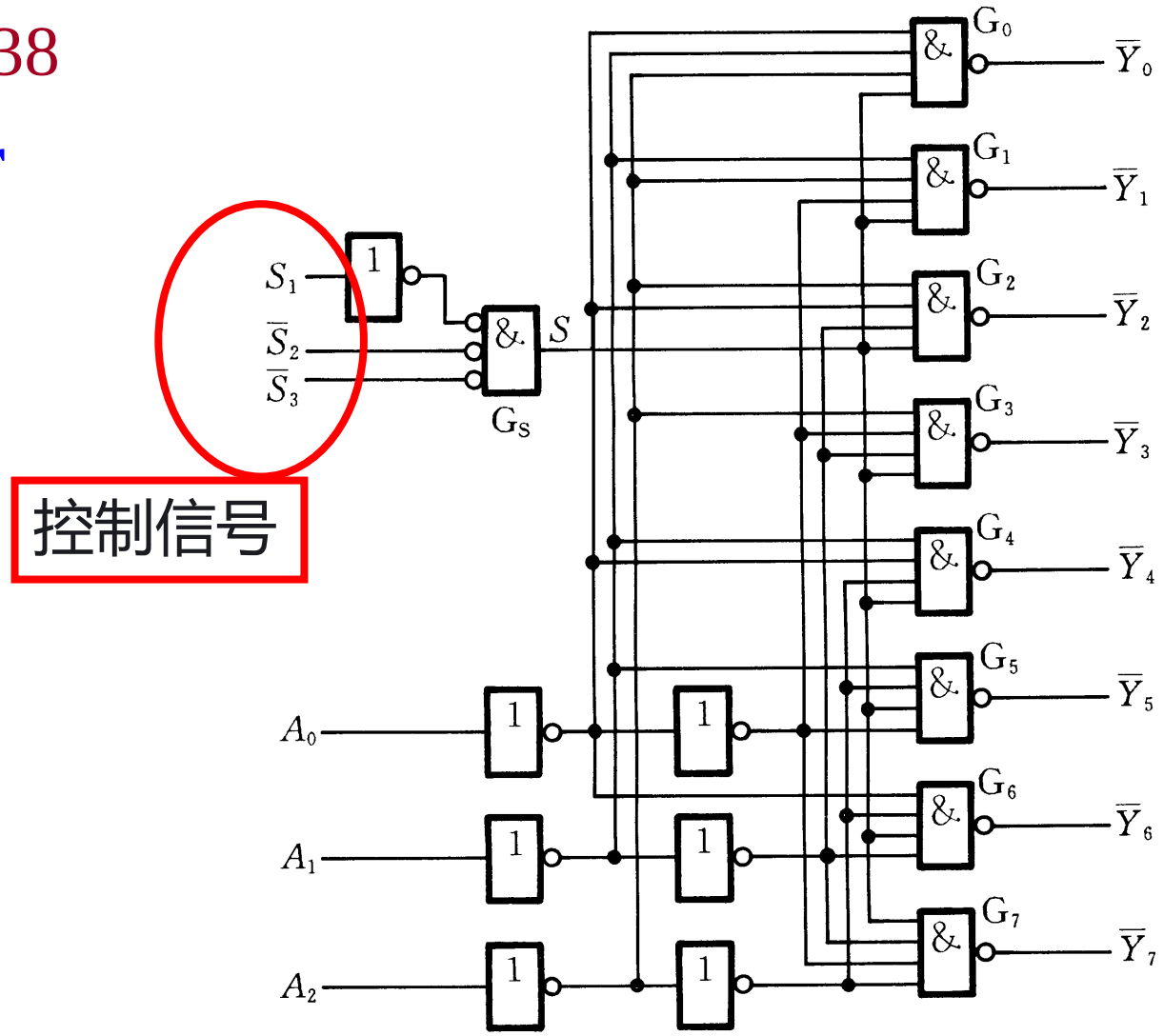
$Y_5 = A_2\bar{A}_1A_0 = m_5$ 译码函数 $\bar{Y}_5 = \overline{A_2\bar{A}_1A_0} = \bar{m}_5$

3线－8线译码器74LS138

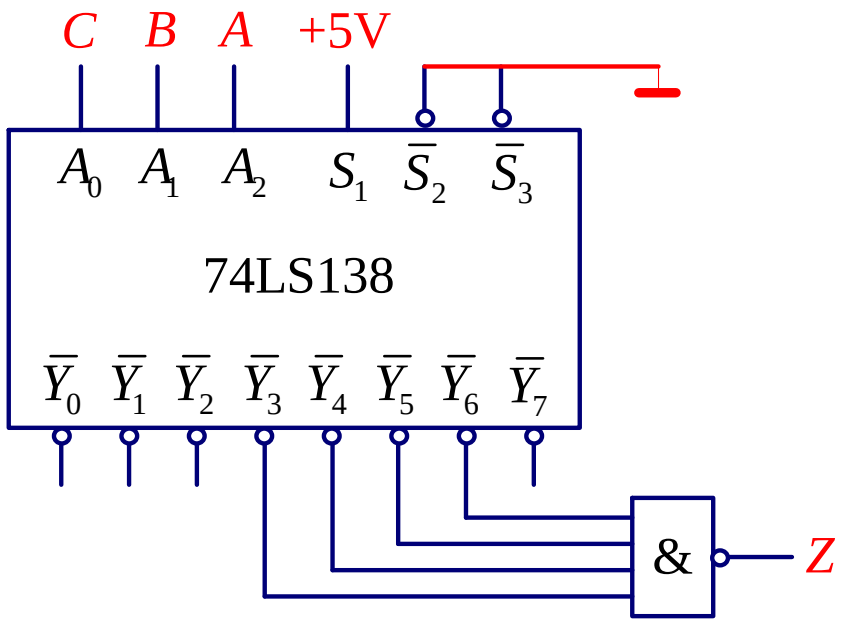
3-line-to-8-line decoder

1-of-8 decoder





例：用3线—8线译码器74LS138实现逻辑函数 $Z = A\overline{C} + \overline{A}BC + A\overline{B}C$

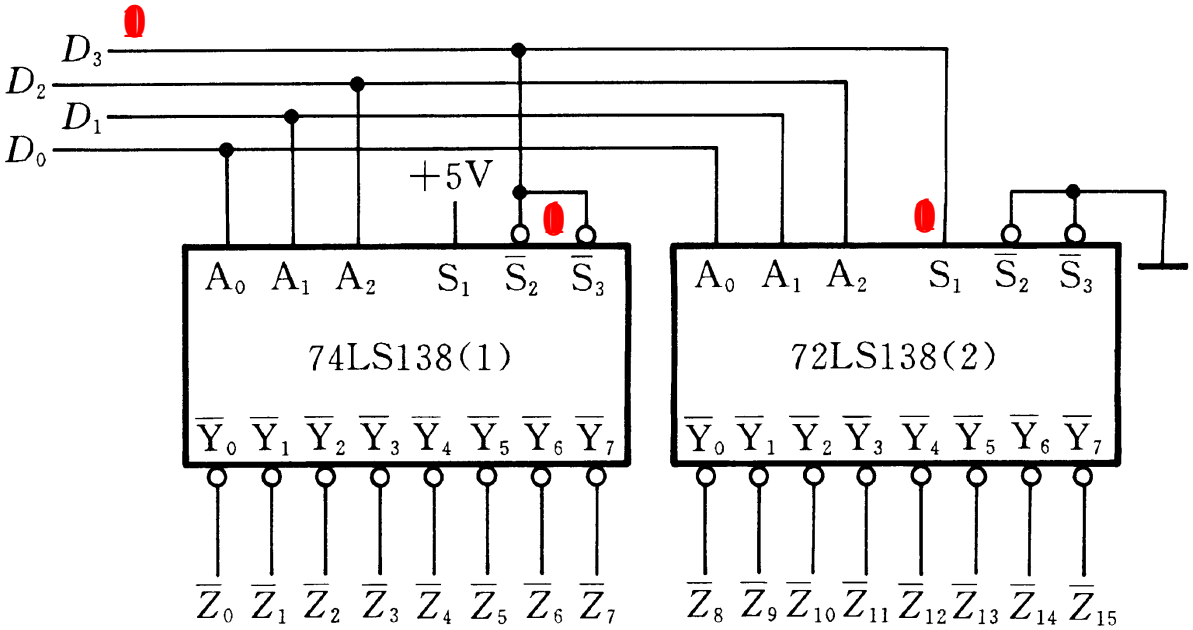


S_1	$\overline{S}_2+\overline{S}_3$	A_2	A_1	A_0	$\overline{Y}_0$	$\overline{Y}_1$	$\overline{Y}_2$	$\overline{Y}_3$	$\overline{Y}_4$	$\overline{Y}_5$	$\overline{Y}_6$	$\overline{Y}_7$
0	×	×	×	×	1	1	1	1	1	1	1	1
×	1	×	×	×	1	1	1	1	1	1	1	1
1	0	0	0	0	0	1	1	1	1	1	1	1
1	0	0	0	1	1	0	1	1	1	1	1	1
1	0	0	1	0	1	1	0	1	1	1	1	1
1	0	0	1	1	1	1	1	0	1	1	1	1
1	0	1	0	0	1	1	1	1	0	1	1	1
1	0	1	0	1	1	1	1	1	1	0	1	1
1	0	1	1	0	1	1	1	1	1	1	0	1
1	0	1	1	1	1	1	1	1	1	1	1	0

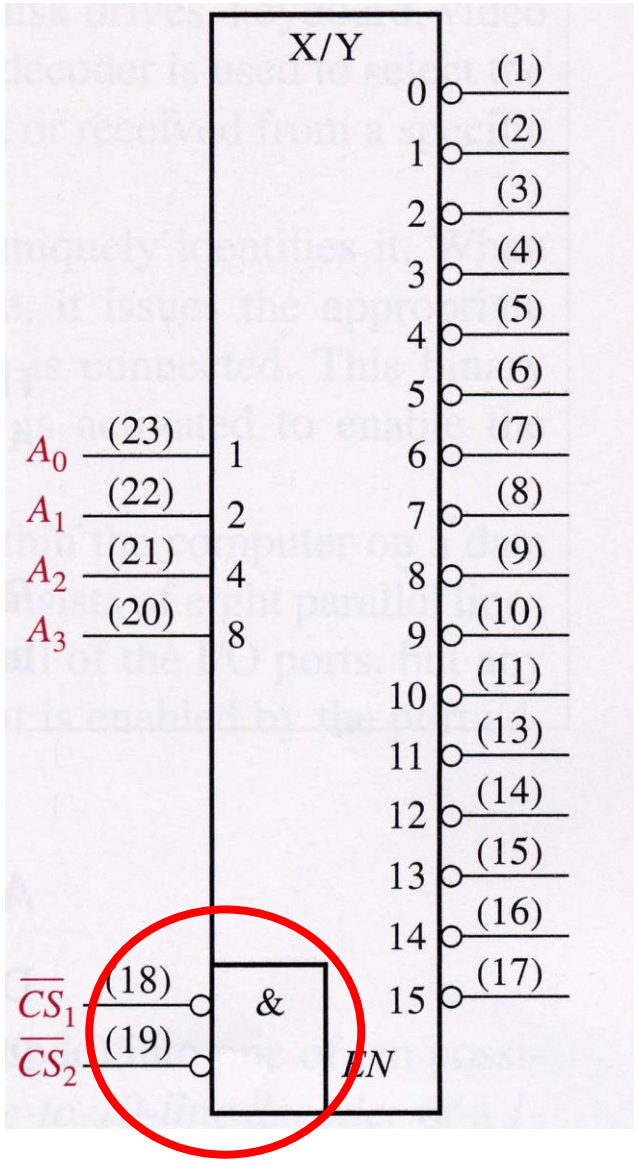
$$Z = AB\overline{C} + A\overline{B}\overline{C} + \overline{A}BC + A\overline{B}C = m_6 + m_4 + m_3 + m_5 = \overline{\overline{m}_3 \cdot \overline{m}_4 \cdot \overline{m}_5 \cdot \overline{m}_6}$$

例：试用两片3线-8线译码器74LS138组成 4线-16线译码器，将输入的4位二进制代码$D_3D_2D_1D_0$译成16个独立的低电平信号 $\overline{Z}_0 \sim \overline{Z}_{15}$

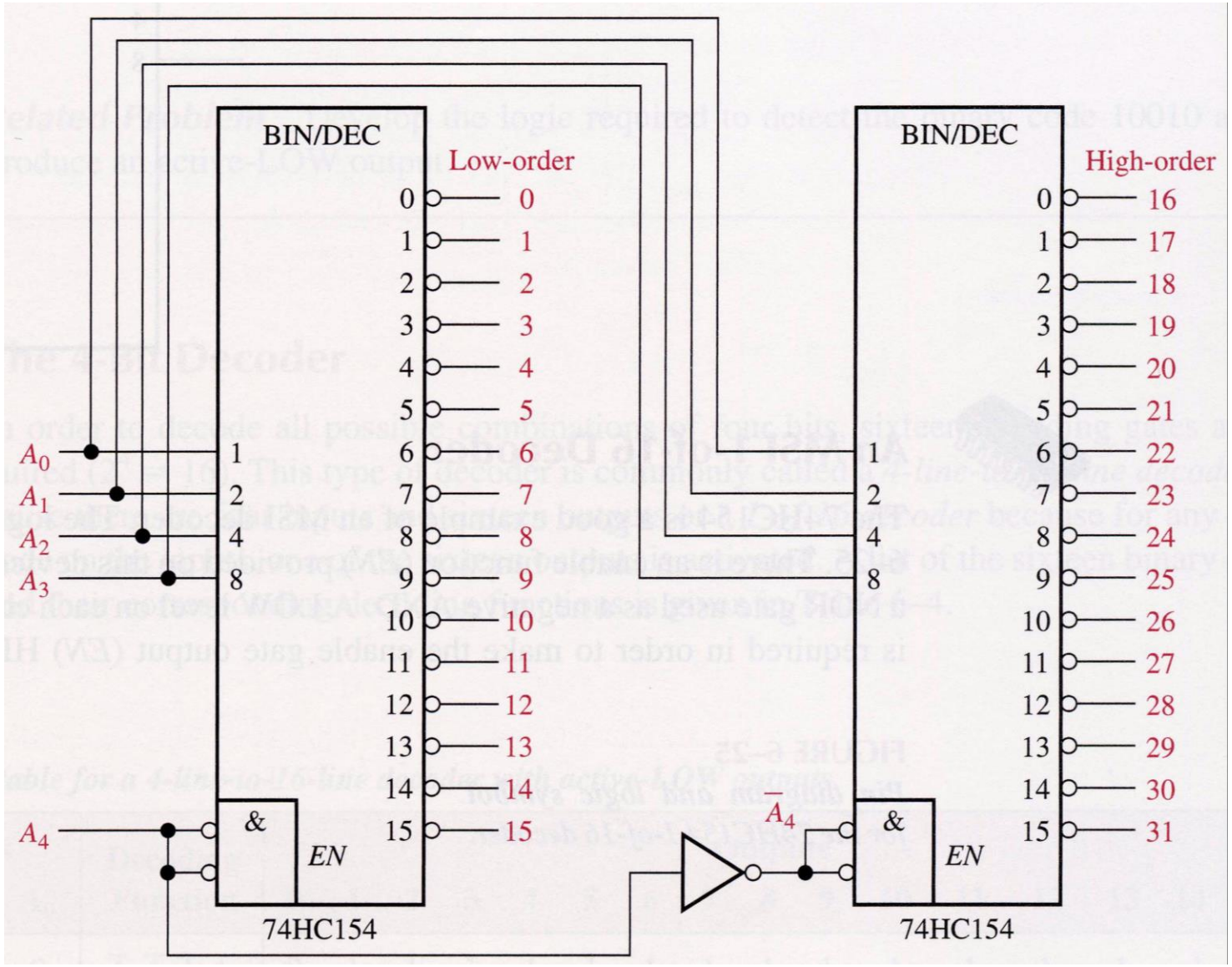
D_3	D_2	D_1	D_0	$\overline{Z}_0$	$\overline{Z}_1$	…	$\overline{Z}_7$	$\overline{Z}_8$	$\overline{Z}_9$	…	$\overline{Z}_{15}$
0	0	0	0	0	1	…	1	1	1	1	1
0	0	0	1	1	0	…	1	1	1	1	1
…	…	…	…	…	…	…	1	1	1	1	1
0	1	1	1	1	1	1	0	1	1	1	1
1	0	0	0	1	1	1	1	0	1	1	1
1	0	0	1	1	1	1	1	1	0	1	1
…	…	…	…	…	…	…	…	…	…	…	…
1	1	1	1	1	1	1	1	1	1	1	0



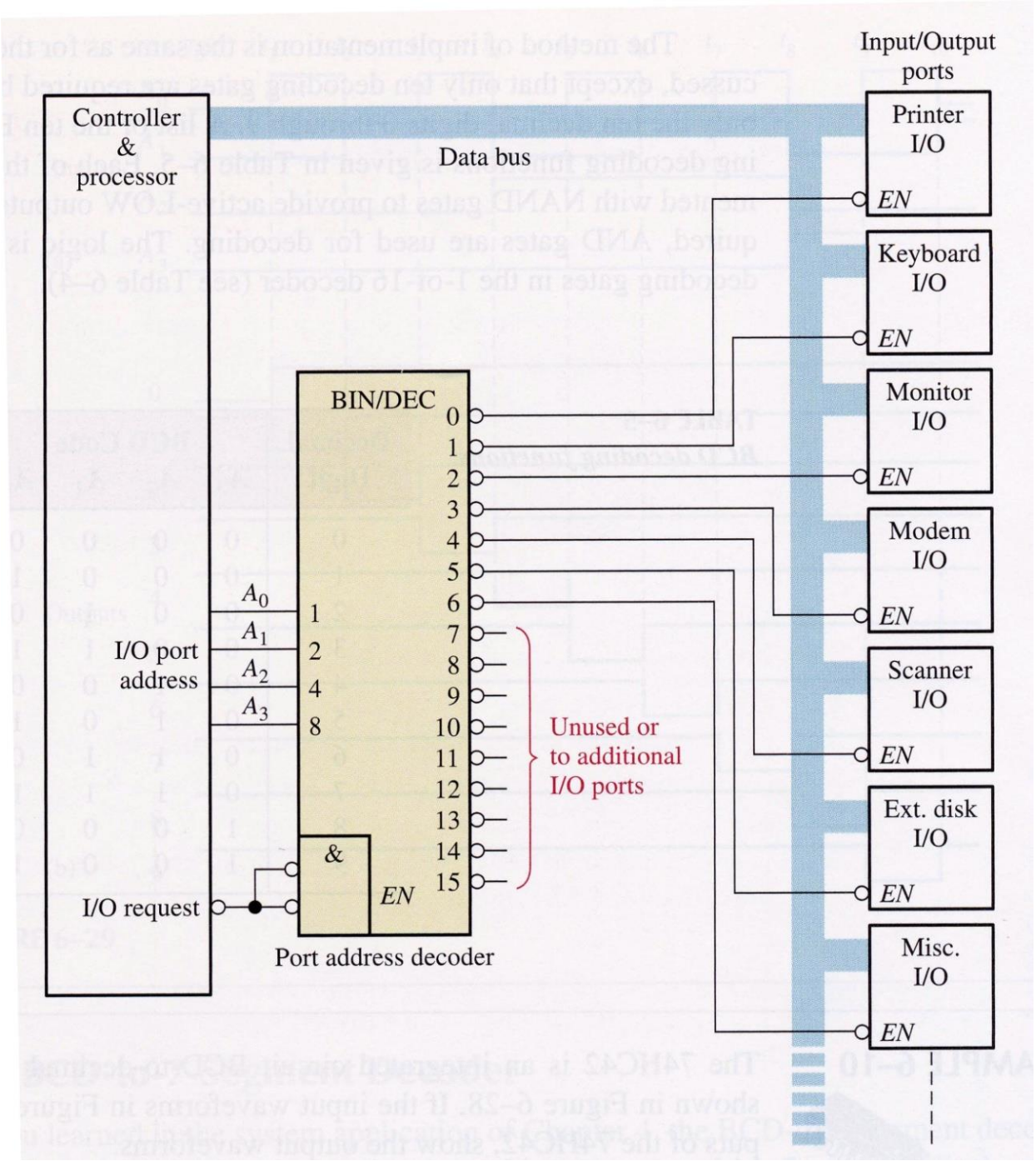
4线－16线译码器74HC154



扩展为5线-32线译码器

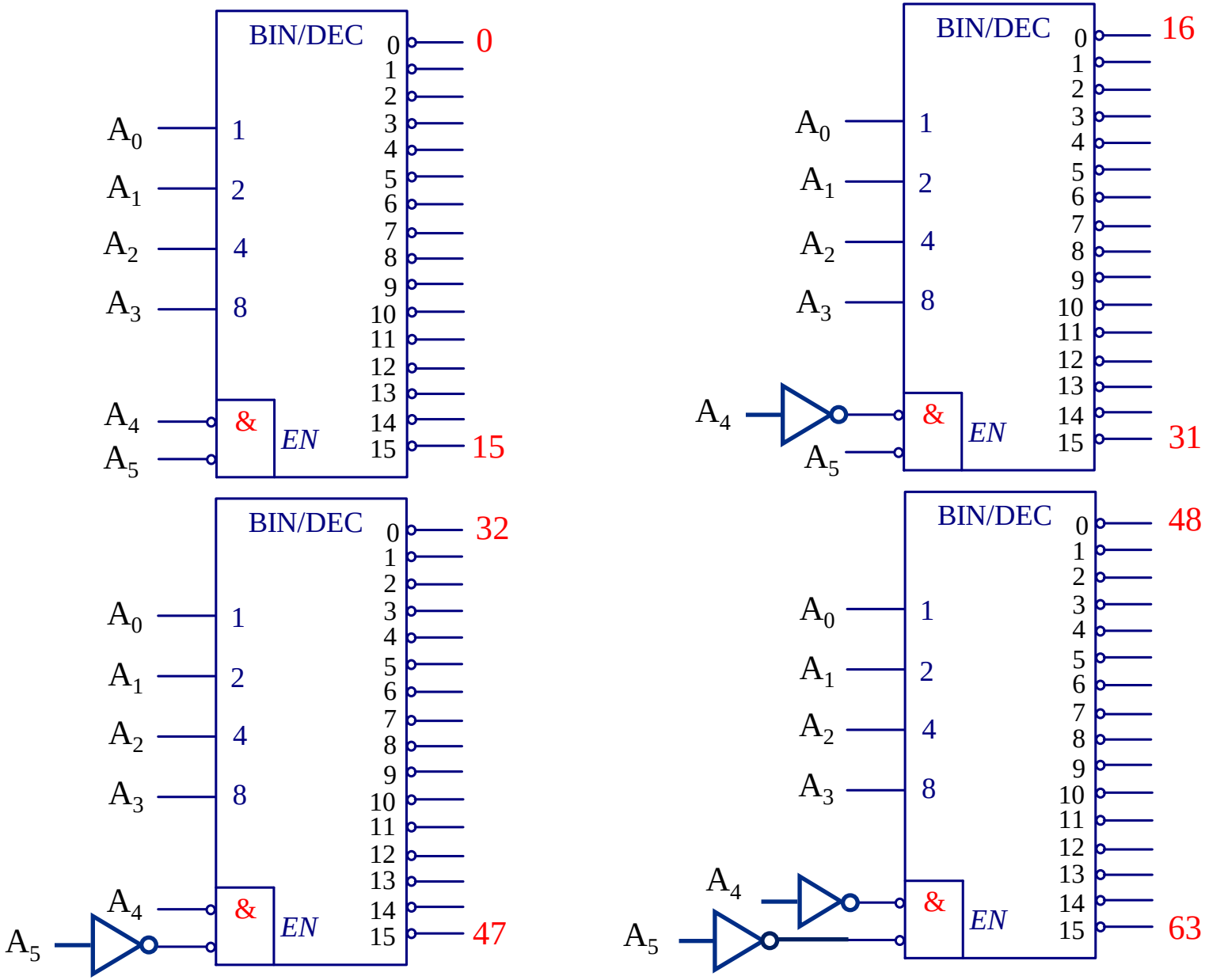


应用：简单计算机I/O端口系统



How many 74HC154 1-of-16 decoders are necessary to decode a 6-bit binary number?

$2^6=2^4\times4=64$



What will be the state of the outputs for the decoder with the input shown?

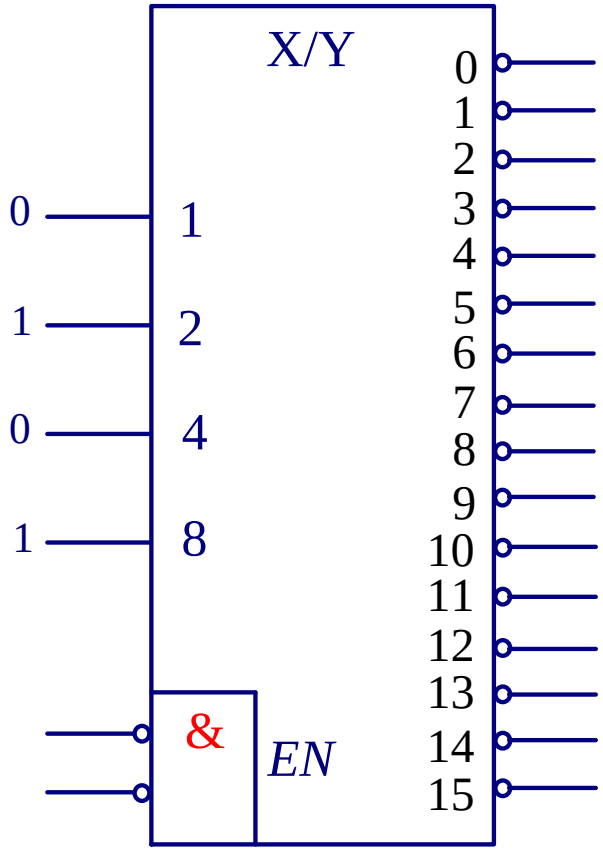
A

A. All output will be HIGH

B. The 10 output will be LOW, all others HIGH

C. The 10 output will be HIGH, all others LOW

D. The 8 and 2 outputs will be HIGH, all others LOW



2、二－十进制译码器

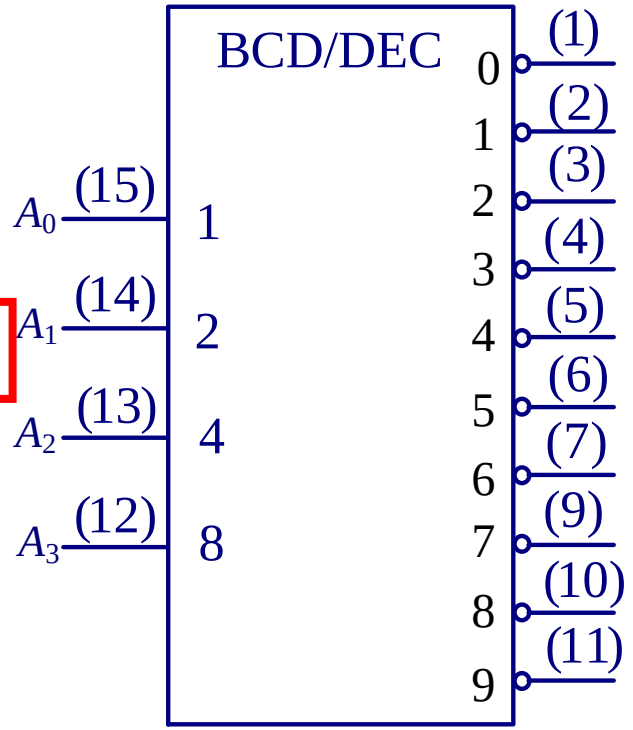
将输入的10个BCD码译成10组不同的高低电平输出

序号	输入 A_3	A_2	A_1	A_0	输出 $\overline{Y}_0$	$\overline{Y}_1$	$\overline{Y}_2$	$\overline{Y}_3$	$\overline{Y}_4$	$\overline{Y}_5$	$\overline{Y}_6$	$\overline{Y}_7$	$\overline{Y}_8$	$\overline{Y}_9$
0	0	0	0	0	0	1	1	1	1	1	1	1	1	1
1	0	0	0	1	1	0	1	1	1	1	1	1	1	1
2	0	0	1	0	1	1	0	1	1	1	1	1	1	1
3	0	0	1	1	1	1	1	0	1	1	1	1	1	1
4	0	1	0	0	1	1	1	1	0	1	1	1	1	1
5	0	1	0	1	1	1	1	1	1	0	1	1	1	1
6	0	1	1	0	1	1	1	1	1	1	0	1	1	1
7	0	1	1	1	1	1	1	1	1	1	1	0	1	1
8	1	0	0	0	1	1	1	1	1	1	1	1	0	1
9	1	0	0	1	1	1	1	1	1	1	1	1	1	0
伪码	1	0	1	0	1	1	1	1	1	1	1	1	1	1
	1	0	1	1	1	1	1	1	1	1	1	1	1	1
	1	1	0	0	1	1	1	1	1	1	1	1	1	1
	1	1	0	1	1	1	1	1	1	1	1	1	1	1
	1	1	1	0	1	1	1	1	1	1	1	1	1	1
	1	1	1	1	1	1	1	1	1	1	1	1	1	1

拒绝伪码

$$\overline{Y}_0 = \overline{\overline{A}_3 \cdot \overline{A}_2 \cdot \overline{A}_1 \cdot \overline{A}_0} = \overline{m}_0$$

二-十进制译码器74LS42



S J T U

There are 4 decoders and two OR gates. The inputs number of each OR gate is less than 8. Using the decoders and OR gates to implement the following two functions expressed in Karnaugh map.

(Hint: implement the four min-terms in a same row/column with a single output of a decoder)

ab \ cd	00	01	11	10
00	1			1
01	1			1
11	1	1		1
10	1			

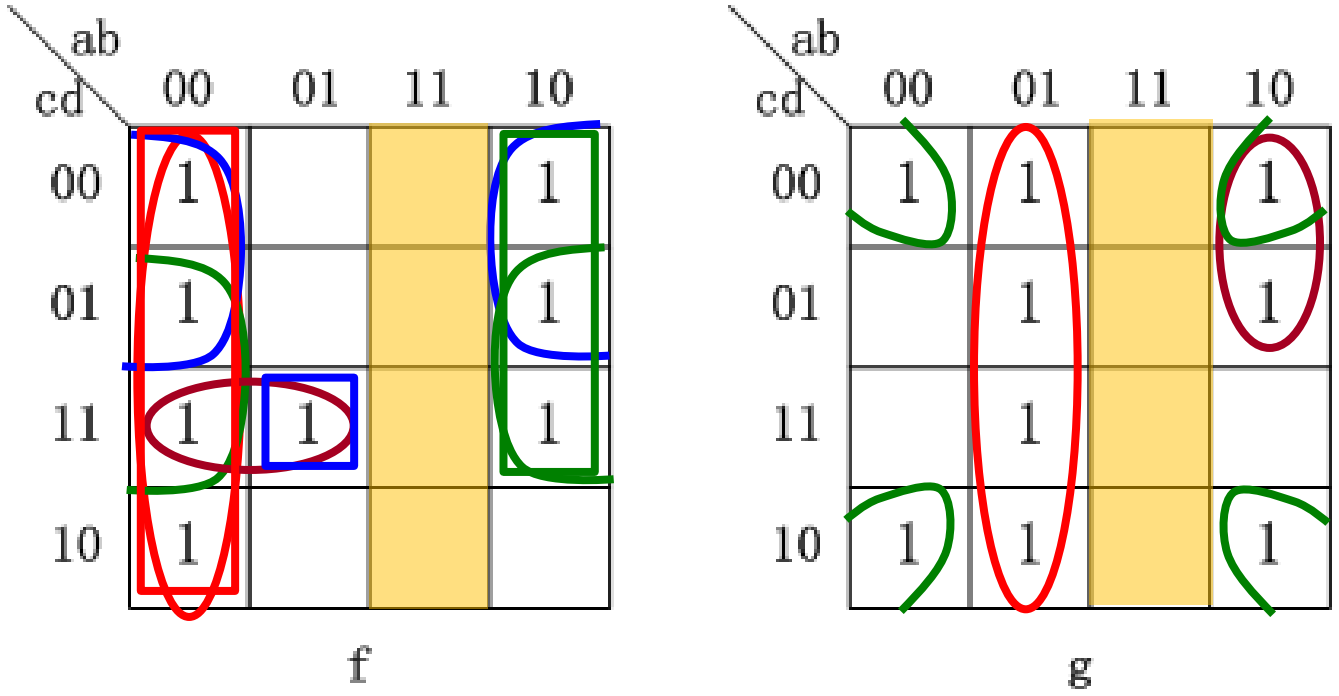
f

ab \ cd	00	01	11	10
00	1	1		1
01		1		1
11		1		
10	1	1		1

g

Decoder's truth table

EN	a	b	0	1	2	3
0	×	×	0	0	0	0
1	0	0	1	0	0	0
1	0	1	0	1	0	0
1	1	0	0	0	1	0
1	1	1	0	0	0	1



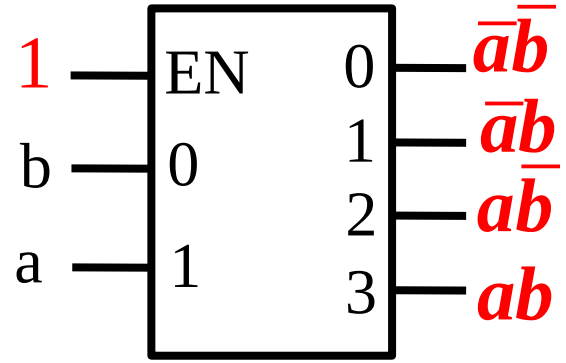
Decoder's truth table

EN	a	b	0	1	2	3
0	×	×	0	0	0	0
1	0	0	1	0	0	0
1	0	1	0	1	0	0
1	1	0	0	0	1	0
1	1	1	0	0	0	1

$$f = \bar{a}\bar{b} + \bar{b}\bar{c} + \bar{b}d + \bar{a}cd \qquad g = \bar{a}b + \bar{b}\bar{d} + a\bar{b}\bar{c}$$

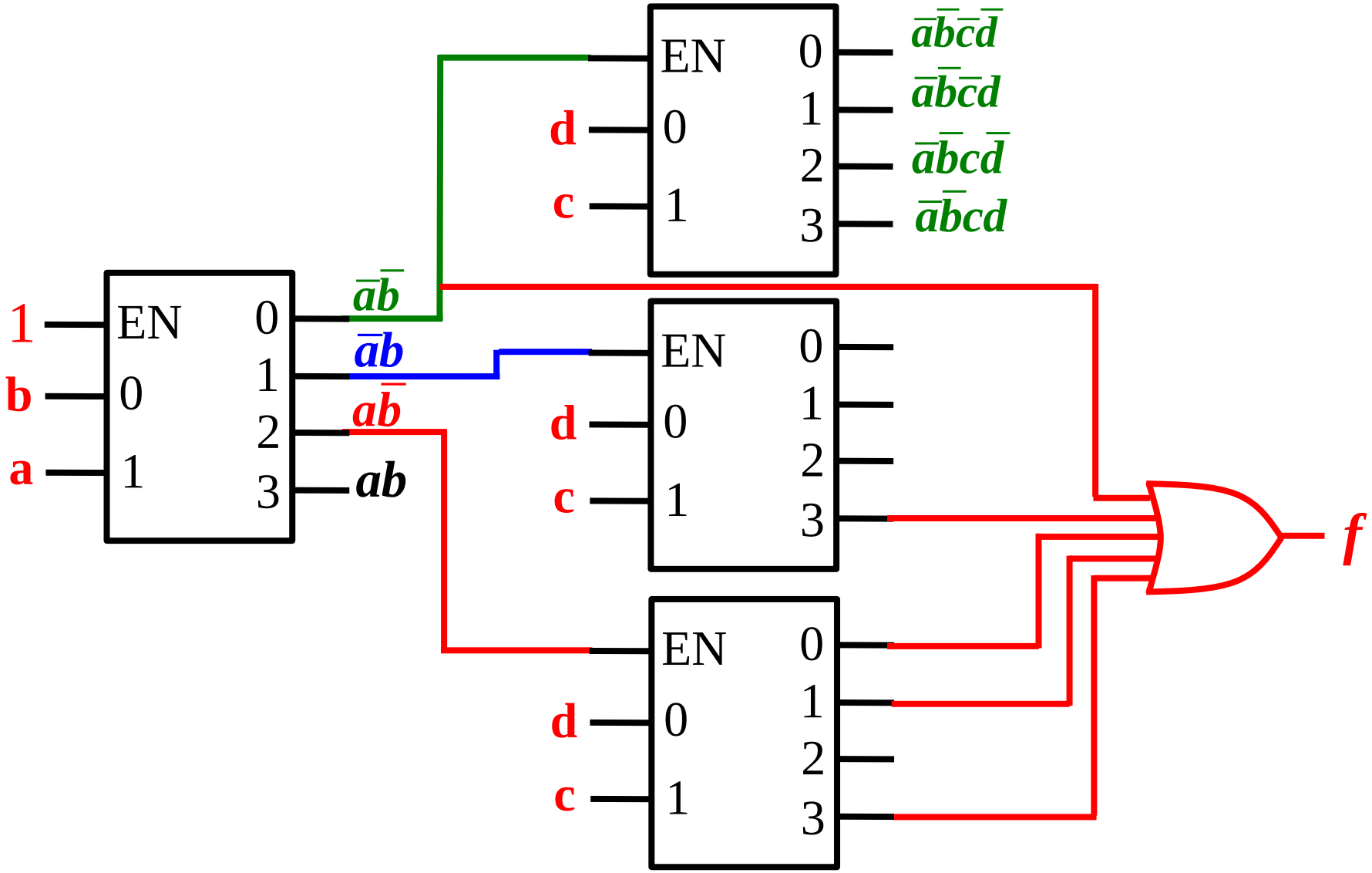
$$f = \bar{a}\bar{b} + \bar{a}bcd + a\bar{b}(\bar{c}d + \bar{c}\bar{d} + cd)$$

$$g = \bar{a}\bar{b}(c\bar{d} + \bar{c}\bar{d}) + \bar{a}b + a\bar{b}(c\bar{d} + \bar{c}\bar{d} + \bar{c}d)$$



$$f = \bar{a}\bar{b} + a\bar{b}(\bar{c}d + c\bar{d} + cd) + \bar{a}bcd$$

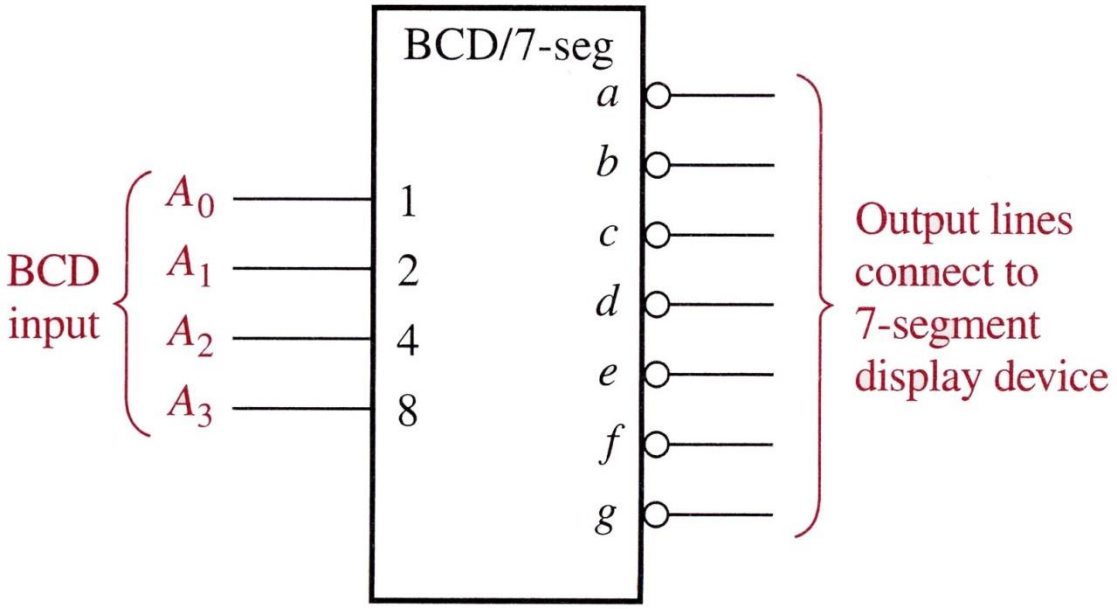
$$g = \bar{a}b + a\bar{b}(c\bar{d} + \bar{c}\bar{d} + \bar{c}d) + \bar{a}\bar{b}(c\bar{d} + \bar{c}\bar{d})$$



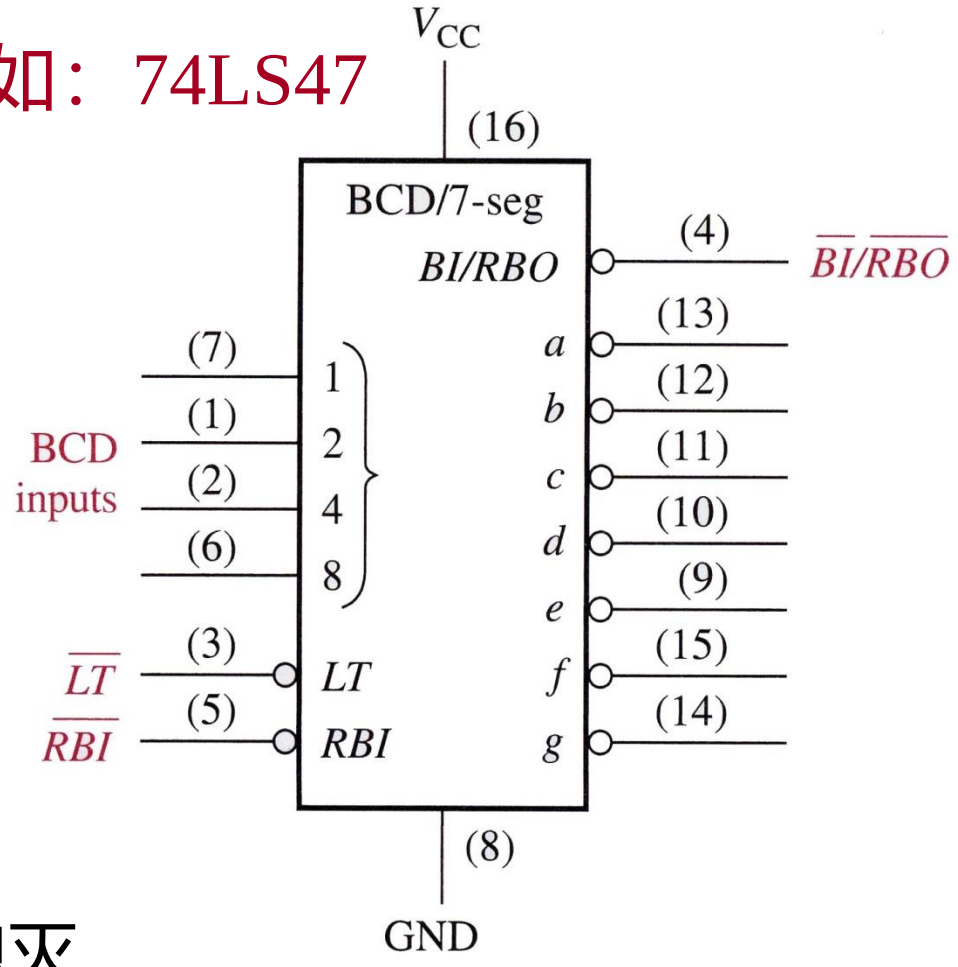
3、显示译码器

作输入用时，若 $\overline{BI}=0$，将被驱动的数码管熄灭

作输出用时，若 $\overline{RBO}=0$表示已将本该显示的零熄灭了



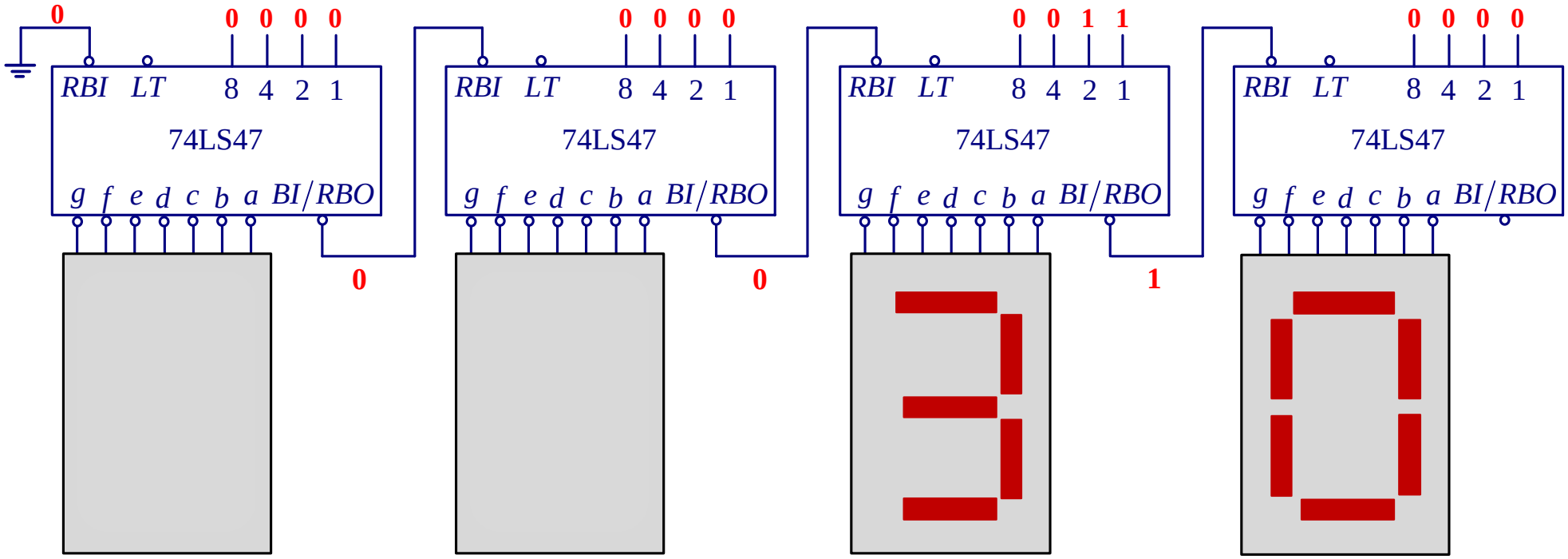
如：74LS47



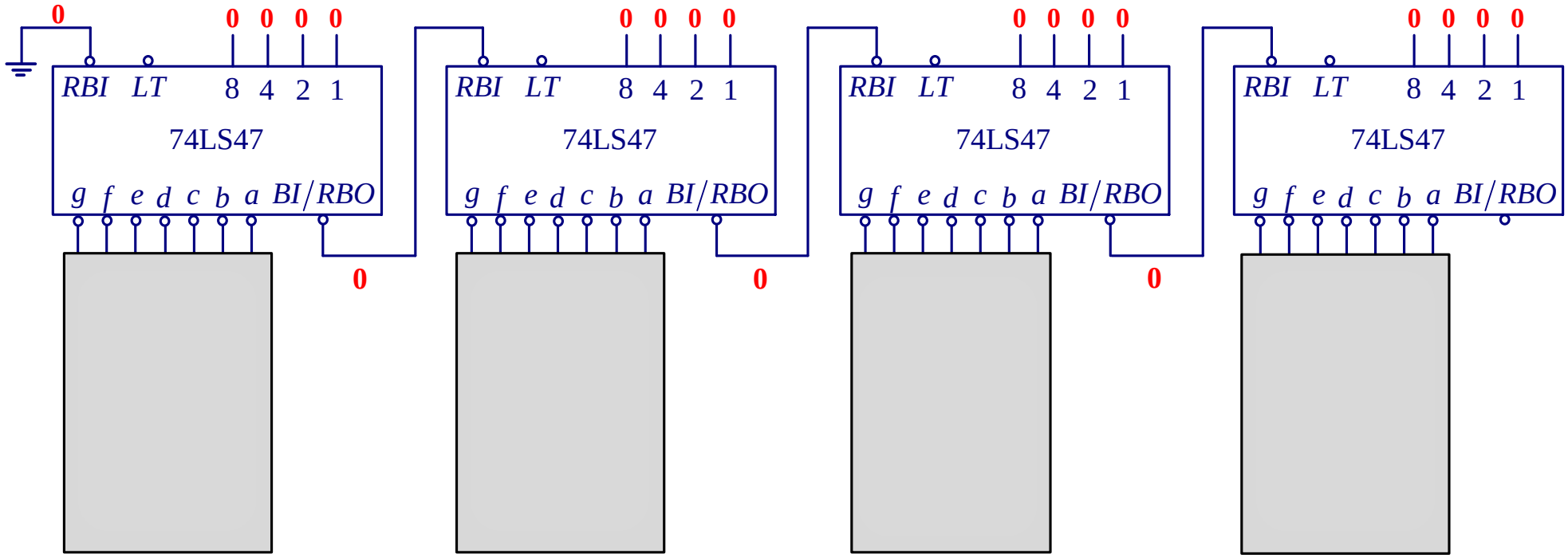
$\overline{BI}=1$，$\overline{LT}=0$ 时，数码管七段同时点亮，可检查是否正常。平时置 $\overline{LT}$ 为高电平。

$\overline{RBI}=0$ 时，将本应显示 **0** 但不希望显示的 **0** 熄灭

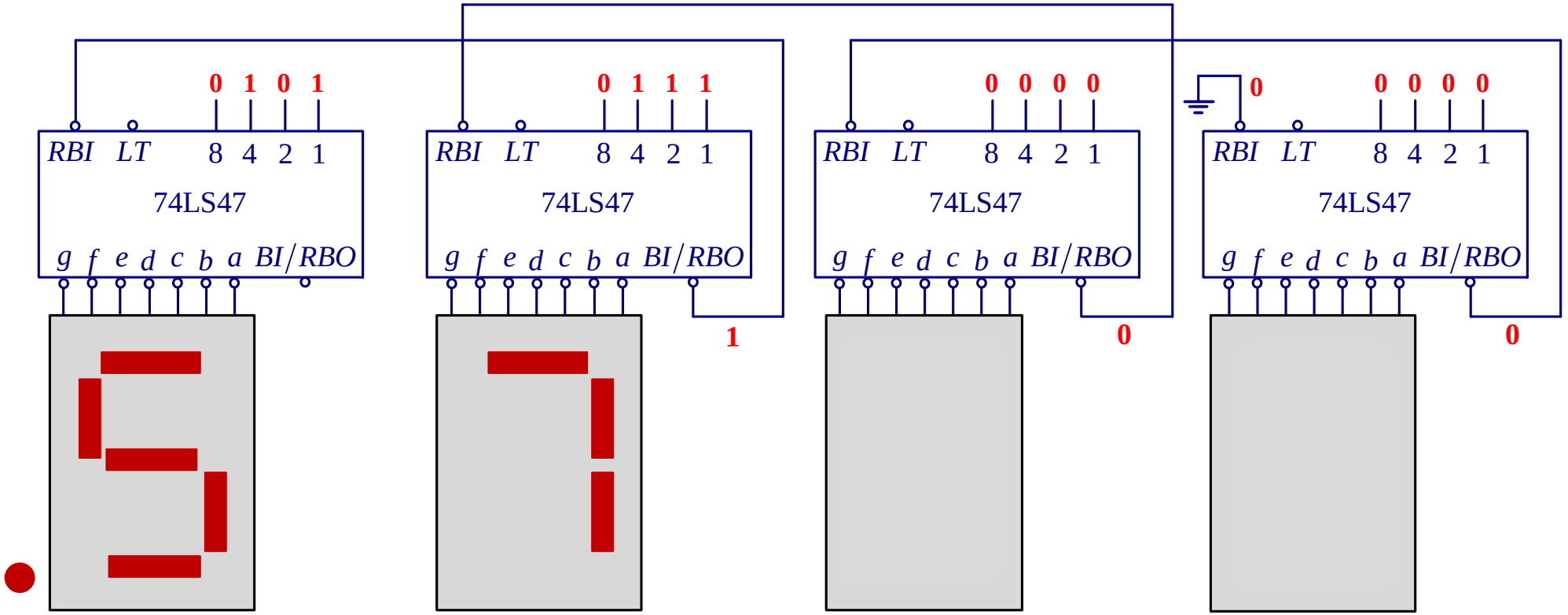
Leading Zero Suppression



Leading Zero Suppression



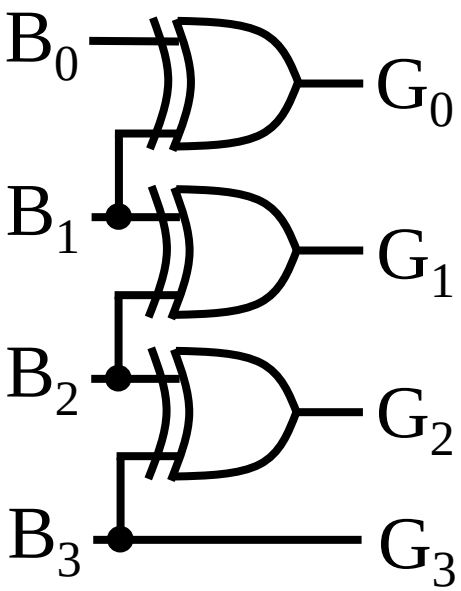
Trailing Zero Suppression



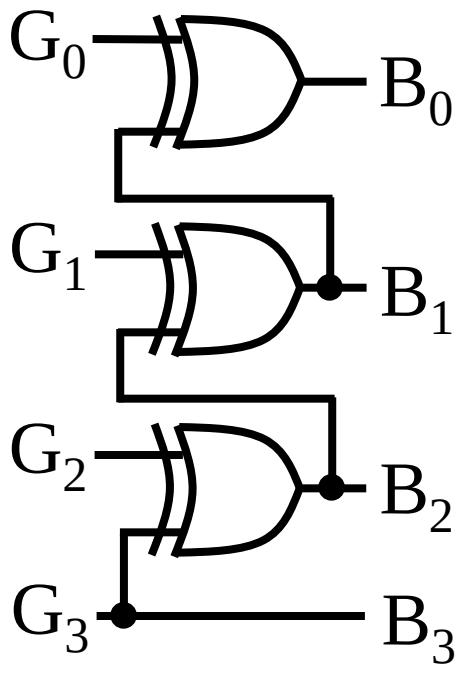
五、代码转换器 Code converters

Binary-Gray Conversion

二进制数 → 格雷码

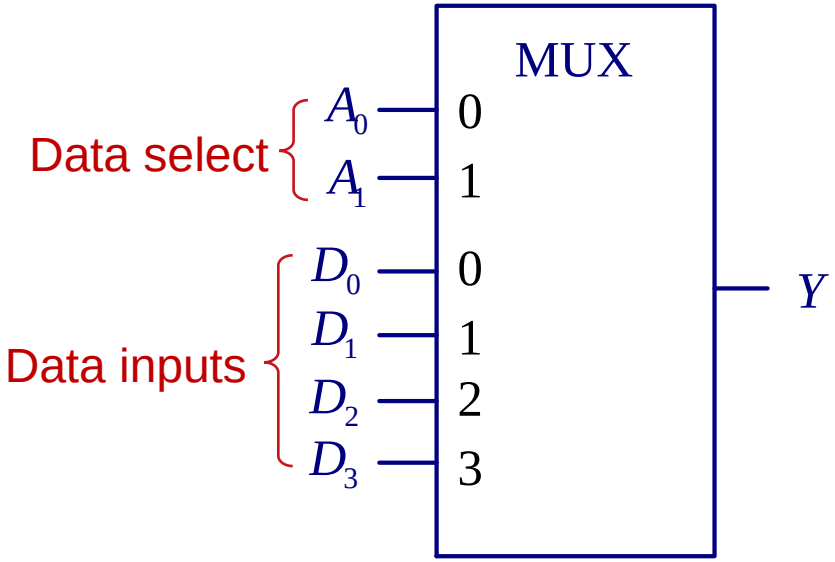


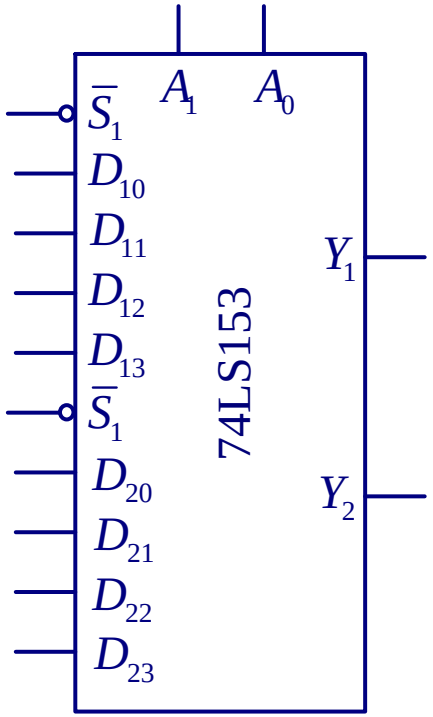
格雷码 → 二进制数

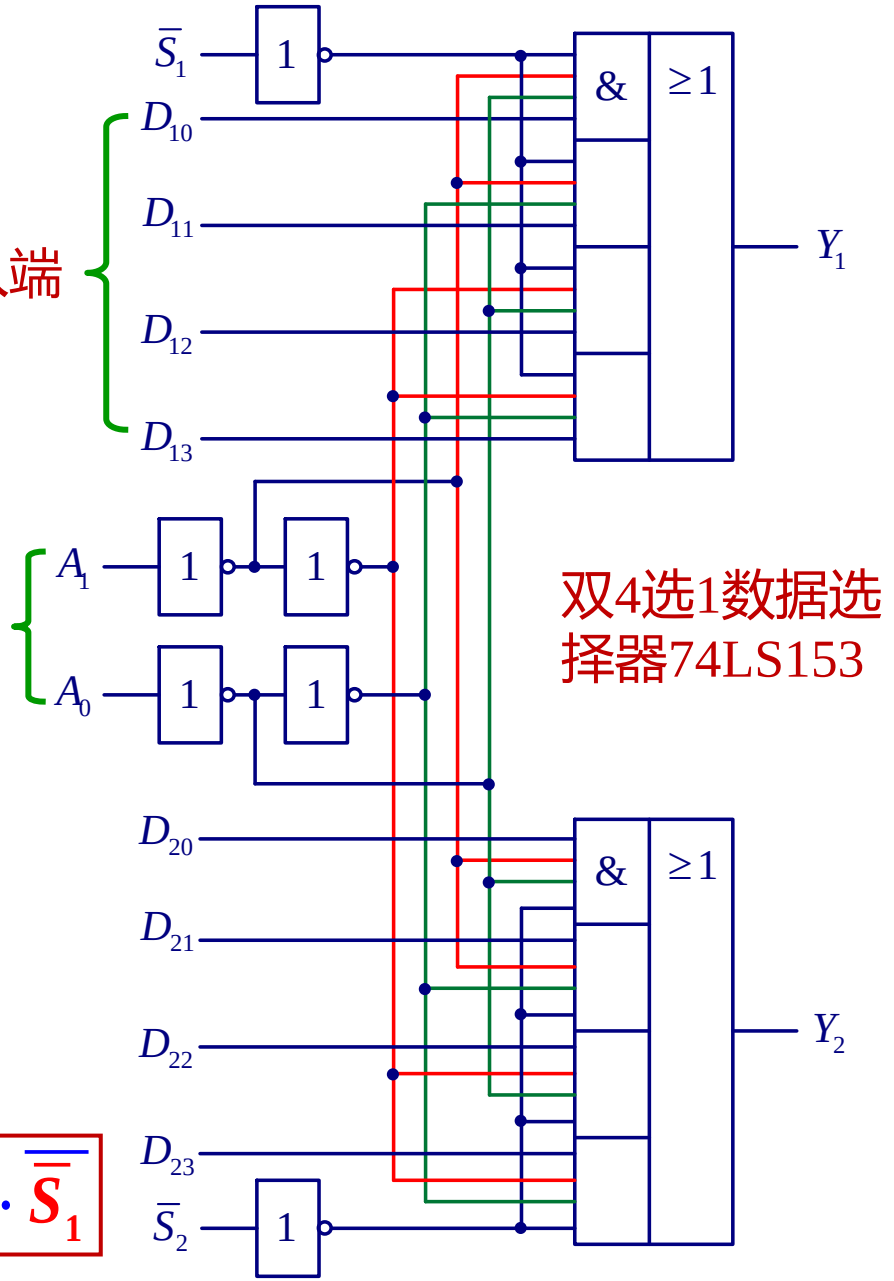


六、数据选择器 Multiplexers

在数字传输过程中，需要从一组数据中选出某一个时，要用到数据选择器，也称多路开关

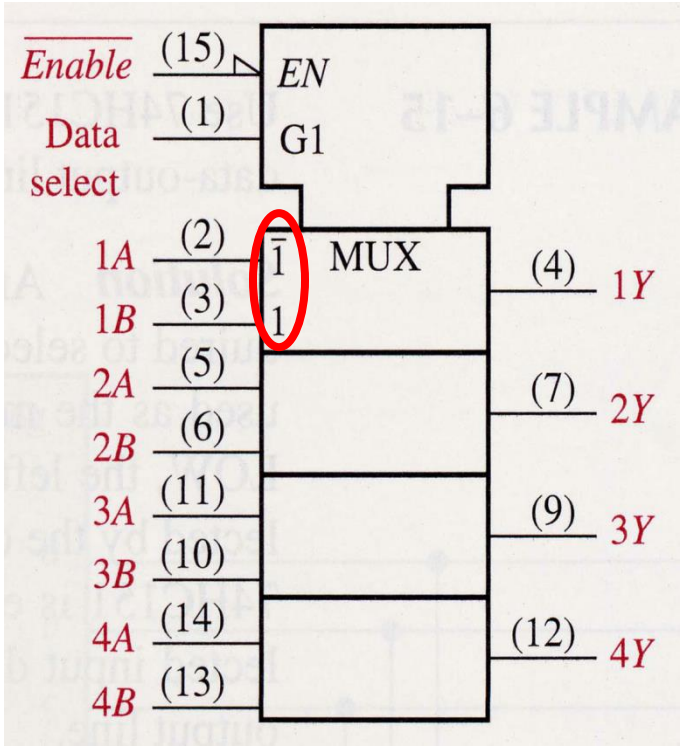




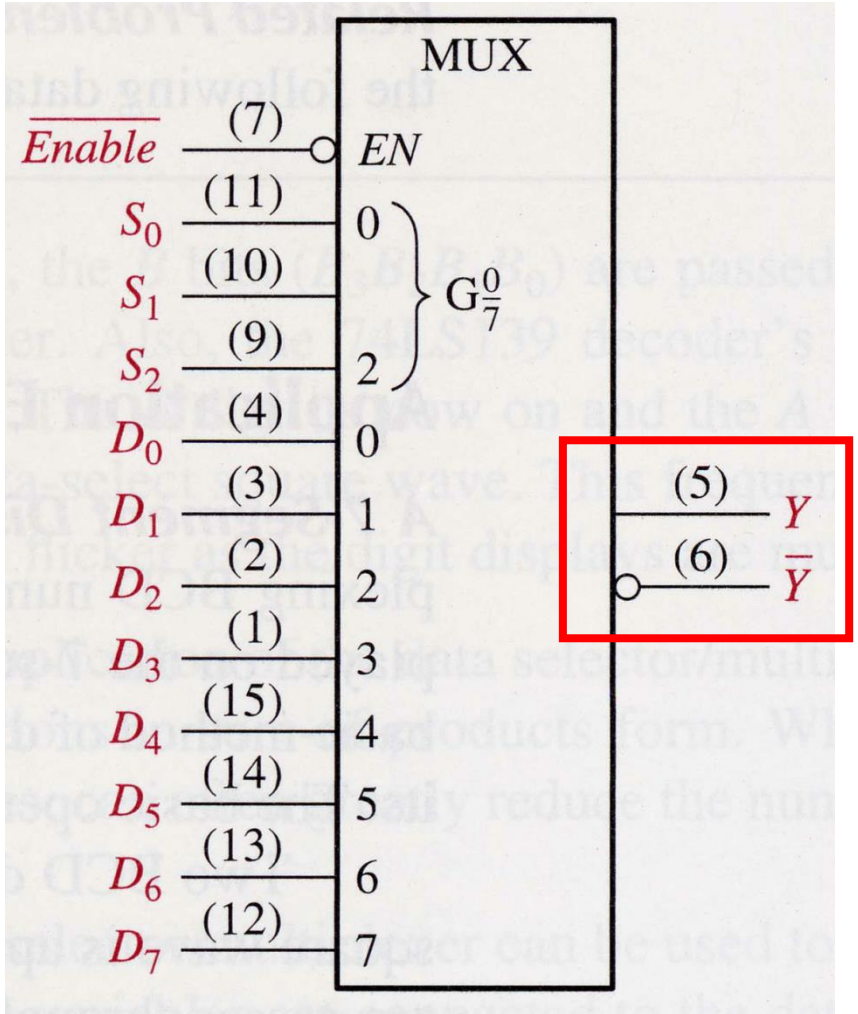


$$Y_1 = [D_{10}(\overline{A}_1\overline{A}_0) + D_{11}(\overline{A}_1 A_0) + D_{12}(A_1\overline{A}_0) + D_{13}(A_1 A_0)] \cdot \overline{\overline{S}_1}$$

74HC157A：四重2输入数据选择器



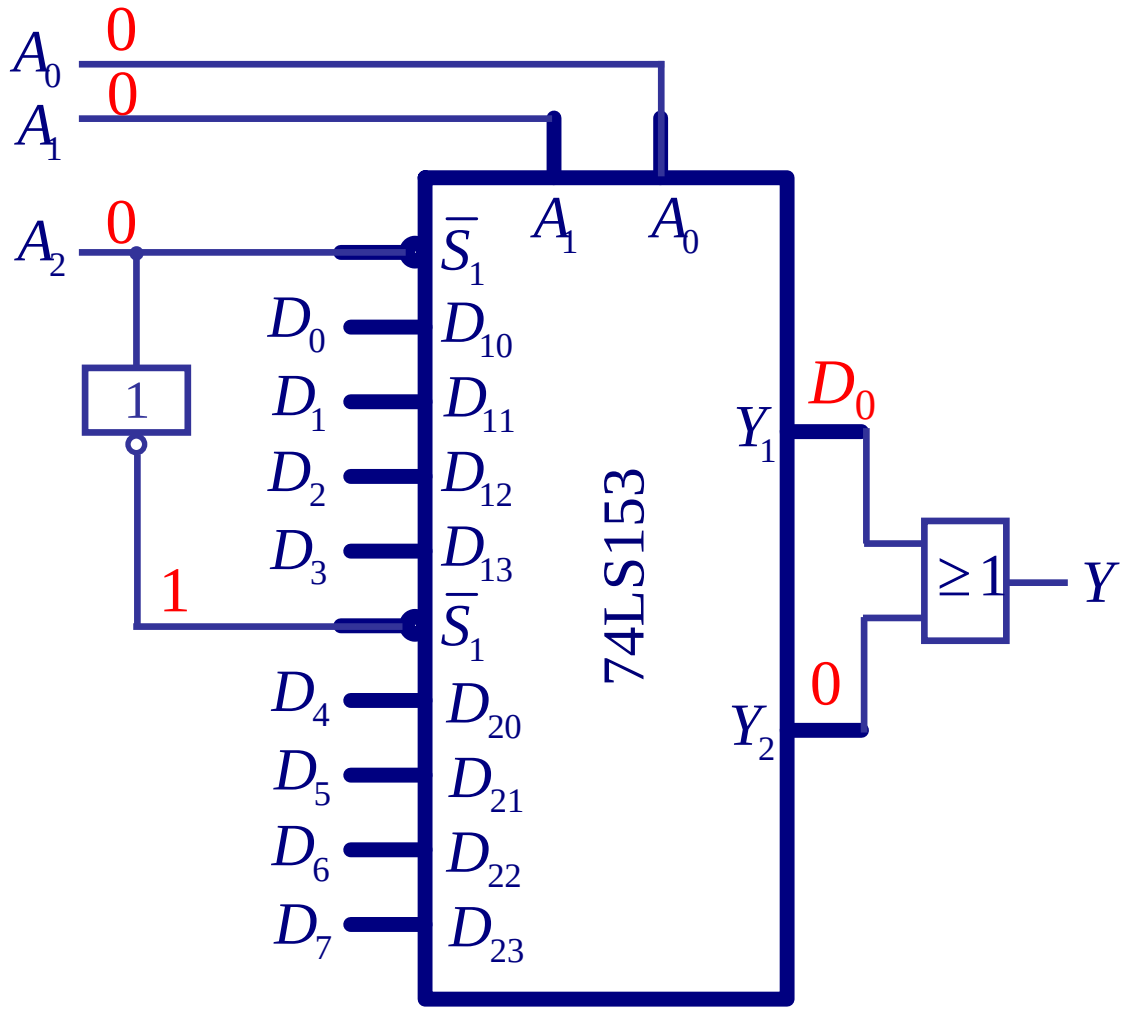
74HC151：8输入数据选择器



扩展：利用控制端实现多选1数据选择器，如用“4选1”实现“8选1”

真值表

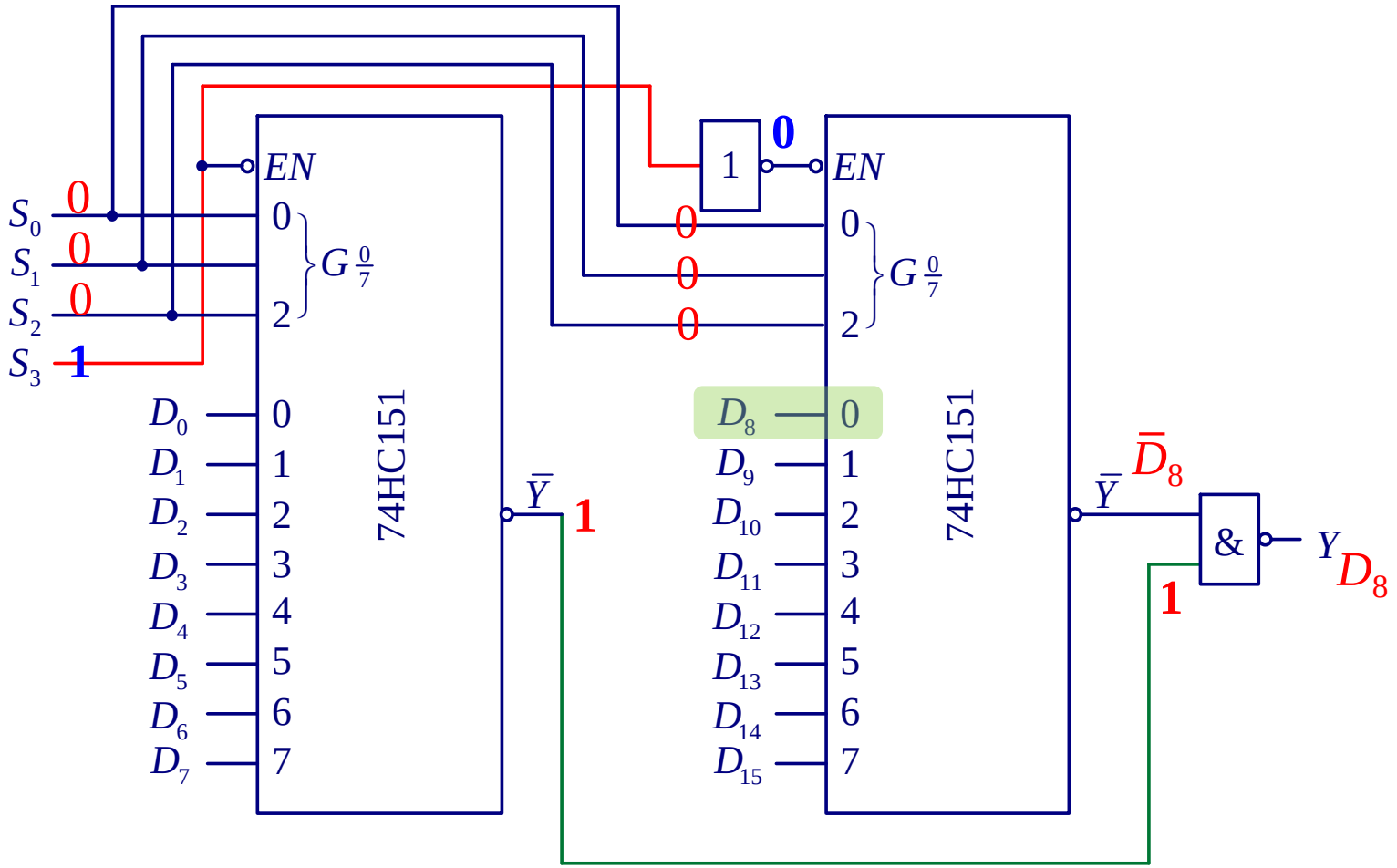
A_2	A_1	A_0	Y
0	0	0	D_0
0	0	1	D_1
0	1	0	D_2
0	1	1	D_3
1	0	0	D_4
1	0	1	D_5
1	1	0	D_6
1	1	1	D_7

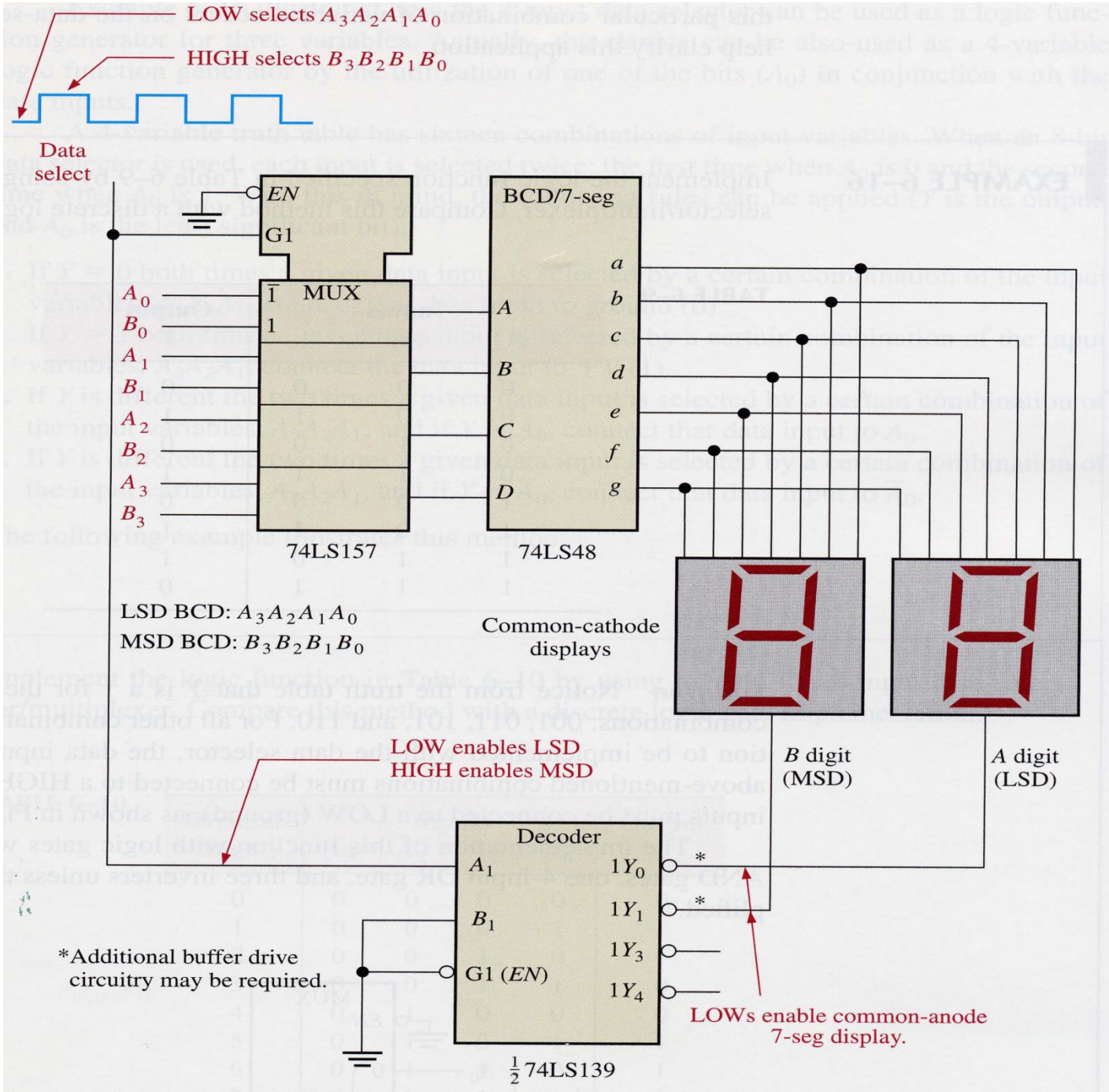


由于数据选择器被禁止时输出低电平

扩展为“16选1”数据选择器

S_3	S_2	S_1	S_0	Y
0	0	0	0	D_0
0	0	0	1	D_1
0	0	1	0	D_2
0	0	1	1	D_3
0	1	0	0	D_4
0	1	0	1	D_5
0	1	1	0	D_6
0	1	1	1	D_7
1	0	0	0	D_8
1	0	0	1	D_9
1	0	1	0	D_{10}
1	0	1	1	D_{11}
1	1	0	0	D_{12}
1	1	0	1	D_{13}
1	1	1	0	D_{14}
1	1	1	1	D_{15}



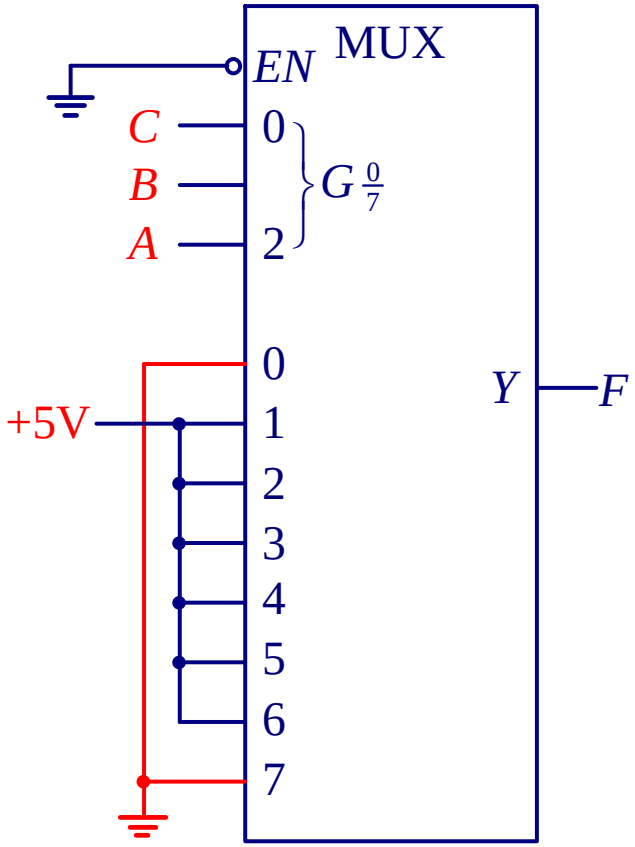

LOW selects $A_3A_2A_1A_0$
HIGH selects $B_3B_2B_1B_0$
Data select
EN
G1
MUX
$\overline{1}$
1
A_0
B_0
A_1
B_1
A_2
B_2
A_3
B_3
74LS157
BCD/7-seg
A
B
C
D
a
b
c
d
e
f
g
74LS48
LSD BCD: $A_3A_2A_1A_0$
MSD BCD: $B_3B_2B_1B_0$
Common-cathode displays
LOW enables LSD
HIGH enables MSD
B digit (MSD)
A digit (LSD)
Decoder
A_1
B_1
G1 (EN)
$1Y_0$
$1Y_1$
$1Y_3$
$1Y_4$
*
*
*Additional buffer drive circuitry may be required.
LOWs enable common-anode 7-seg display.
$\frac{1}{2}$ 74LS139

S J T U

用数据选择器设计组合逻辑电路

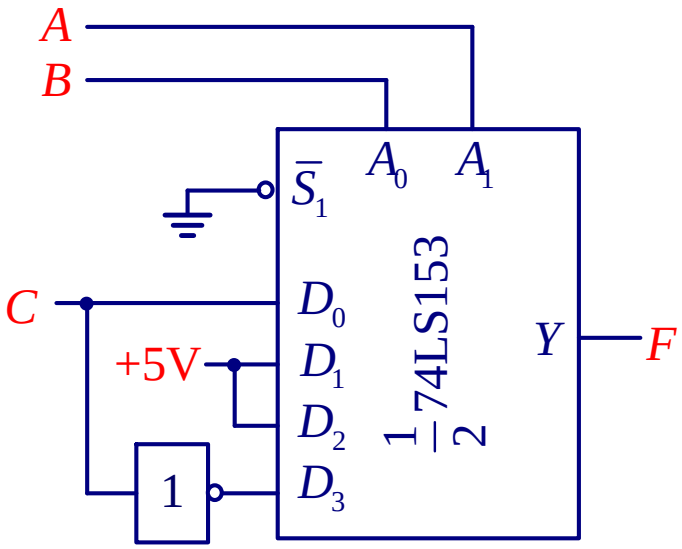
例：用8选1数据选择器实现函数 $F = A\bar{B} + \bar{A}C + B\bar{C}$

$$F = A\bar{B}C + A\bar{B}\bar{C} + \bar{A}BC + \bar{A}\bar{B}C + AB\bar{C} + \bar{A}B\bar{C} + \bar{A}\bar{B}\bar{C}\cdot 0 + ABC\cdot 0$$



方法二：用4选1数据选择器实现

$$F = \bar{A}\bar{B}\cdot C + \bar{A}B\cdot 1 + A\bar{B}\cdot 1 + AB\cdot \bar{C}$$



例：设计用三个开关（A、B、C）控制一个电灯的电路，要求改变任何一个开关的状态都能控制电灯由亮变暗或由灭变亮，设初始状态是000，电灯是暗的。要求用数据选择器实现。

1. 首先指明逻辑符号取“0”、“1”的含义；

三个开关A、B、C按下时为1，不按时为0。
用逻辑变量F表示电灯，灯亮是1，灯灭是0。

2. 根据题意列出真值表；

A	B	C	F
0	0	0	0
0	0	1	1
0	1	1	0
0	1	0	1
1	1	0	0
1	1	1	1
1	0	1	0
1	0	0	1

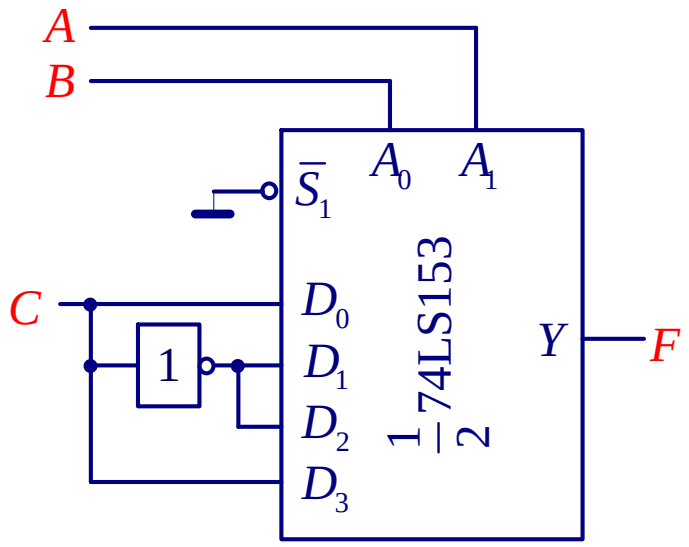
3. 列出逻辑表达式，并化简

$$F=\bar{A}\bar{B}C+\bar{A}B\bar{C}+ABC+A\bar{B}\bar{C}$$

$$=C\left(\bar{A}\bar{B}\right)+\bar{C}\left(\bar{A}B\right)+\bar{C}\left(A\bar{B}\right)+C\left(AB\right)$$

D_0 D_1 D_2 D_3

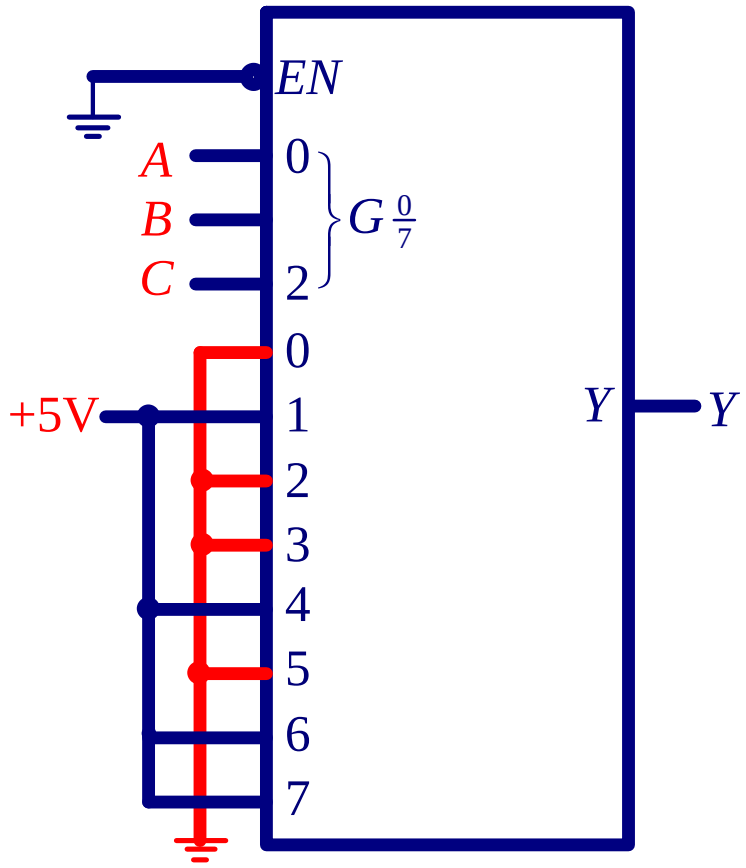
4. 画出逻辑图



课堂练习

1. The diagram shows a ___being used as a/an ___and Y= ___ .

C



A. Multiplier, divider, 2.25

B. multiplexer, encoder, 7641

C. multiplexer, logic function generator,

$Y=\overline{C}\,\overline{B}A+C\overline{B}\,\overline{A}+CB\overline{A}+CBA$

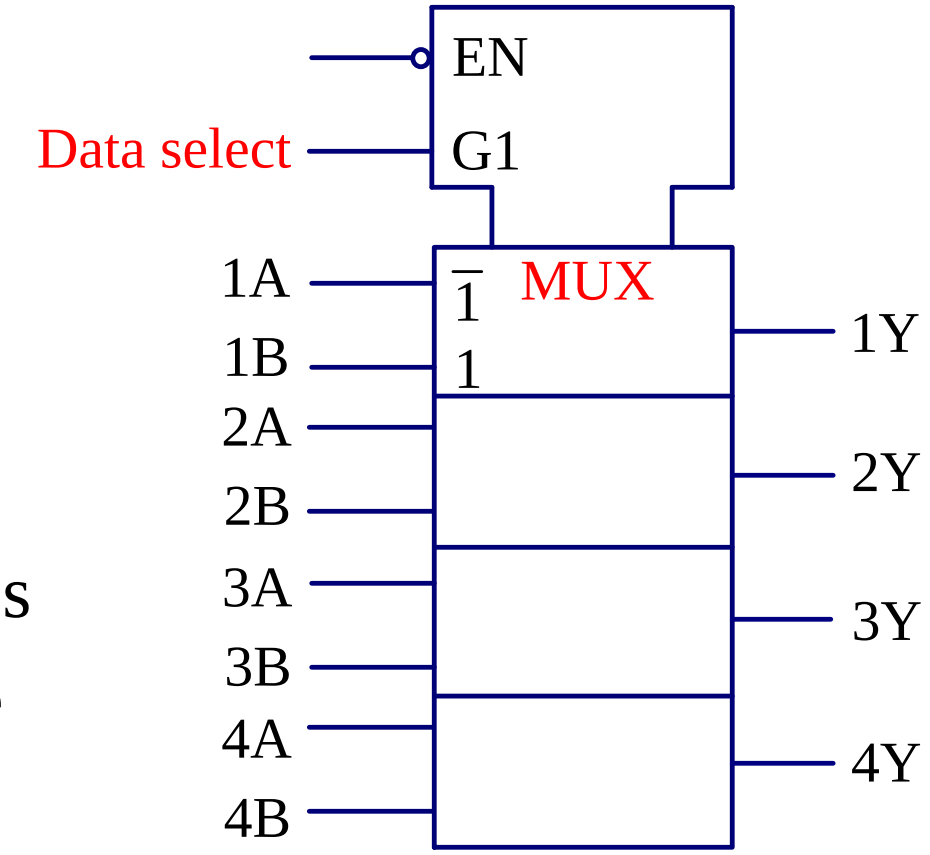
D. Logic function generator,

multiplexer,1467

2. Which statement best describes the following figure, and what is the function of the terminal labeled EN? A

A. Quad two-input multiplexer. EN is the enable input which requires an active LOW for the device to work.

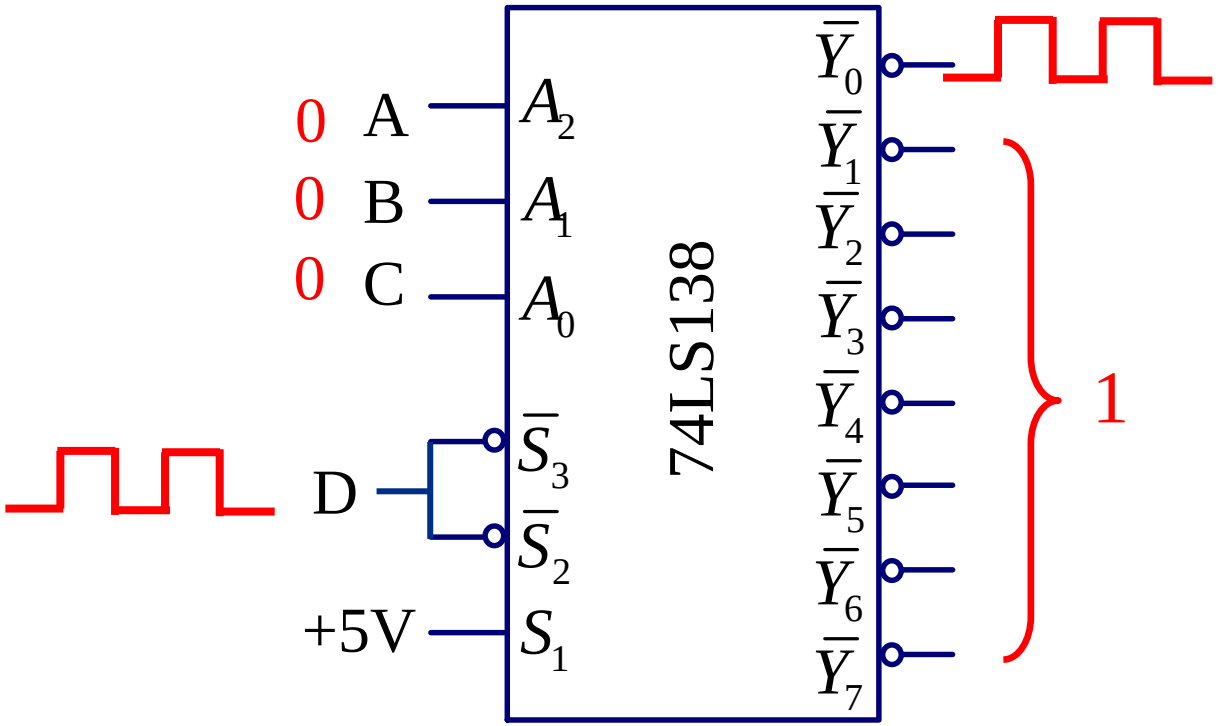
B. Quad two-input AND gate which requires an active LOW on the EN input to enable all the gate.

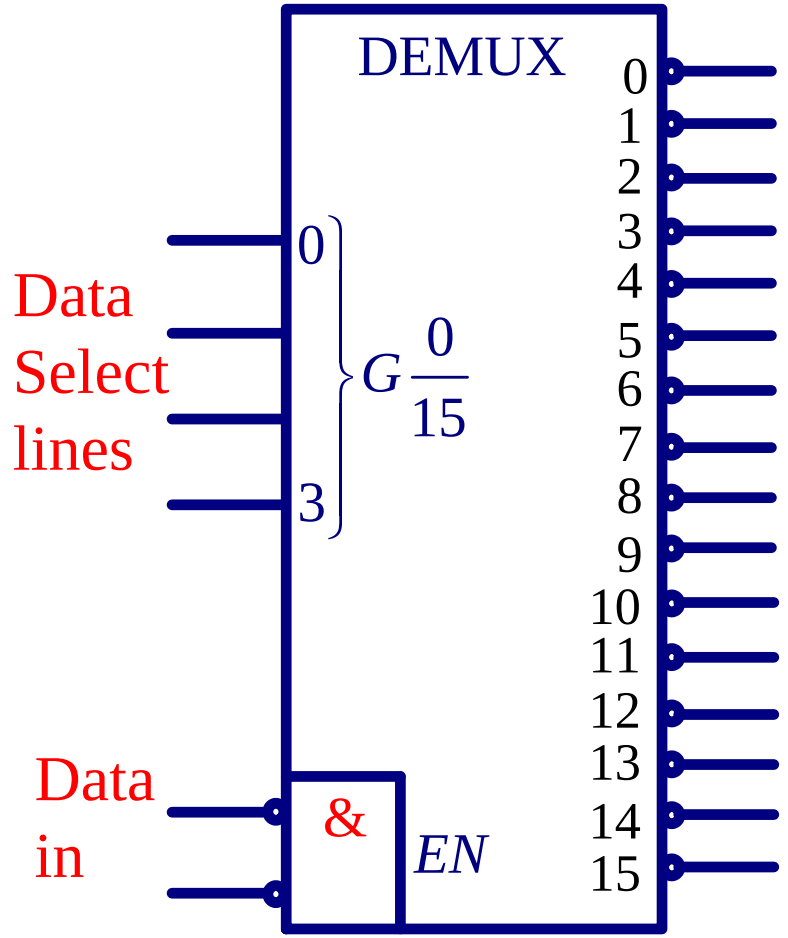


七、数据分配器 Demultiplexer

——将一路输入数据分配到多路输出

一般用译码器改接，不单独生产



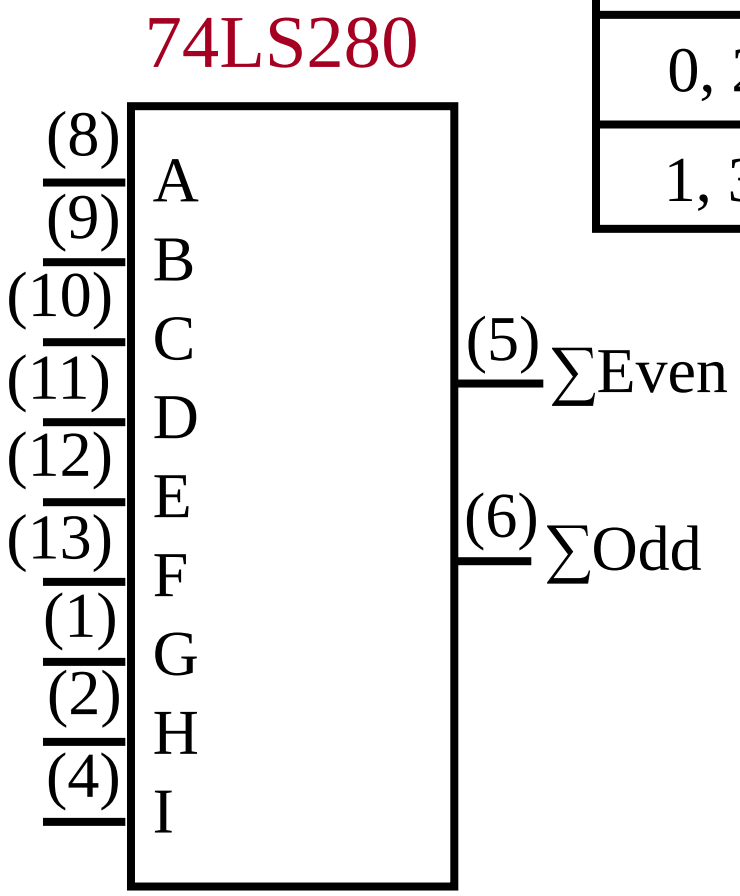


八、奇偶发生器/校验器 Parity generators/checkers

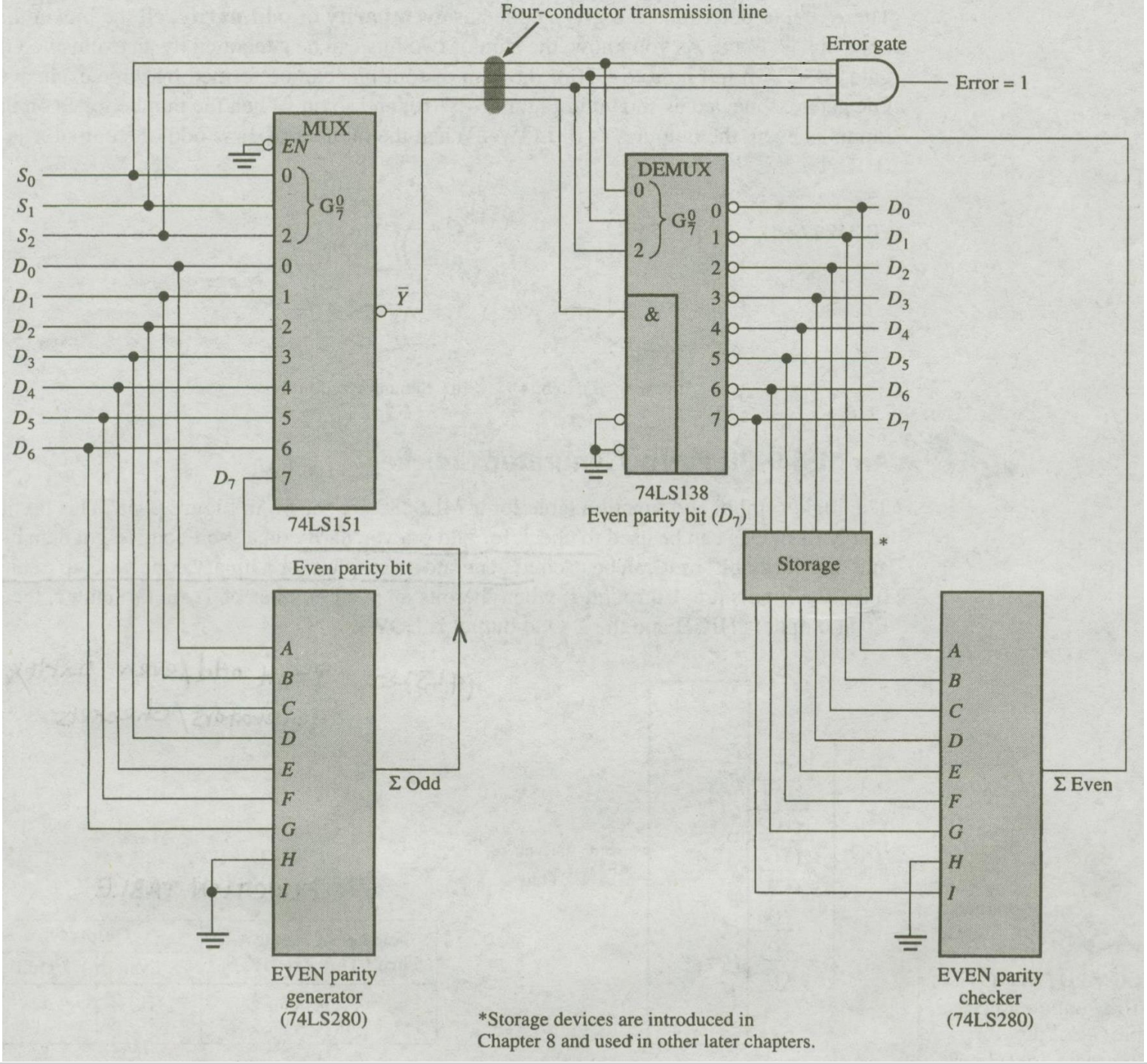
偶数个1之和为0。奇数个1之和为1

高电平输入数	∑Even	∑Odd
0, 2, 4, 6, 8	H	L
1, 3, 5, 7, 9	L	H

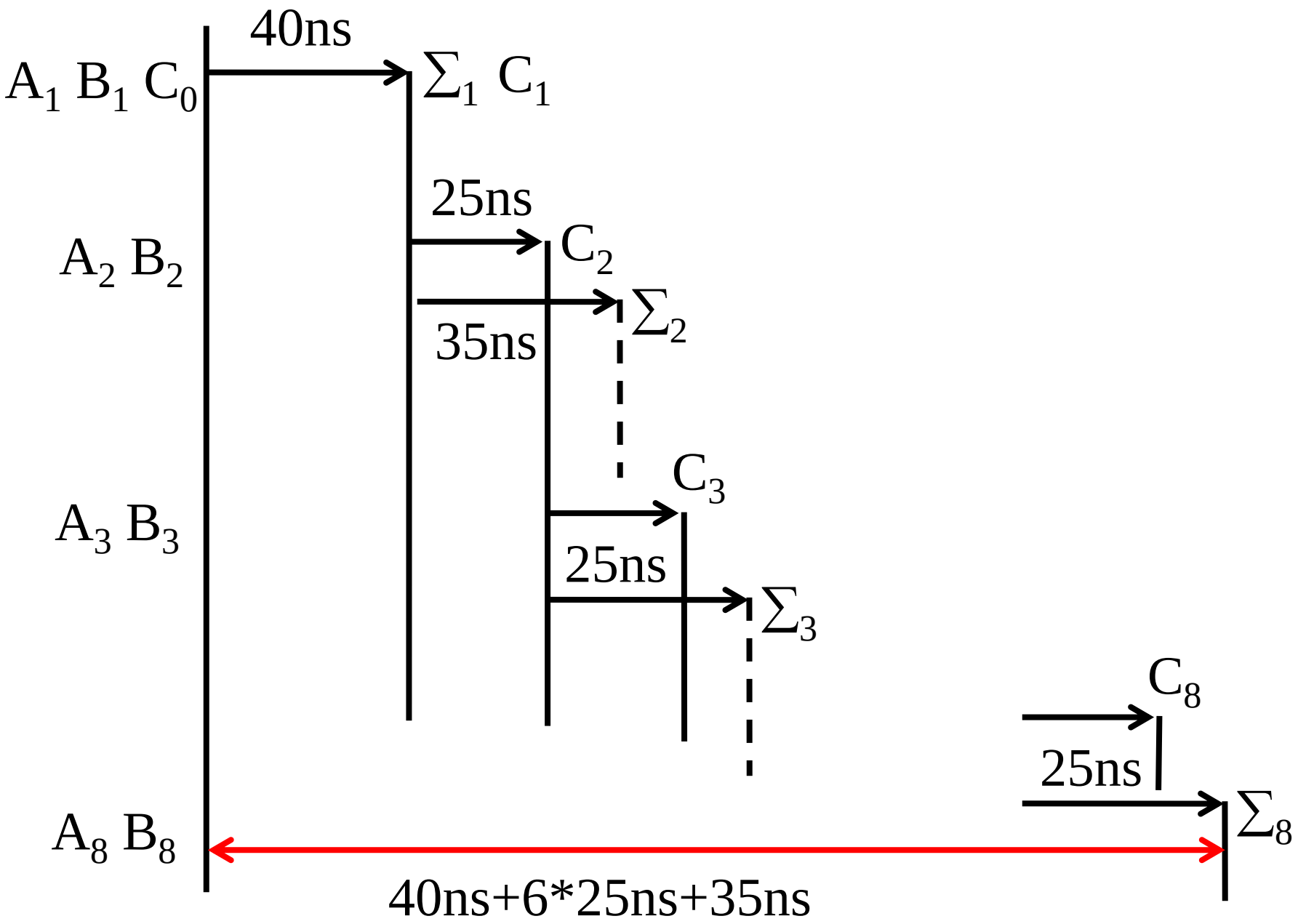
用于奇偶发生器时，可对一个9位二进制代码产生一个奇偶校验位。实现偶校验时，∑奇数(6引脚)作为校验位输出；实现奇校验时，∑偶数(5引脚)作为校验位输出。



用于奇偶校验时，可对9位代码（8个数据位，1个校验位）进行校验。


Four-conductor transmission line
Error gate
Error = 1
MUX
EN
S_0
S_1
S_2
D_0
D_1
D_2
D_3
D_4
D_5
D_6
D_7
$\bar{Y}$
74LS151
DEMUX
&
74LS138
Even parity bit (D_7)
Storage
*
Even parity bit
Σ Odd
Σ Even
A
B
C
D
E
F
G
H
I
EVEN parity generator (74LS280)
EVEN parity checker (74LS280)
*Storage devices are introduced in Chapter 8 and used in other later chapters.

6-11参考答案


40ns
A_1 B_1 C_0
$\sum_1$ C_1
25ns
A_2 B_2
C_2
$\sum_2$
35ns
C_3
A_3 B_3
25ns
$\sum_3$
C_8
25ns
$\sum_8$
A_8 B_8
40ns+6*25ns+35ns